

# DNA Replication, Repair, and Recombination

## CHAPTER

# 5

The ability of cells to maintain a high degree of order in a chaotic universe depends on the accurate duplication of vast quantities of genetic information carried in chemical form as DNA. This process, called *DNA replication*, must occur before a cell can produce two genetically identical daughter cells. Maintaining order also requires the continued surveillance and repair of this genetic information, because DNA inside cells is repeatedly damaged by chemicals and radiation from the environment, as well as by thermal accidents and reactive molecules generated inside the cell. In this chapter, we describe the protein machines that replicate and repair the cell's DNA. These machines catalyze some of the most rapid and accurate processes that take place within cells, and their mechanisms provide clear illustrations of the elegance and efficiency of cell chemistry.

The short-term survival of a cell depends on preventing harmful changes in its DNA. But the long-term survival of a species requires that these same DNA sequences be changeable over many generations to permit evolutionary adaptation to changing circumstances. We shall see that, despite the great efforts that cells make to protect their DNA, occasional changes in DNA sequences are unavoidable. These changes produce the genetic variation that is required for natural selection to drive the evolution of organisms.

We begin this chapter with a brief discussion of the changes that occur in DNA as it is passed down from generation to generation. Next, we discuss the mechanisms—DNA replication and DNA repair—that are responsible for minimizing these changes. Finally, we consider some of the most intriguing pathways that alter DNA sequences—those of *DNA recombination*. These pathways include the movement within chromosomes of special DNA sequences called transposable elements.

## THE MAINTENANCE OF DNA SEQUENCES

The survival of an individual organism demands a high degree of genetic stability. Only rarely do the cell's DNA-maintenance processes fail, resulting in permanent change in the DNA. Such a change is called a *mutation*, and it can destroy an organism if it occurs in a vital position in the DNA sequence.

### Mutation Rates Are Extremely Low

The **mutation rate**, the rate at which changes occur in DNA sequences, can be determined directly from experiments carried out with a bacterium such as *Escherichia coli*—a resident of our intestinal tract and a commonly used laboratory organism (see Figure 1-38). Under laboratory conditions, an *E. coli* cell divides about once every 30 minutes; as a result, a single cell can generate a very large population—several billion—in less than a day. In such a population, it is possible to detect the small fraction of bacteria that have suffered a damaging mutation in a particular gene. For example, the mutation rate of a gene specifically required for cells to use the sugar lactose as an energy source can be determined by growing the cells in the presence of a different sugar, such as

## IN THIS CHAPTER

The Maintenance of DNA Sequences

DNA Replication Mechanisms

The Initiation and Completion of DNA Replication in Chromosomes

DNA Repair

Homologous Recombination

Transposition and Conservative Site-specific Recombination

glucose, and testing them subsequently to see how many have lost the ability to survive on a lactose diet. The fraction of damaged genes will underestimate the actual mutation rate because many mutations are *silent* (for example, those that change a codon but not the amino acid it specifies or those that change an amino acid without affecting the activity of the protein coded for by the gene). After correcting for these silent mutations, one finds that bacteria display a mutation rate of about three nucleotide changes per  $10^{10}$  nucleotides copied.

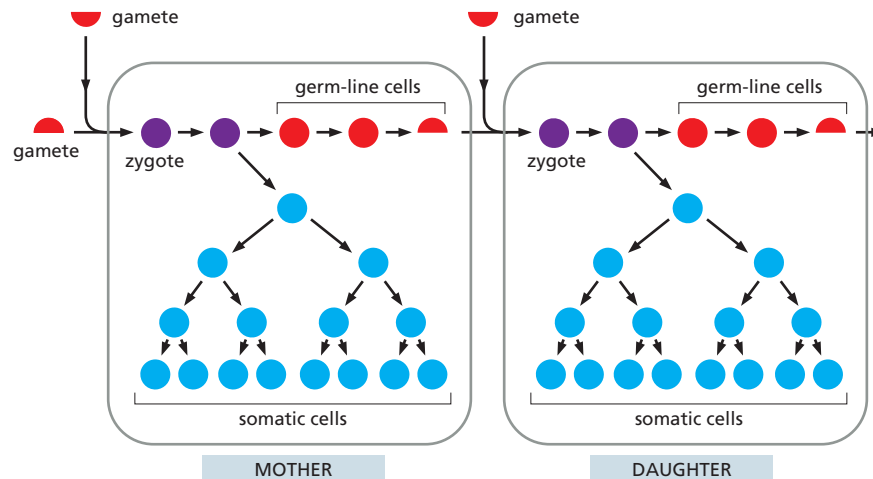
It is also possible to measure the germ-line mutation rate in more complex, sexually reproducing organisms such as humans. In this case, the complete genomes from a family—parents and offspring—are directly sequenced, and a careful comparison reveals that approximately 70 new single-nucleotide mutations typically arise in the germ lines of each offspring. Normalized to the size of the human genome, the mutation rate is one nucleotide change per  $10^8$  nucleotides per human generation. (This is a slight underestimate because some germ-line mutations will be lethal and will therefore be absent from progeny; however, because relatively little of the human genome carries critical information, this consideration has a negligible effect on the true mutation rate.) It is estimated that approximately 100 cell divisions occur in the germ line from the time of conception to the time of production of the eggs and sperm that go on to make the next generation. Thus, the human mutation rate, expressed in terms of cell divisions (instead of human generations), is approximately one nucleotide change per  $10^{10}$  nucleotides copied.

Although *E. coli* and humans differ greatly in their modes of reproduction and in their generation times, when the mutation rates of each are normalized to a single round of DNA replication, they are both extremely low and within a factor of 3 of each other. We shall see later in the chapter that the basic mechanisms that ensure these low rates of mutation have been conserved since the very early history of cells on Earth.

### Low Mutation Rates Are Necessary for Life as We Know It

Because many mutations are deleterious, no species can afford to allow them to accumulate at a high rate in its germ cells. Even though the observed mutation frequency is very low, it is thought to limit the number of essential genes that any organism can rely on to perhaps 30,000. More essential genes than this, and the probability that at least one critical component will suffer a damaging mutation becomes catastrophically high. By an extension of the same argument, a mutation frequency tenfold higher would limit an organism to about 3000 essential genes. In this case, evolution would have been limited to organisms considerably less complex than a fruit fly.

The cells of a sexually reproducing animal or plant are of two types: **germ cells** and **somatic cells**. The germ cells transmit genetic information from parent to offspring; the somatic cells form the body of the organism ([Figure 5-1](#)). We have seen that germ cells must be protected against high rates of mutation to maintain the species. However, the somatic cells of multicellular organisms must also be protected from genetic change to properly maintain the organized structure of the body. Nucleotide changes in somatic cells can give rise to variant cells, some of which, through “local” natural selection, proliferate rapidly at the expense of the rest of the organism. In an extreme case, the result is the uncontrolled cell proliferation that we know as cancer. This condition is due largely to an accumulation of changes in the DNA sequences of somatic cells, as discussed in Chapter 20. Any significant increase in the mutation frequency would presumably cause a disastrous increase in the incidence of cancer by accelerating the rate at which dangerous somatic-cell variants arise. Thus, both for the perpetuation of a species with a large number of genes (germ-cell stability) and for the prevention of cancer resulting from mutations in somatic cells (somatic-cell stability), multicellular organisms like ourselves absolutely depend on the remarkably high fidelity with which their DNA sequences are replicated and maintained.



**Figure 5-1 Germ-line cells and somatic cells carry out fundamentally different functions.** In sexually reproducing organisms, genetic information is propagated into the next generation exclusively by germ-line cells (red). This cell lineage includes the specialized reproductive cells—the gametes (eggs and sperm, half circles)—which contain only half the number of chromosomes as that contained in the other cells in the body (full circles). When two gametes come together during fertilization, they form a fertilized egg, or zygote (purple), which once again contains a full set of chromosomes. The zygote gives rise to both germ-line cells and somatic cells (blue). Somatic cells form the body of the organism but do not contribute their DNA to the next generation.

## Summary

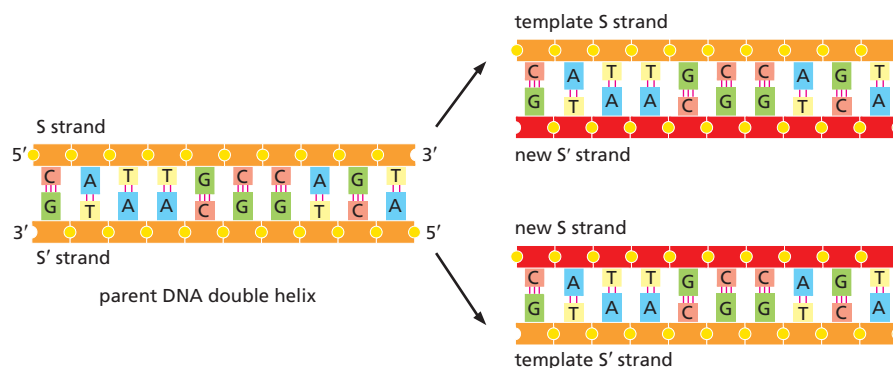
*In all cells, DNA sequences are maintained and replicated with extremely high fidelity. The mutation rate, approximately one nucleotide change per  $10^{10}$  nucleotides each time the DNA is replicated, is very similar for organisms as different as bacteria and humans. Because of this remarkable accuracy, the sequence of the human genome (approximately  $3.1 \times 10^9$  nucleotide pairs) is unchanged or changed by only a few nucleotides each time a typical human cell divides. This allows humans to pass accurate genetic instructions from one generation to the next and also—for most of us—to avoid the changes in somatic cells that lead to cancer.*

## DNA REPLICATION MECHANISMS

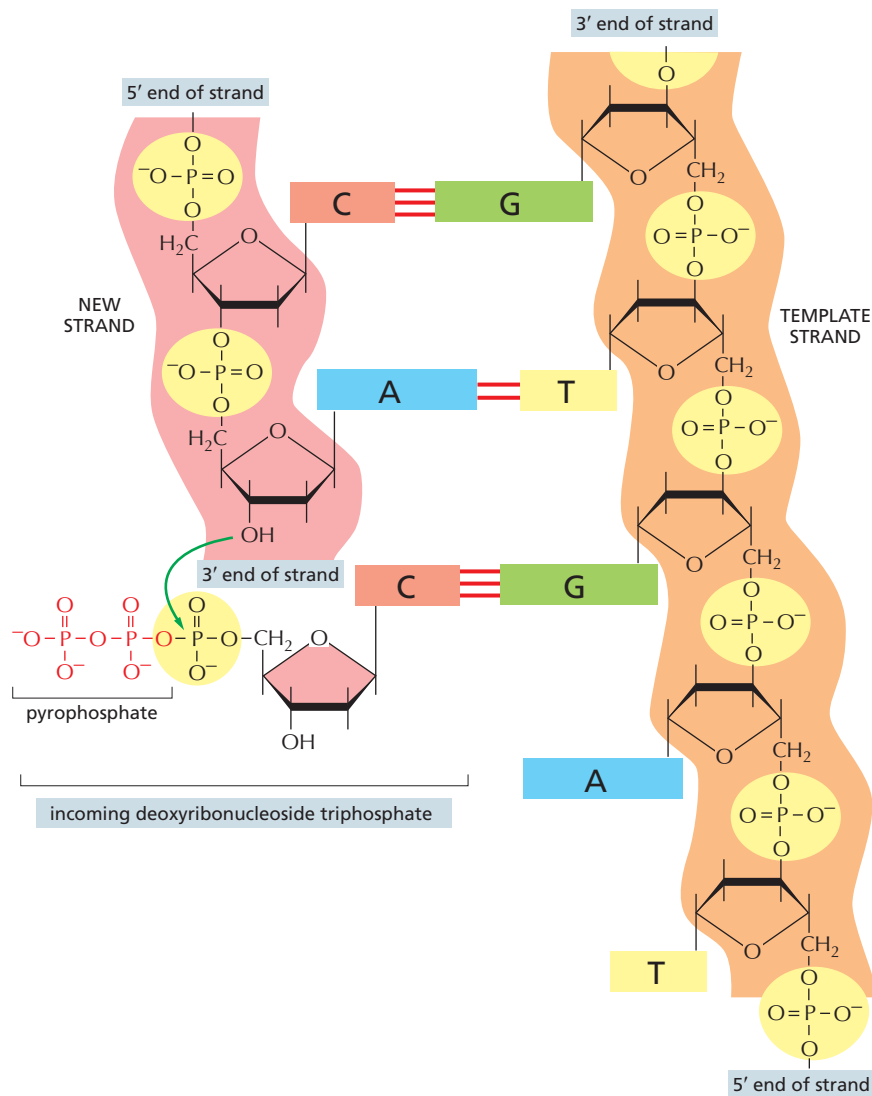
All organisms duplicate their DNA with extraordinary accuracy before each cell division. In this part of the chapter, we explore how an elaborate “replication machine” achieves this accuracy, while duplicating DNA at rates as high as 1000 nucleotides per second.

### Base-pairing Underlies DNA Replication and DNA Repair

As introduced in Chapter 1, *DNA templating* is the mechanism the cell uses to copy the nucleotide sequence of one DNA strand into a complementary DNA sequence (Figure 5-2). This process requires the separation of the DNA helix into two template strands, and it entails the recognition of each nucleotide in the DNA *template strands* by a free (unpolymerized) complementary nucleotide. The separation of the DNA helix exposes the hydrogen-bond donor and acceptor groups on each DNA base to allow its base-pairing with the appropriate incoming free nucleotide, aligning it for its enzyme-catalyzed polymerization into a new DNA chain.



**Figure 5-2 DNA acts as a template for its own replication.** Because the nucleotide A will successfully pair only with T, and G with C, each strand of a DNA double helix—labeled here as the S strand and its complement, the S' strand—can serve as a template to specify the sequence of nucleotides in a complementary strand. In this way, both strands of a DNA double helix can be copied with precision, producing two exact copies of the original double helix. How complementary nucleotides base-pair is shown in Figure 4-5.



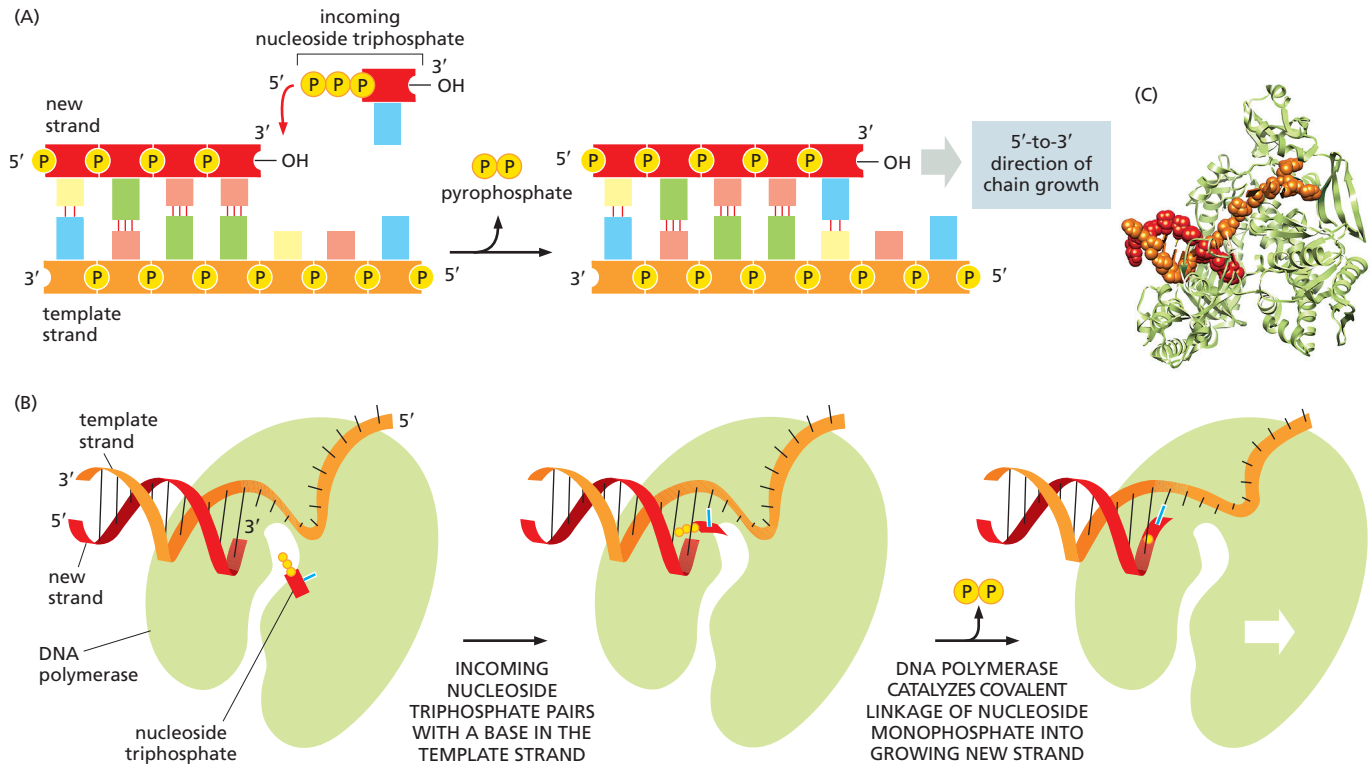
**Figure 5-3 The chemistry of DNA synthesis.** Nucleotides enter the reaction as deoxyribonucleoside triphosphates, and the addition of a deoxyribonucleotide to the 3' end of a polynucleotide chain is the fundamental reaction by which DNA is synthesized. As shown, base-pairing between an incoming deoxyribonucleoside triphosphate and an existing strand of DNA (the *template strand*) guides the formation of the new strand of DNA and ensures that its nucleotide sequence is complementary to that of the template.

The first nucleotide-polymerizing enzyme, **DNA polymerase**, was discovered in 1957. The free nucleotides that serve as substrates for this enzyme were found to be deoxyribonucleoside triphosphates, and their polymerization into DNA required a single-strand DNA template. **Figure 5-3** and **Figure 5-4** illustrate the stepwise mechanism of this reaction.

### The DNA Replication Fork Is Asymmetrical

During DNA replication inside a cell, each of the two original DNA strands serves as a template for the formation of an entire new strand. Because each of the two daughters of a dividing cell inherits a new DNA double helix containing one original and one new strand (**Figure 5-5**), the DNA double helix is said to be replicated *semiconservatively*. How is this feat actually accomplished?

Analyses carried out in the early 1960s on the whole replicating chromosome of an *E. coli* bacterium revealed a localized region of replication that moves progressively along the parent DNA double helix. Because of its Y-shaped structure,

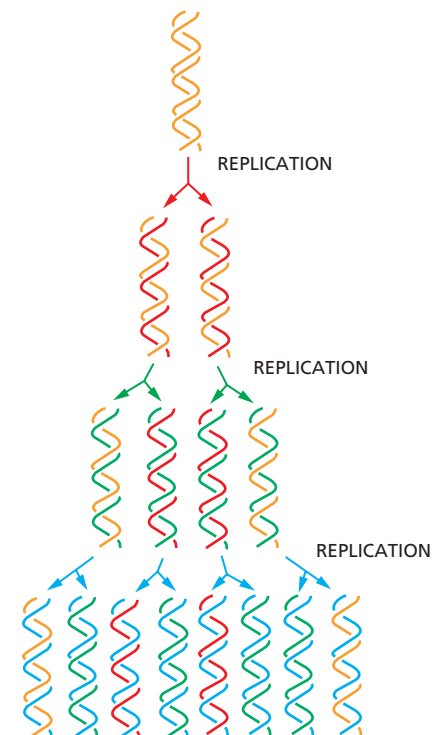


**Figure 5-4** How DNA polymerase adds a deoxyribonucleotide to the end of a growing DNA strand. (A) An incoming deoxynucleoside triphosphate forms a base pair with its partner in the template strand. It is then covalently attached to the free 3'-hydroxyl (3'-OH) end of the growing DNA strand. The new DNA strand is therefore synthesized in the 5'-to-3' direction. The energy for the polymerization reaction comes from the hydrolysis of a high-energy phosphate bond in the incoming nucleoside triphosphate and the release of pyrophosphate, which is subsequently hydrolyzed to yield two molecules of inorganic phosphate (not shown). (B) The reaction is catalyzed by the enzyme DNA polymerase (light green). The polymerase guides the incoming nucleoside triphosphate to the template strand and positions it such that its 5' triphosphate will be able to react with the 3'-hydroxyl group on the newly synthesized strand. The white arrow indicates the direction of polymerase movement. (C) Structure of DNA polymerase, as determined by x-ray crystallography, also showing the replicating DNA. The template strand is the longer, orange strand, and the DNA strand being synthesized is colored red. See [Movie 5.1](#). (C, PDB code: 1KRP)

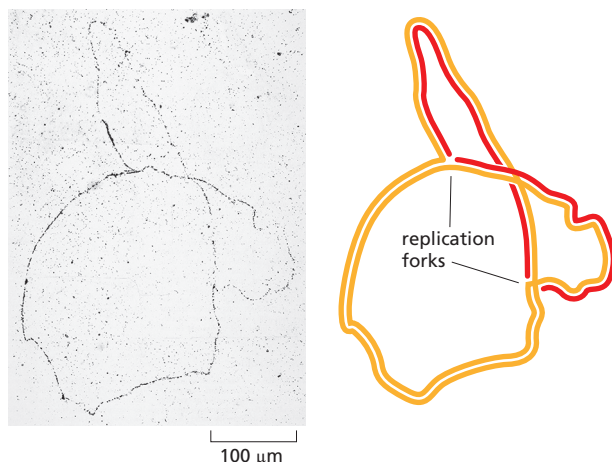
this active region is called a **replication fork** ([Figure 5-6](#)). At the replication fork, a multienzyme complex that contains the DNA polymerase synthesizes the DNA of both new daughter strands.

Initially, the simplest mechanism of DNA replication seemed to be the continuous growth of both new strands, nucleotide by nucleotide, at the replication fork as it moves from one end of a DNA molecule to the other. But because of the antiparallel orientation of the two DNA strands in the DNA double helix (see [Figure 5-2](#)), this mechanism would require one daughter strand to polymerize in the 5'-to-3' direction and the other in the 3'-to-5' direction. Such a replication fork would require two distinct types of DNA polymerase enzymes. However, as attractive as this model might seem, the DNA polymerases at replication forks can synthesize only in the 5'-to-3' direction.

How, then, can a DNA strand grow in the 3'-to-5' direction? The answer came from an experiment performed in the late 1960s. Researchers added highly radioactive  $^3\text{H}$ -thymidine to dividing bacteria for a few seconds, so that only the most recently replicated DNA—that just behind the replication fork—became radiolabeled. This experiment revealed the transient existence of pieces of



**Figure 5-5** In each round of DNA replication, each of the two strands of DNA is used as a template for the formation of a new, complementary strand. DNA replication is *semiconservative* because each daughter DNA double helix is composed of one conserved (old) strand and one newly synthesized strand.



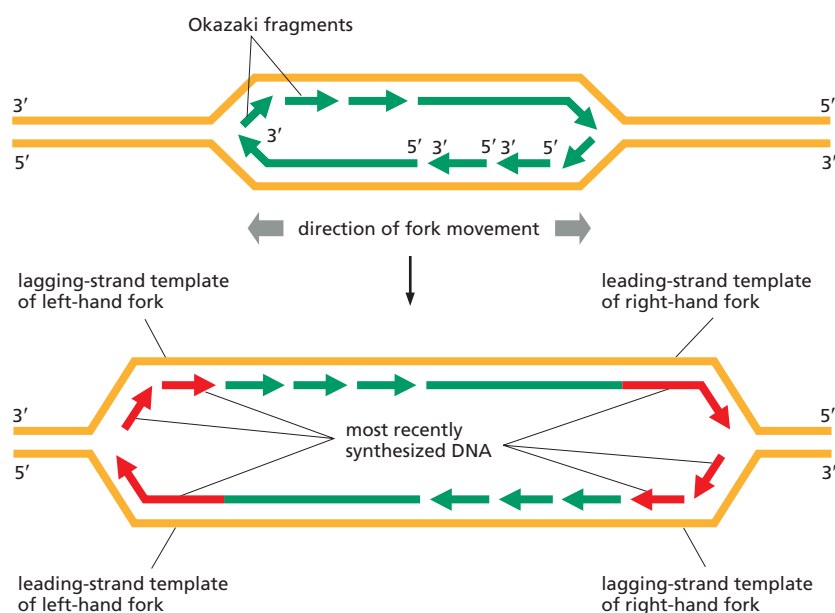
**Figure 5-6** Two replication forks moving in opposite directions on the *E. coli* chromosome, a large circular DNA molecule. Each replication fork has a Y-shaped structure and moves progressively along the DNA, spinning out newly replicated DNA behind it. The stem of the Y is the parent DNA double helix, and the two arms of the Y contain the newly synthesized DNA. The image on the left was obtained by feeding *E. coli* radioactive thymine for several hours, gently isolating the DNA on filter paper, and placing a piece of photographic film next to the DNA. Because radioactivity exposes photographic film, an image of the DNA was captured when the film was developed and viewed under a light microscope. The diagram on the right is an interpretation of the result, with parent DNA strands in orange and newly synthesized DNA strands in red. During its isolation for this experiment, the *E. coli* DNA folded on itself, accounting for the crossing of the double helix. (From J. Cairns, *Cold Spring Harb. Symp. Quant. Biol.* 38:43–46, 1963. With permission from Cold Spring Harbor Laboratory Press.)

DNA that were 1000–2000 nucleotides long, now commonly known as *Okazaki fragments* (named for their discoverer), at the growing replication fork. Similar replication intermediates were later found in eukaryotes, where they are only 100–200 nucleotides long. The Okazaki fragments were shown to be synthesized only in the 5′-to-3′ chain direction and to be joined together after their synthesis to create long DNA chains.

Each replication fork therefore has an asymmetric structure (Figure 5-7). The DNA daughter strand that is synthesized continuously is known as the **leading strand**. Its synthesis slightly precedes the synthesis of the daughter strand that is synthesized discontinuously, known as the **lagging strand**. For the lagging strand, the direction of nucleotide polymerization is opposite to the overall direction of DNA chain growth. The synthesis of this strand by a discontinuous “backstitching” mechanism means that DNA replication requires only the 5′-to-3′ type of DNA polymerase.

### The High Fidelity of DNA Replication Requires Several Proofreading Mechanisms

As discussed at the beginning of the chapter, the fidelity of copying DNA during replication is such that only about one mistake occurs for every  $10^{10}$  nucleotides copied. This accuracy is much higher than one would expect solely from the



**Figure 5-7** At each replication fork, the lagging DNA strand is synthesized in pieces. The upper diagram shows two replication forks moving in opposite directions on a double-helical DNA molecule, as in Figure 5-6; the lower diagram shows the same two forks a short time later. Because both of the new strands at a replication fork are synthesized in the 5′-to-3′ direction, the lagging strand of DNA must be made initially as a series of short DNA strands, which are later joined together. To replicate the lagging strand, the DNA polymerase molecule on that side of the fork uses a backstitching mechanism: it synthesizes a short piece of DNA in the 5′-to-3′ direction, stops, and is then moved by its protein machine back toward the fork in order to synthesize the next fragment.

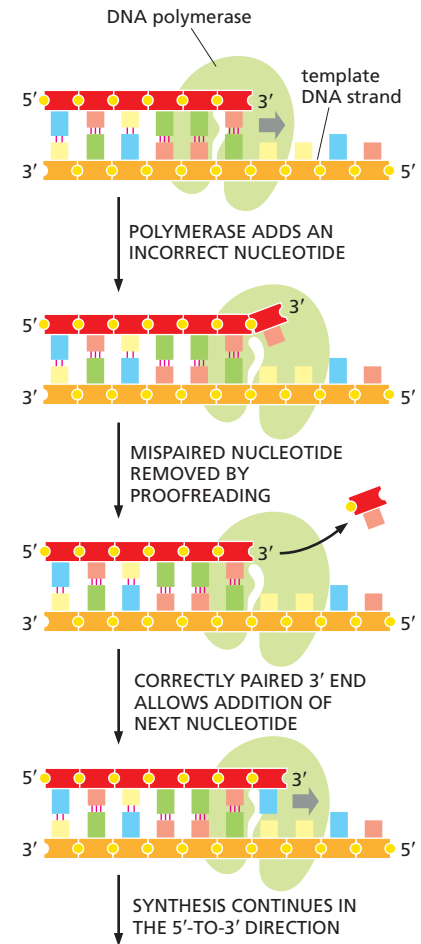
properties of complementary base-pairing. The standard complementary base pairs (see Figure 4-5) are not the only ones possible. For example, with small changes in helix geometry, two hydrogen bonds can form between G and T in DNA. In addition, rare configurations of the four DNA bases (known as tautomers) occur transiently in ratios of 1 part to  $10^4$  or  $10^5$ . These forms mispair without a change in helix geometry: the rare tautomeric form of C pairs with A instead of G, for example.

If the DNA polymerase did nothing special when a mispairing occurred between an incoming deoxyribonucleoside triphosphate and the DNA template, the wrong nucleotide would often be incorporated into the new DNA chain, producing frequent mutations. The high fidelity of DNA replication, however, depends not only on the initial base-pairing, but also on several “proofreading” mechanisms that act sequentially to correct any initial mispairings that might have occurred.

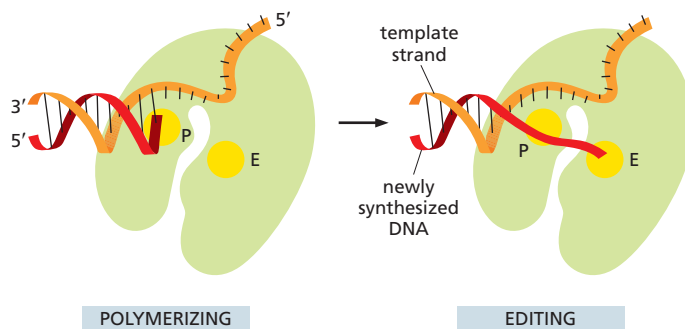
DNA polymerase performs the first proofreading step just before a new nucleotide is covalently added to the growing chain. After complementary nucleotide binding, but before the nucleotide is covalently added to the growing chain, the enzyme must undergo a conformational change in which its “grip” tightens around the active site. Because this change occurs more readily with correct than incorrect base-pairing, it allows the polymerase to “double-check” the exact base-pair geometry before it catalyzes the addition of the nucleotide. Incorrectly paired nucleotides are harder to add and therefore more likely to diffuse away before the polymerase can mistakenly add them.

The next error-correcting reaction, known as *exonucleolytic proofreading*, takes place immediately after those rare instances in which an incorrect nucleotide is covalently added to the growing chain. DNA polymerase enzymes are highly discriminating in the types of DNA chains they will elongate: they require a previously formed, base-paired 3'-OH end of a *primer strand* (see Figure 5-4). Those DNA molecules with a mismatched (improperly base-paired) nucleotide at the 3'-OH end of the primer strand are not effective as templates because the polymerase has difficulty extending such a strand. DNA polymerase molecules correct such a mismatched primer strand by means of a separate catalytic site (either in a separate subunit or in a separate protein domain of the polymerase molecule, depending on the polymerase). This 3'-to-5' *proofreading exonuclease* clips off any unpaired or mispaired residues at the primer terminus, continuing until enough nucleotides have been removed to regenerate a correctly base-paired 3'-OH terminus that can prime DNA synthesis. In this way, DNA polymerase functions as a “self-correcting” enzyme that removes its own polymerization errors as it moves along the DNA (Figure 5-8 and Figure 5-9).

Because the self-correcting properties of DNA polymerase depend on its requirement for a perfectly base-paired primer terminus, it is apparently not possible for such an enzyme to start DNA synthesis *de novo* without an existing primer. By contrast, the RNA polymerase enzymes involved in gene transcription do not need such an efficient exonucleolytic proofreading mechanism: errors in making RNA are not passed on to the next generation, and the occasional defective RNA molecule that is produced has no long-term significance. As a result,



**Figure 5-8** During DNA synthesis, DNA polymerase proofreads its own work. If an incorrect nucleotide is accidentally added to a growing strand, the DNA polymerase stops, cleaves it from the strand, and replaces it with the correct nucleotide before continuing.



**Figure 5-9** DNA polymerase contains separate sites for DNA synthesis and proofreading. The DNA polymerase, which cradles the DNA molecule being replicated, is shown in the polymerizing mode (*left*) and in the proofreading, or editing, mode (*right*). The catalytic sites for the polymerization activity (P) and editing activity (E) are indicated. When the polymerase adds an incorrect nucleotide, the newly synthesized DNA strand (red) transiently unpairs from the template strand (orange), and its 3' end moves into the editing site (E) to allow the incorrect nucleotide to be removed. These diagrams are based on the structure of an *E. coli* DNA polymerase molecule, as determined by x-ray crystallography.

**TABLE 5–1 The Three Steps That Give Rise to High-Fidelity DNA Synthesis**

| Replication step                  | Errors per nucleotide added |
|-----------------------------------|-----------------------------|
| 5'→3' polymerization              | 1 in 10 <sup>5</sup>        |
| 3'→5' exonucleolytic proofreading | 1 in 10 <sup>2</sup>        |
| Strand-directed mismatch repair   | 1 in 10 <sup>3</sup>        |
| Combined                          | 1 in 10 <sup>10</sup>       |

The third step, strand-directed mismatch repair, is described later in this chapter. For the polymerization step, “errors per nucleotide added” describes the probability that an incorrect nucleotide will be added to the growing chain. For the other two steps, “errors per nucleotide added” describes the probability that an error will not be corrected. Each step therefore reduces the chance of a final error by the factor shown.

RNA polymerases do not require a base-paired end 3'-OH for nucleotide addition and are able to start new polynucleotide chains without a primer.

On average, about one mistake is made for every 10<sup>4</sup> polymerization events both in RNA synthesis and in the separate process of translating mRNA sequences into protein sequences. This error rate is over 100,000 times greater than that in DNA replication, where, as we have seen, a series of proofreading processes makes the process unusually accurate (Table 5–1).

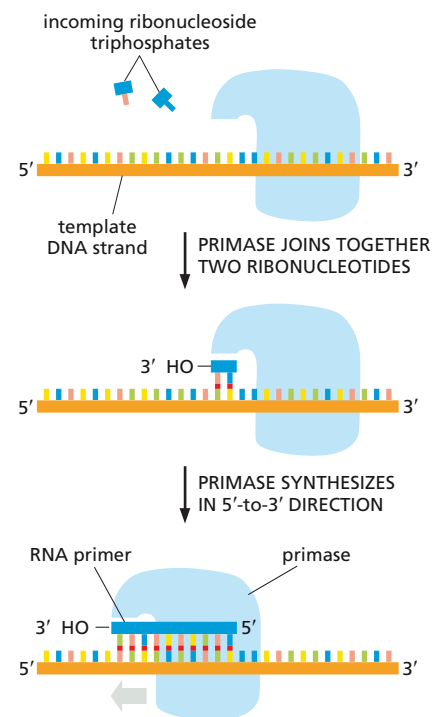
### DNA Replication in the 5'-to-3' Direction Allows Efficient Error Correction

The need for accuracy probably explains why DNA replication occurs only in the 5'-to-3' direction. If there were a DNA polymerase that added deoxyribonucleoside triphosphates in the 3'-to-5' direction, the growing 5' end of the chain, rather than the incoming mononucleotide, would have to provide the activating triphosphate needed for the covalent linkage (see Figure 5–3). In this case, the mistakes in polymerization could not be simply hydrolyzed away, because the bare 5' end of the chain thus created would immediately terminate DNA synthesis. It is therefore possible to correct a mismatched base only if it has been added to the 3' end of a DNA chain. Although the backstitching mechanism for DNA replication seems complex, it preserves the 5'-to-3' direction of polymerization that is required for exonucleolytic proofreading.

Despite these safeguards against DNA replication errors, DNA polymerases occasionally leave mistakes behind in the DNA that they produce. However, as we shall see later in this chapter, cells have yet another chance to correct these errors by a process called *strand-directed mismatch repair*. Before discussing this mechanism, however, we describe the other types of proteins that function at the replication fork as part of a large protein machine that replicates DNA.

### A Special Nucleotide-polymerizing Enzyme Synthesizes Short RNA Primer Molecules

For the leading strand, a primer is needed only at the start of replication: once a replication fork is established, the DNA polymerase is continuously presented with a base-paired chain end on which to add new nucleotides. On the lagging side of the fork, however, each time the DNA polymerase completes a short DNA Okazaki fragment (which takes a few seconds), it must start synthesizing a completely new fragment at a site further along the template strand (see Figure 5–7). Each time this occurs, a special mechanism is required to produce a base-paired primer strand for the DNA polymerase to elongate. This requires an enzyme called **DNA primase** that uses ribonucleoside triphosphates to synthesize short **RNA primers** on the lagging strand (Figure 5–10). In eukaryotes, these primers are about 10 nucleotides long and are made at intervals of 100–200 nucleotides



**Figure 5–10** RNA primers are synthesized by an RNA polymerase called DNA primase, which uses a DNA strand as a template. Like DNA polymerase, primase synthesizes in the 5'-to-3' direction. Unlike DNA polymerase, however, primase can start a new polynucleotide chain by joining together two nucleoside triphosphates without the need for a base-paired 3' end as a starting point. A DNA primase uses ribonucleoside triphosphates rather than deoxyribonucleoside triphosphates, and it is much less accurate than a DNA polymerase.

**Figure 5–11** Different enzymes act in series to synthesize DNA on the lagging strand. In eukaryotes, RNA primers are made at intervals of about 200 nucleotides on the lagging strand, and each RNA primer is approximately 10 nucleotides long. These primers are extended by DNA polymerases at the replication fork to produce Okazaki fragments. The primers are subsequently removed by nucleases that recognize the RNA strand in an RNA–DNA hybrid helix and destroy it; this leaves gaps that are filled in by an accurate “repair” DNA polymerase that proofreads as it fills in the gaps. The completed DNA fragments are finally joined together by an enzyme called DNA ligase, which catalyzes the formation of a phosphodiester bond between the 3′-hydroxyl end of one fragment and the 5′-phosphate end of the next, thus linking up the sugar–phosphate backbones. This nick-sealing reaction requires an input of energy in the form of ATP (see Figure 5–12).

on the lagging strand. The synthesis of the leading strand also requires an RNA primer, but only at its very beginning.

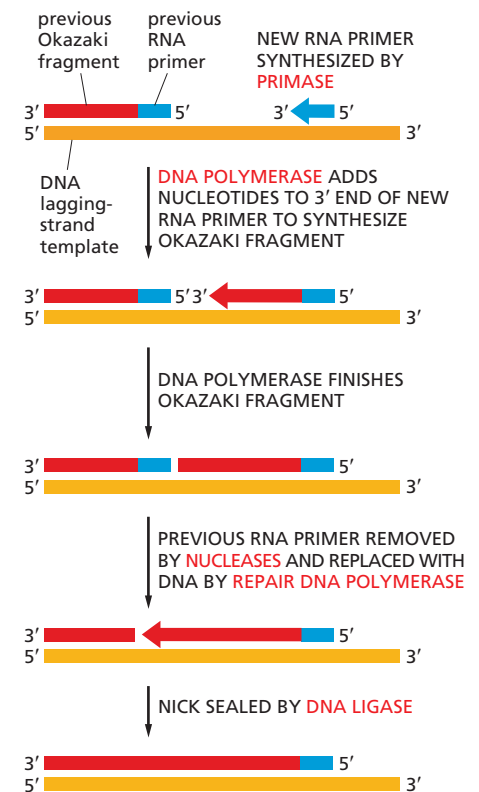
RNA was introduced in Chapter 1 and is described in detail in Chapter 6. Here, we note only that RNA is very similar in structure to DNA. A strand of RNA can form base pairs with a strand of DNA, generating a DNA–RNA hybrid double helix if the two nucleotide sequences are complementary. Thus, the same templating principle used for DNA synthesis guides the synthesis of RNA primers. Because an RNA primer contains a properly base-paired nucleotide with a 3′-OH group at one end, it can be elongated by the DNA polymerase at this end to begin an Okazaki fragment.

The synthesis of each Okazaki fragment ends when this DNA polymerase runs into the RNA primer attached to the 5′ end of the previous fragment. To produce a continuous DNA chain from the many DNA fragments made on the lagging strand, a special DNA repair system acts quickly to remove the RNA primers and replace them with DNA. An enzyme called **DNA ligase** then joins the 3′ end of the new DNA fragment to the 5′ end of the previous one to complete the process (**Figure 5–11** and **Figure 5–12**).

Why might an erasable RNA primer be used instead of a DNA primer? The argument that a self-correcting polymerase cannot start chains *de novo* also implies the converse: an enzyme that starts chains anew cannot be efficient at self-correction. Thus, any enzyme that primes the synthesis of Okazaki fragments will of necessity make a relatively inaccurate copy. If these inaccurate copies were allowed to remain, the resulting increase in the overall mutation rate would be enormous. It therefore seems likely that the use of RNA rather than DNA for priming brings a powerful advantage to the cell: the ribonucleotides in the primer automatically mark these sequences as “suspect copy” to be efficiently removed and replaced by DNA produced by a highly accurate DNA polymerase.

### Special Proteins Help to Open Up the DNA Double Helix in Front of the Replication Fork

For DNA synthesis to proceed, the DNA double helix must be opened up ahead of the replication fork so that the incoming deoxyribonucleoside triphosphates can form base pairs with the template strand. The DNA double helix is very stable under physiological conditions: the base pairs are locked in place so strongly that it requires temperatures approaching that of boiling water to separate the two strands in a test tube. For this reason, two additional types of replication proteins—DNA helicases and single-strand DNA-binding proteins—are needed



**Figure 5–12** DNA ligase joins together Okazaki fragments on the lagging strand during DNA synthesis. The ligase enzyme uses a molecule of ATP to activate the 5′ phosphate of one fragment (step 1) before forming a new bond with the 3′ hydroxyl of the other fragment (step 2).



**Figure 5–13** How DNA helicase enzymes can separate strands as they move along a DNA single strand. An experiment is diagrammed, in which a short, complementary DNA fragment is base-paired to a longer DNA strand to form a region of DNA double helix. Because the purified DNA helicase added acts as a “moving wedge,” the double helix is pulled apart as the helicase runs unidirectionally along the DNA single strand, releasing the short DNA strand in a reaction that requires the presence of both the helicase protein and ATP. The rapid stepwise movement of the helicase is powered by its ATP hydrolysis (shown schematically in Figure 3–71A). As indicated, many DNA helicases are composed of a ring of six subunits.

to open the double helix and present an appropriate single-stranded DNA template for the DNA polymerase to copy.

**DNA helicases** were first isolated as proteins that hydrolyze ATP when they are bound to single strands of DNA. As described in Chapter 3, the binding and hydrolysis of ATP can change the shape of a protein molecule in a cyclical manner that allows the protein to perform mechanical work. DNA helicases use this principle to propel themselves rapidly along a single DNA strand. When they encounter a region of double helix, they continue to move along their strand, thereby prying apart the helix ahead of them. This unidirectional movement can occur at rates of up to 1000 nucleotides per second (**Figure 5–13** and **Figure 5–14**).

The two strands of DNA have opposite polarities, and, in principle, a helicase could unwind the DNA double helix in front of a replication fork by moving either in the 5′-to-3′ direction along one strand or in the 3′-to-5′ direction along the other strand. In fact, both types of DNA helicase exist. In the best-understood replication systems in bacteria, a helicase moving 5′-to-3′ along the lagging-strand template has the predominant role.

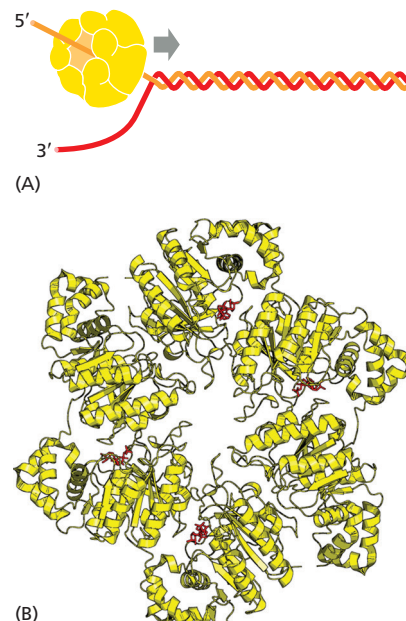
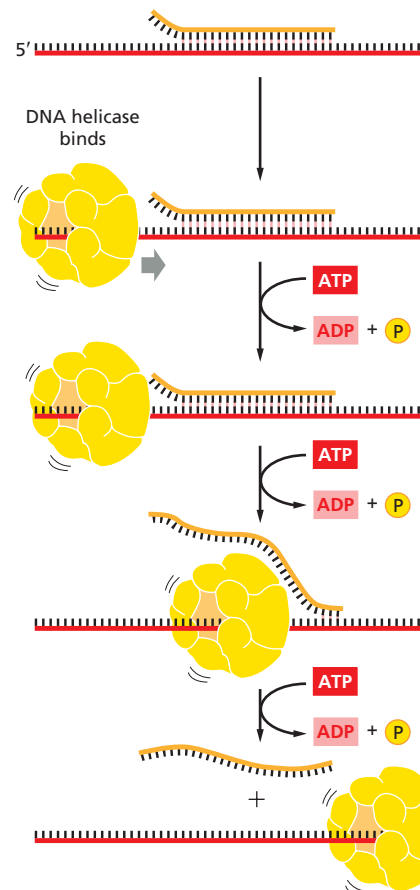
**Single-strand DNA-binding (SSB) proteins** bind tightly and cooperatively to the single-stranded DNA that is produced by helicases. Through cooperative binding, SSB proteins coat and straighten out all regions of single-stranded DNA, thereby preventing the formation of the short hairpin helices that otherwise form in these single strands (**Figure 5–15** and **Figure 5–16**). These regions occur routinely on the lagging-strand template, and if not removed, they can impede the DNA synthesis catalyzed by DNA polymerase.

### A Sliding Ring Holds a Moving DNA Polymerase onto the DNA

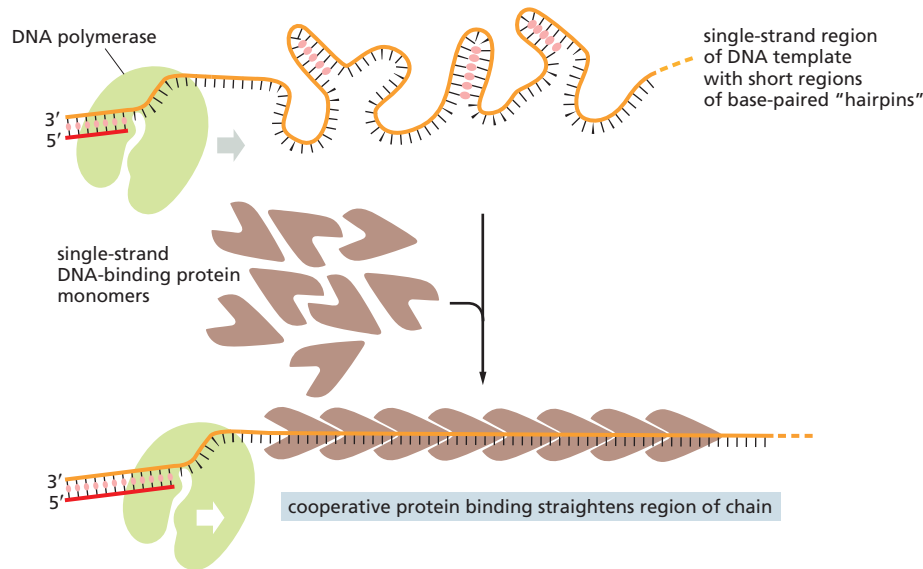
On their own, most DNA polymerase molecules will synthesize only a short string of nucleotides before falling off the DNA template. However, an accessory protein (called PCNA in eukaryotes) forms a **sliding clamp** that keeps the polymerase firmly on the DNA when it is moving but releases the polymerase as soon as it runs into a double-strand region of DNA.

How can a sliding clamp prevent the polymerase from dissociating without impeding the polymerase’s rapid movement along DNA? The three-dimensional structure of the clamp protein revealed that it forms a large ring around the DNA double helix. One face of the ring binds to the back of the DNA polymerase, and the whole ring slides freely along the DNA as the polymerase moves. The assembly of the clamp around the DNA requires a special protein complex, the **clamp loader**, that can open and close the ring in a regulated manner.

The moving DNA polymerase is tightly bound to the clamp, and, on the leading strand, the two remain associated for a very long time. The DNA polymerase on the lagging-strand template also makes use of the clamp, but each time the polymerase reaches the 5′ end of the preceding Okazaki fragment, the polymerase



**Figure 5–14** The structure of a DNA helicase. (A) Diagram of the protein, a hexameric ring, drawn to scale with a replication fork. (B) Detailed structure of the bacteriophage T7 replicative helicase, as determined by x-ray diffraction. Six identical subunits bind and hydrolyze ATP in an ordered fashion to propel this molecule, like a rotary engine, along a DNA single strand that passes through the central hole. Bound ATP molecules in the structure are indicated in red (**Movie 5.2**). (PDB code: 1EQJ.)

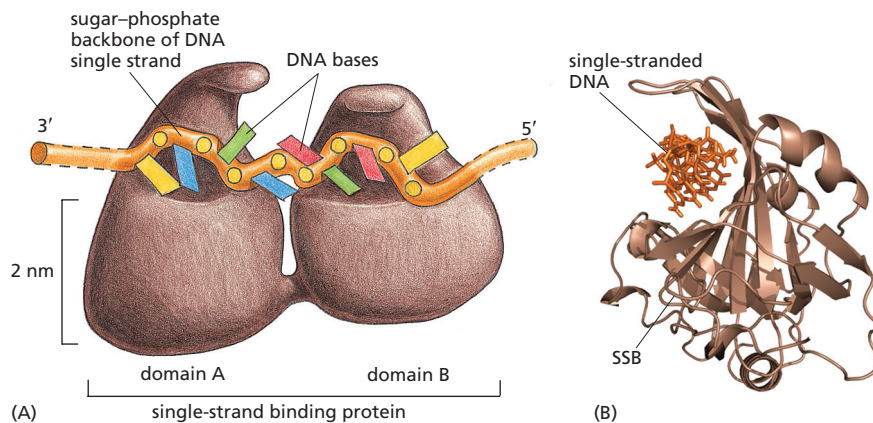


**Figure 5-15** The effect of single-strand DNA-binding proteins (SSB proteins) on the structure of single-stranded DNA. Because each protein molecule prefers to bind next to a previously bound molecule, long rows of this protein form on a DNA single strand. This *cooperative binding* straightens out the DNA template and facilitates the DNA polymerization process. The "hairpin helices" shown in the bare, single-stranded DNA result from a chance matching of short regions of complementary nucleotide sequence.

releases itself from the clamp and dissociates from the template. With the help of the clamp loader, which hydrolyzes ATP as it loads a new clamp onto a primer-template junction (**Figure 5-17**), this lagging-strand polymerase molecule then associates with the new clamp that is assembled on the RNA primer of the next Okazaki fragment.

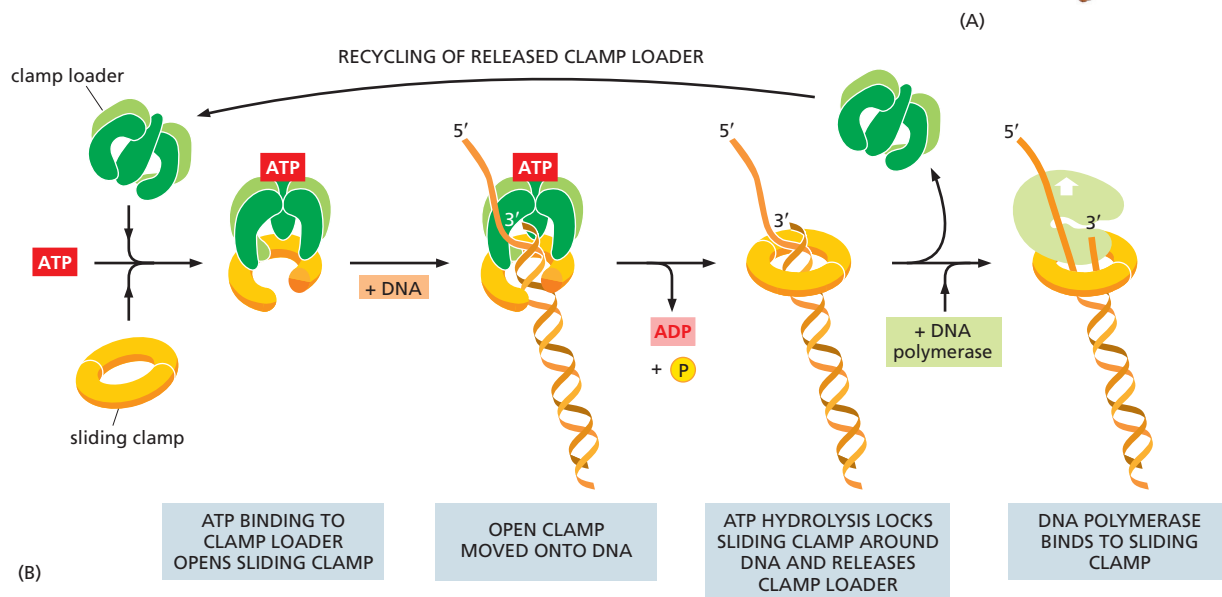
### The Proteins at a Replication Fork Cooperate to Form a Replication Machine

Although we have discussed DNA replication as though it were performed by a set of proteins all acting independently, in reality most of these proteins are held together in a large and orderly multienzyme complex that rapidly synthesizes DNA. This complex can be likened to a tiny sewing machine composed of protein



**Figure 5-16** Human single-strand binding protein bound to DNA. (A) Front view of the two DNA-binding domains of the protein (called RPA), which cover a total of eight nucleotides. Note that the DNA bases remain exposed in this protein-DNA complex. (B) Diagram showing the three-dimensional structure, with the DNA strand (orange) viewed end on. (PDB code: 1JMC.)

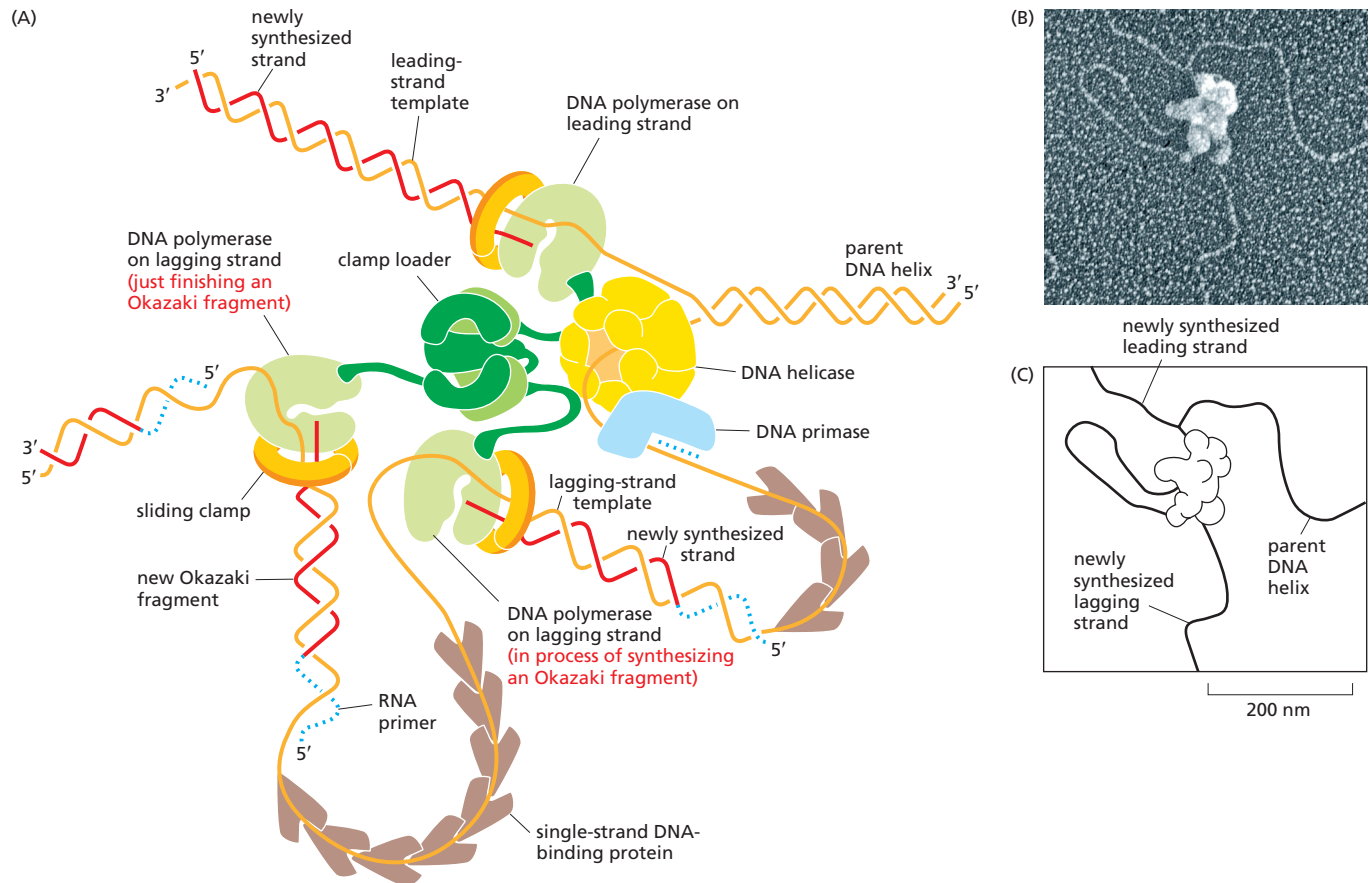
**Figure 5–17 The sliding clamp that holds DNA polymerase on the DNA.** (A) The structure of the clamp protein from *E. coli*, as determined by x-ray crystallography, with a DNA helix added to indicate how the protein fits around DNA (**Movie 5.3**). (B) Schematic illustration showing how the clamp is loaded onto DNA. The structure of the clamp loader (green) resembles a screw nut, with its threads matching the grooves of double-stranded DNA. The loader binds to a free clamp molecule, forcing a gap in its ring of subunits, which enables it to slip around DNA. The loader then “screws” the open clamp onto double-stranded DNA until it encounters the 3′ end of a primer, at which point the loader hydrolyzes ATP and releases the clamp, allowing it to close around the DNA. In the simplified reaction shown here, the clamp loader dissociates once the clamp has been assembled. At bacterial replication forks, the clamp loader remains bound to the polymerase so that, on the lagging strand, it is ready to assemble a new clamp at the start of each new Okazaki fragment. (A, from X.P. Kong et al., *Cell* 69:425–437, 1992; PDB code: 3BEP; B, adapted from B.A. Kelch et al., *Science* 334:1675–1680, 2011.)



parts and powered by nucleoside triphosphate hydrolysis. Like a sewing machine, the replication complex probably remains stationary with respect to its immediate surroundings; the DNA can be thought of as a long strip of cloth being rapidly threaded through it. Although the replication complex has been most intensively studied in *E. coli* and several of its viruses, a very similar complex also operates in eukaryotes, as we shall see below.

How the different proteins at the replication fork work together in bacteria is shown in **Figure 5–18**. At the front of the replication fork, DNA helicase opens the DNA helix. Several identical DNA polymerase molecules work at the fork, one on the leading strand and two on the lagging strand. Whereas the DNA polymerase molecule on the leading strand can operate in a continuous fashion, the DNA polymerase molecules on the lagging strand alternate at short intervals, using the short RNA primers made by DNA primase to begin each Okazaki fragment. The close association of all these protein components increases the efficiency of replication, and it is made possible by a folding back of the lagging strand as shown in the figure. This arrangement facilitates the loading of the polymerase clamp each time that an Okazaki fragment is synthesized: the clamp loader and the lagging-strand DNA polymerase molecule are kept in place at the replication fork even when they detach from their DNA template. The replication proteins are thus linked together into a single large unit (total molecular mass  $>10^6$  daltons), enabling DNA to be synthesized on both sides of the replication fork in a coordinated and efficient manner.

On the lagging strand, the DNA replication machine leaves behind a series of unsealed Okazaki fragments, which still contain the RNA that primed their synthesis at their 5′ ends. As discussed earlier, this RNA is removed, and the resulting

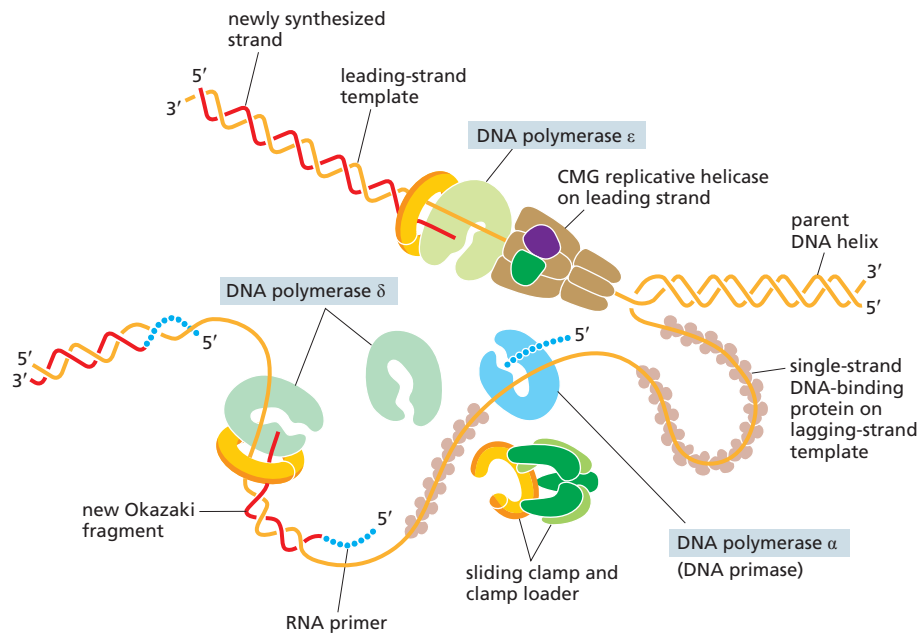


**Figure 5-18 A bacterial replication fork.** (A) In this case, a single DNA polymerase molecule synthesizes the leading strand while two DNA polymerase molecules are used—in alternating fashion—for lagging-strand DNA synthesis. All of these polymerase molecules, which are identical, are held in place at the fork by flexible “arms” that extend from the clamp loader. Additional interactions (for example, between the DNA helicase and primase) ensure that all the individual components function together as a well-coordinated protein machine (**Movie 5.4**). (B) An electron micrograph showing the replication machine from the bacteriophage T4 as it moves along a template synthesizing DNA behind it. (C) An interpretation of the micrograph is given in the sketch: note especially the DNA loop on the lagging strand. Apparently, during the preparation of this sample for electron microscopy, the replication proteins became partly detached from the very front of the replication fork. (B, from P.D. Chastain et al., *J. Biol. Chem.* 278:21276–21825, 2003. With permission from American Society for Biochemistry and Molecular Biology.)

gap is filled in by DNA repair enzymes that operate behind the replication fork (see Figure 5-11).

### DNA Replication Is Fundamentally Similar in Eukaryotes and Bacteria

Much of what we know about DNA replication was first derived from studies of purified bacterial and bacteriophage multienzyme systems capable of DNA replication *in vitro*. The development of these systems in the 1970s was greatly facilitated by the prior isolation of mutants in a variety of replication genes; these mutants were exploited to identify and purify the corresponding replication proteins. The first eukaryotic replication system that accurately replicated DNA *in vitro* was described in the mid-1980s, and mutations in genes encoding nearly all of the replication components have now been isolated and analyzed in the yeast *Saccharomyces cerevisiae*. As a result, much is known about the detailed enzymology of DNA replication in eukaryotes, and it is clear that the fundamental features of DNA replication—including replication-fork geometry and the use of 5'→3' DNA polymerases, helicases, clamps, clamp loaders, and single-strand binding proteins—are similar.



**Figure 5–19 Schematic diagram of a eukaryotic replication fork.** Unlike the bacterial replication proteins, those from eukaryotes are thought to function largely independently, perhaps accounting for the slower speed of the eukaryotic replication fork (**Movie 5.5**). Note that the eukaryotic CMG helicase moves unidirectionally along the leading-strand template, whereas the bacterial helicase discussed earlier moves in one direction along the lagging-strand template (see Figure 5–18). In both cases, the DNA duplex is rapidly pried apart at the front of the moving replication fork by harnessing the energy of ATP hydrolysis.

However, there are some important differences in how bacteria and eukaryotes replicate their DNA. Perhaps most important, eukaryotes use three different kinds of DNA polymerase at each replication fork (**Figure 5–19**). Polymerase  $\epsilon$  (Pol $\epsilon$ ) synthesizes the leading strand, whereas Pol $\alpha$  and Pol $\delta$  synthesize the lagging-strand Okazaki fragments. Each type of polymerase has special properties that make it well suited for its job. Pol $\epsilon$  binds to both the sliding clamp and the replicative helicase, allowing it to synthesize very long stretches of leading-strand DNA without dissociating. Pol $\alpha$  includes DNA primase as one of its subunits, which begins all new chains by synthesizing a short length of RNA. This RNA is extended by a different subunit of Pol $\alpha$ , which adds only about 20 nucleotides of DNA before dissociating. Finally, Pol $\delta$ , which is loaded in conjunction with a sliding clamp, takes over and completes synthesis of each Okazaki fragment to produce a total length of about 200 nucleotides.

The use of three different kinds of DNA polymerase at the replication fork is part of a trend toward higher complexity observed for eukaryotic DNA replication compared to that of bacteria. As another example, the eukaryotic single-strand binding protein is formed from three different subunits, while only a single subunit is found in bacteria. Likewise, the eukaryotic replicative helicase (known as the CMG helicase) is composed of 11 different protein subunits, while the bacterial enzyme is a hexamer of 6 identical subunits. We do not know why the eukaryotic replication machinery is so much more complex than that of bacteria; however, there are several possibilities. In eukaryotes, DNA replication must be coordinated with the elaborate process of mitosis; it must also deal with DNA packaged into nucleosomes, topics we discuss in the next part of the chapter. It is also possible that the difference in complexity between bacteria and eukaryotes largely reflects evolutionary pressure for bacteria to make do with fewer genes.

Another important distinction between eukaryotic and bacterial replication protein complexes lies in the detailed structures of their individual protein

components. With the exception of the sliding clamp, the replication proteins in bacteria have completely different structures and amino acid sequences than those of their eukaryotic counterparts. The simplest interpretation of this surprising fact is that, over hundreds of millions of years, the DNA replication machinery in eukaryotes and bacteria evolved independently, yet converged on the same basic mechanisms. This situation is in contrast to other fundamental processes in the cell, such as transcription and translation, where the fundamental components (RNA polymerase and the ribosome) are very similar between bacteria and eukaryotes—and where the structures are conserved from an ancient, common ancestor.

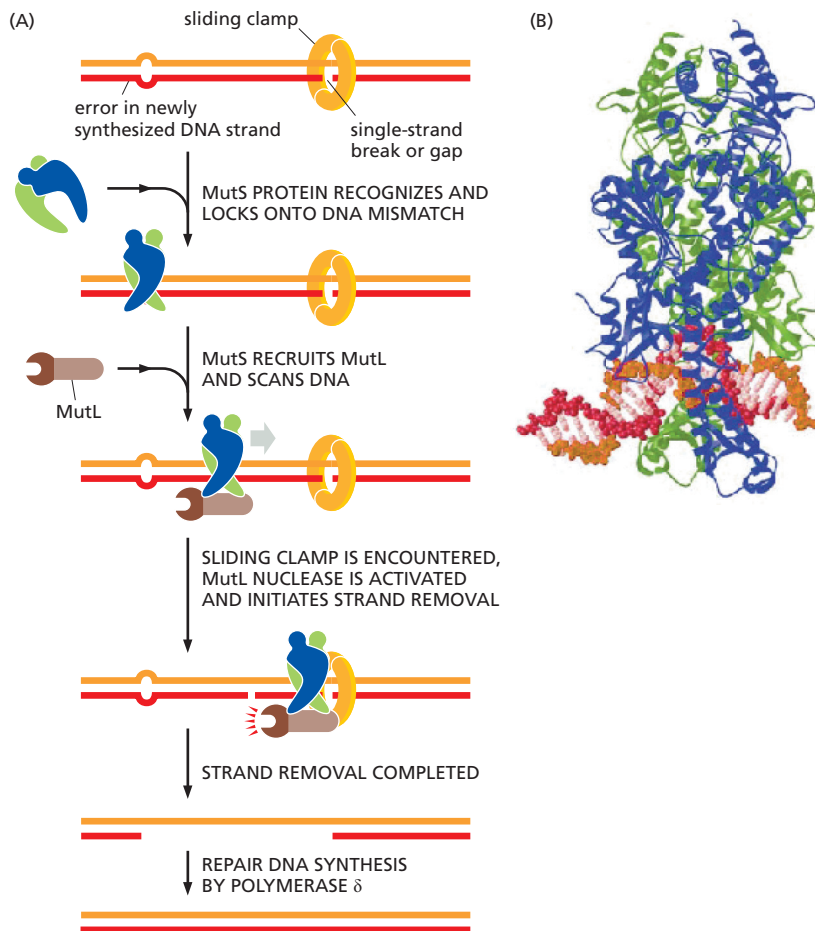
### A Strand-directed Mismatch Repair System Removes Replication Errors That Remain in the Wake of the Replication Machine

Because bacteria such as *E. coli* are capable of dividing once every 30 minutes, it is relatively easy to screen large populations to find a rare mutant cell that is altered in a specific process. One interesting class of mutants consists of those with alterations in so-called *mutator genes*, which greatly increase the rate of spontaneous mutation. Not surprisingly, one such mutant makes a defective form of the 3'-to-5' proofreading exonuclease that is a part of the DNA polymerase enzyme (see Figures 5-8 and 5-9). The mutant DNA polymerase no longer proofreads effectively, and many replication errors that would otherwise have been removed accumulate in the DNA.

The study of other *E. coli* mutants exhibiting abnormally high mutation rates uncovered an additional proofreading system, common to all cells on Earth, that removes those rare replication errors that were made by the polymerase and missed by its proofreading exonuclease. These errors leave mismatched base pairs behind the replication fork, which are subsequently recognized and corrected by a **strand-directed mismatch repair** system. This system picks out mismatches from normal DNA by monitoring their potential to distort the DNA double helix, which is greatly increased by the misfit between noncomplementary base pairs. However, if the repair system simply recognized a mismatch in newly replicated DNA and randomly corrected one of the two mismatched nucleotides, it would mistakenly “correct” the original template strand to match the error exactly half the time, thereby failing to lower the overall error rate. To be effective, such a proofreading system must be able to remove only the nucleotide on the newly synthesized strand, where the error occurred.

The strand-distinction mechanism used by the mismatch proofreading system in *E. coli* depends on the methylation of selected A residues in the DNA. Methyl groups are added to all A residues in the sequence GATC, but not until some time after the GATC has been synthesized. As a result, the only unmethylated GATC sequences lie in the newly synthesized strands just behind a replication fork. The recognition of these unmethylated GATCs (which are base-paired to methylated GATCs) allows the new DNA strands to be transiently distinguished from old ones, as required if their mismatches are to be selectively removed. The five-step error-correction process involves recognition of a mismatch, identification of the newly synthesized strand, excision of the portion containing the misincorporated nucleotide, resynthesis of the excised segment using the old strand as a template, and ligation to seal the DNA backbone. This strand-directed mismatch repair system reduces the number of errors made during DNA replication by an additional factor of 100–1000 (see Table 5-1, p. 260).

A similar mismatch proofreading system functions in eukaryotic cells, but it uses a different way to distinguish the newly synthesized DNA strands from the parent strands. On the lagging strand, the newly synthesized DNA will contain transient single-strand gaps before the series of Okazaki fragments are processed and ligated together. Each gap will usually carry a sliding clamp, which remains on the DNA after the DNA polymerase has dissociated from it to begin the next fragment. Together, the clamp and the single-strand break signal to the mismatch



**Figure 5-20 Strand-directed mismatch repair in eukaryotes.** (A) The MutS protein binds to a mismatched base pair, recruits the MutL protein, and the complex scans the nearby DNA for a gap and a sliding clamp whose orientation determines which strand is to be cut and its nucleotides replaced. When these are encountered, MutL is activated and begins to cleave the DNA. In most organisms, MutL is joined by another nuclease and, together, they remove the newly synthesized DNA starting at the gap and extending past the mismatch. The gap is then filled in by DNA polymerase  $\delta$  and sealed by DNA ligase. (B) The structure of the MutS protein bound to a DNA mismatch. This protein is a dimer, which grips the DNA double helix as shown, kinking the DNA at the mismatched base pair. It seems that the MutS protein scans the DNA for mismatches by testing for sites that can be readily kinked, which are those with an abnormal base pair. (PDB code: 1EWQ.)

repair proteins to correct the mismatch using the parent DNA strand as the template (Figure 5-20).

The two faces of the clamp differ, and the clamp loader always loads the clamp in the same orientation with respect to the 3' end of the previously synthesized Okazaki fragment. Because all the clamps on the DNA “face” in the same direction relative to the replication process, the oriented clamps can be used by the mismatch repair machinery to distinguish newly synthesized DNA from parent DNA. It is not known for certain how strand discrimination occurs on the leading strand (where gaps in newly synthesized DNA should be rare), but because oriented sliding clamps are also left behind by the leading-strand polymerase, they can signal old from new DNA in the same way that they do on the lagging strand. The recent discovery of a correction system that removes misincorporated ribonucleotides suggests a further possibility for distinguishing newly synthesized DNA from parent DNA, as we discuss in the next section.

Mismatch correction is crucial for all cells; its importance for humans is seen in individuals who inherit one defective copy of a mismatch repair gene (along

with a functional gene on the other copy of the chromosome). These individuals have a marked predisposition for certain types of cancers. For example, in a type of colon cancer called *hereditary nonpolyposis colorectal cancer (HNPCC)*, a spontaneous deleterious mutation of the one functional gene will produce a clone of somatic cells that, because they are deficient in mismatch proofreading, accumulate mutations unusually rapidly. Because most cancers arise in cells that have accumulated many mutations (as discussed in Chapter 20), cells deficient in mismatch proofreading have a greatly enhanced chance of becoming cancerous. Fortunately, most of us inherit two good copies of each gene that encodes a mismatch proofreading protein; this protects us, because it is highly unlikely that both copies will become mutated in the same cell.

### The Accidental Incorporation of Ribonucleotides During DNA Replication Is Corrected

We have seen that cells have several ways to correct mistakes where the wrong deoxynucleotide has been incorporated in newly replicated DNA. Occasionally, however, DNA polymerases make a different kind of mistake, one that is not caused by improper base-pairing: in this case, they accidentally incorporate a ribonucleotide instead of a deoxyribonucleotide. These molecules differ by a single  $-OH$  group in the sugar portion of the nucleotide. Yet, when incorporated into DNA, they weaken the DNA chain at that point, rendering it highly susceptible to breakage. If left unrepaired, these “weak links” would cause high mutation rates and genome rearrangements. Even if it does not cause a break, an incorporated ribonucleotide distorts the DNA double helix and can stall some polymerases during the next cycle of DNA replication.

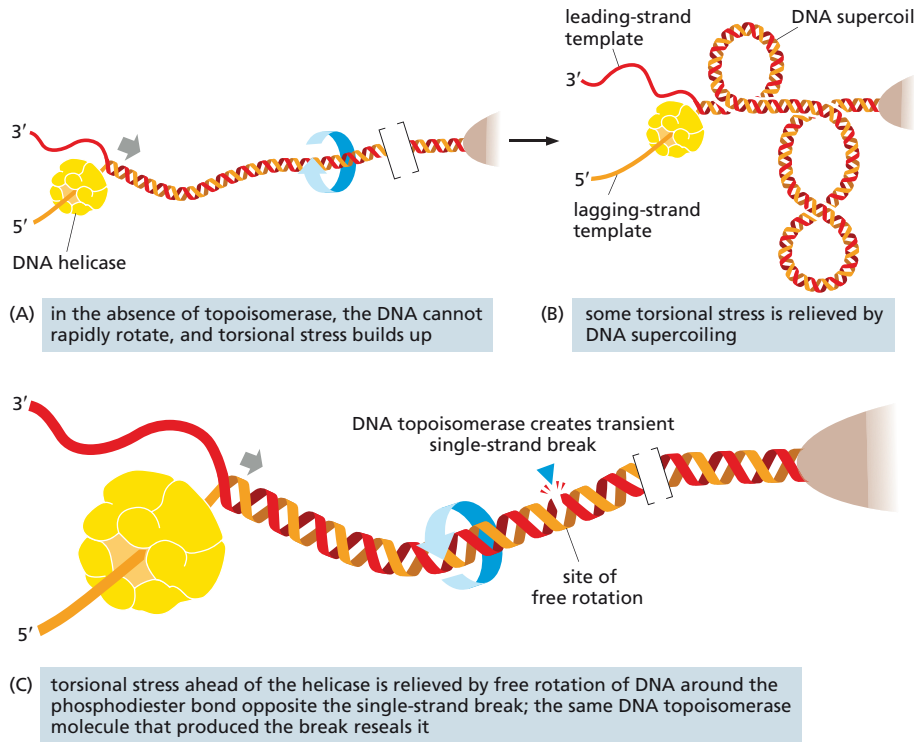
Although DNA polymerases much prefer deoxyribonucleotides over ribonucleotides (by a factor of about a million), the concentration of ribonucleotides in the cell is much higher than that of their deoxy counterparts, as much as 500-fold for ATP, which has many different uses in the cell. This concentration imbalance means that a ribonucleotide is accidentally incorporated approximately once per several thousand nucleotides of DNA synthesized. These mistakes are corrected by specific nucleases that cleave the DNA chain when they encounter a ribonucleotide, leading to the excision of the ribonucleotide and its replacement by DNA, much in the same way that RNA primers are replaced by DNA to complete lagging-strand synthesis (see Figure 5-11). Because this repair process produces gaps only in newly synthesized DNA, it has been proposed that these transient lesions help the mismatch repair system “know” which strand to repair; in particular, these cues may be especially important on the leading strand.

### DNA Topoisomerases Prevent DNA Tangling During Replication

As a replication fork moves along double-stranded DNA, it creates what has been called the “winding problem.” The two parent strands that are wound around each other must be unwound and separated for replication to occur. For every 10 nucleotide pairs replicated at the fork, one complete turn of the parent double helix must be unwound. In principle, this unwinding can be achieved by rapidly rotating the entire chromosome ahead of a moving fork; however, this is energetically highly unfavorable (particularly for long chromosomes). Instead, the DNA in front of a replication fork becomes overwound (Figure 5-21). This overwinding is continually relieved by enzymes known as *DNA topoisomerases*.

A **DNA topoisomerase** can be viewed as a reversible nuclease that adds itself covalently to a DNA backbone phosphate, thereby breaking a phosphodiester bond in a DNA strand. This reaction is reversible, and the phosphodiester bond re-forms as the protein leaves.

One type of topoisomerase, called *topoisomerase I*, produces a transient single-strand break; this break in the phosphodiester backbone allows the

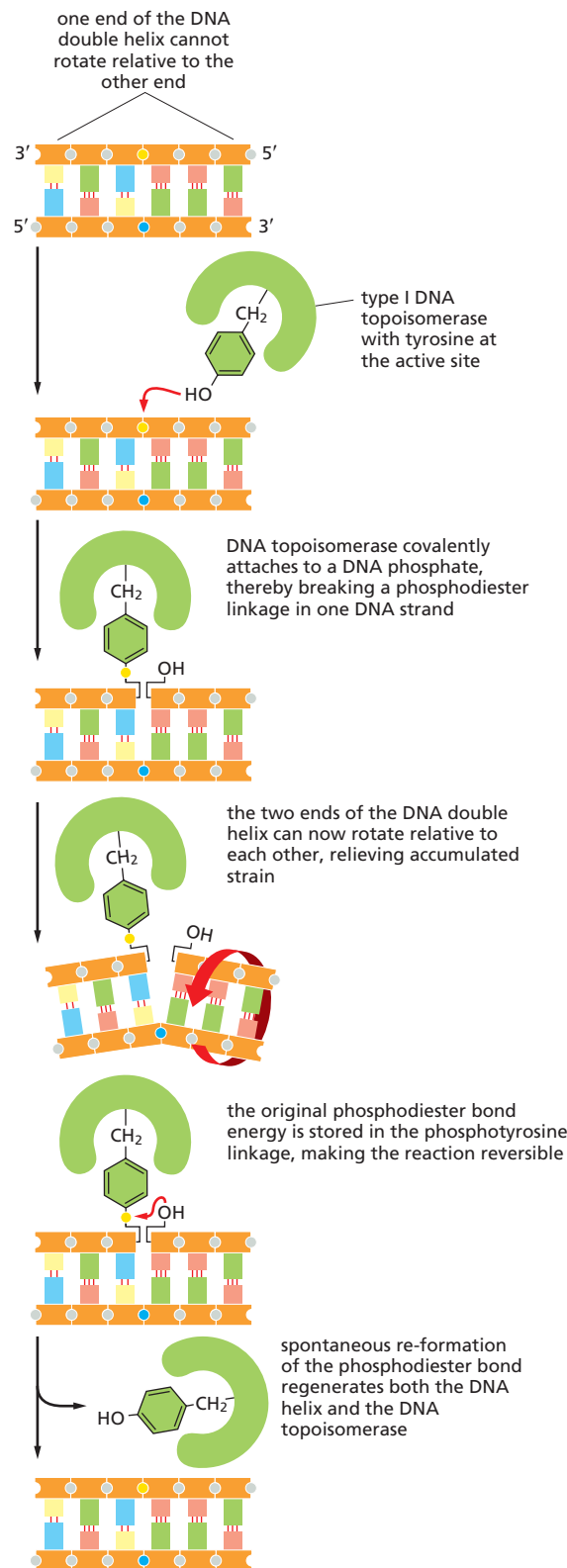


**Figure 5-21** The “winding problem” that arises during DNA replication. (A) For a bacterial replication fork moving at 500 nucleotides per second, the parent DNA helix ahead of the fork must rotate at about 50 revolutions per second. The brackets represent about 20 turns of DNA. (B) If the ends of the DNA double helix remain fixed (or difficult to rotate), tension builds up in front of the replication fork as it becomes overwound. Some of this tension can be taken up by supercoiling, whereby the DNA double helix twists around itself. However, if the tension continues to build up, the replication fork will eventually stop because further unwinding requires more energy than the DNA helicase at the fork can provide. (C) DNA topoisomerases relieve this stress by generating temporary single-strand breaks in the DNA, which allow rapid rotation around the single strands opposite the break.

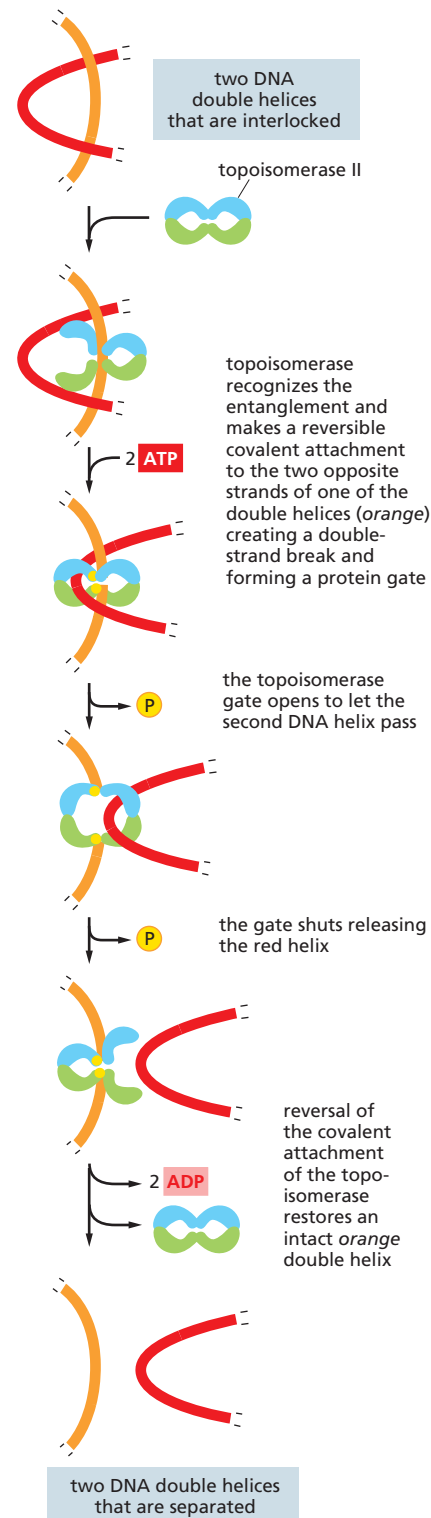
two sections of DNA helix on either side of the nick to rotate freely relative to each other, using the phosphodiester bond in the strand opposite the nick as a swivel point (Figure 5-22). Any tension in the DNA helix will drive this rotation in the direction that relieves the tension. As a result, DNA replication can occur with the rotation of only a short length of helix—the part just ahead of the fork. Because the covalent linkage that joins the DNA topoisomerase protein to a DNA phosphate retains the energy of the cleaved phosphodiester bond, resealing is rapid and does not require additional energy input. In this respect, the rejoining mechanism differs from that catalyzed by the enzyme DNA ligase, discussed previously (see Figure 5-12).

A second type of DNA topoisomerase, *topoisomerase II*, forms a covalent linkage to both strands of the DNA helix at the same time, making a transient *double-strand break* in the helix. These enzymes are activated by sites on chromosomes where two double helices cross over each other, such as those generated by supercoiling in front of a replication fork (see Figure 5-21B). As illustrated in Figure 5-23, once a topoisomerase II molecule binds to such a crossing site, the protein uses ATP hydrolysis to perform the following set of reactions: (1) it breaks one double helix reversibly to create a DNA “gate”; (2) it causes the second, nearby double helix to pass through this opening; and (3) it then reseals the break and dissociates from the DNA. At crossover points generated by supercoiling, passage of the double helix through the gate occurs in the direction that will reduce supercoiling. In this way, type II topoisomerases—like type I topoisomerases—can relieve the overwinding tension generated in front of a replication fork.

Their reaction mechanism also allows type II DNA topoisomerases to efficiently separate any intertwined DNA molecules. This ability of topoisomerase II is especially important for preventing the severe DNA tangling problems that would otherwise arise from DNA replication. This role is nicely illustrated by mutant yeast cells that produce, in place of the normal topoisomerase II, a version that is inactive above 37°C. When the mutant cells are warmed to this temperature, their daughter chromosomes remain intertwined after DNA replication and are unable to separate. The enormous usefulness of topoisomerase II for untangling



**Figure 5–22** The reversible DNA nicking reaction catalyzed by a DNA topoisomerase I enzyme. As indicated, these enzymes transiently form a single covalent bond with DNA; this allows free rotation of the DNA around the covalent backbone bonds linked to the *blue* phosphate. On reversal of the reaction, the enzyme and the DNA are restored, the only difference being the relaxation of tension in the DNA.



**Figure 5–23** The DNA-helix-passing reaction catalyzed by DNA topoisomerase II. Unlike type I topoisomerases, type II enzymes hydrolyze ATP, which is needed to release and reset the enzyme after each cycle. The small *yellow circles* represent the 5' phosphates in the DNA backbone that become covalently bonded to the topoisomerase. Type II topoisomerases are especially important for rapidly dividing cells; partly for that reason, they are effective targets for a large class of antibiotics, the fluoroquinolones, used to treat many different kinds of bacterial infections. These drugs inhibit bacterial topoisomerase II at the third step in the figure and thereby produce high levels of double-strand breaks that are lethal to rapidly dividing cells.

chromosomes before mitosis begins can readily be appreciated by anyone who has struggled to remove a severe tangle from a fishing line—or from a large ball of thread—without the aid of scissors.

## Summary

DNA replication takes place at a Y-shaped structure called a replication fork. Self-correcting DNA polymerase enzymes catalyze nucleotide polymerization in a 5′-to-3′ direction, copying a DNA template strand with remarkable fidelity. Because the two strands of a DNA double helix are antiparallel, this 5′-to-3′ DNA synthesis can take place continuously on only one of the strands at a replication fork (the leading strand). On the lagging strand, short DNA fragments must be made by a “backstitching” process. Because the self-correcting DNA polymerases cannot start a new chain, these lagging-strand DNA fragments are primed by short RNA primer molecules that are subsequently erased and replaced with DNA.

DNA replication requires the cooperation of many proteins. These include (1) DNA polymerases and DNA primases to catalyze nucleoside triphosphate polymerization; (2) DNA helicases and single-strand DNA-binding (SSB) proteins to help in opening up the DNA helix so that it can be copied; (3) clamps and clamp loaders to enable DNA polymerases to copy longer stretches of DNA; (4) DNA ligases and enzymes that degrade RNA primers to seal together the discontinuously synthesized lagging-strand DNA fragments; and (5) DNA topoisomerases to help to relieve helical winding and DNA tangling problems. Many of these proteins associate with each other at a replication fork to form a highly efficient “replication machine,” through which the activities and spatial movements of the individual components are coordinated.

The self-correcting DNA polymerases make mistakes only rarely when copying DNA; when they do, a variety of enzymes inspect the DNA shortly after it is made and correct any mishaps. Given the number of proteins dedicated to the task, copying DNA with extreme accuracy is clearly of great importance to all cells on Earth.

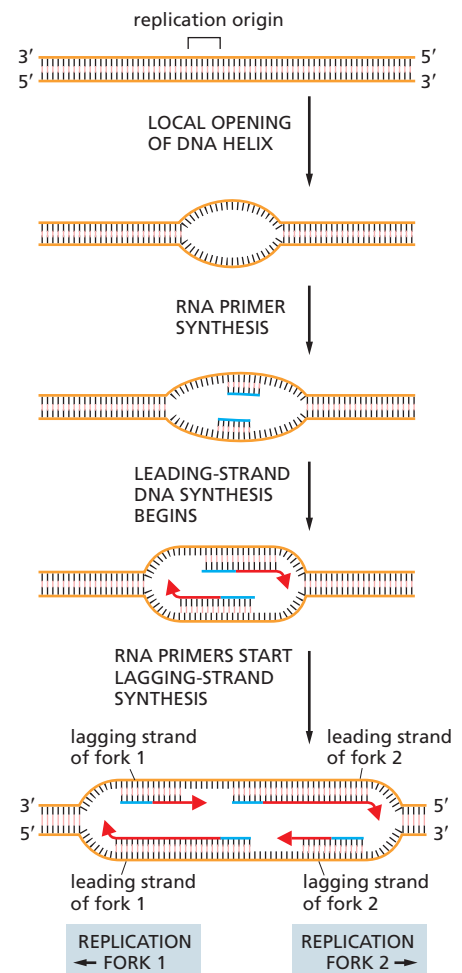
## THE INITIATION AND COMPLETION OF DNA REPLICATION IN CHROMOSOMES

We have seen how a set of replication proteins rapidly and accurately generates two daughter DNA double helices behind a replication fork. But how is this replication machinery assembled in the first place, and how are replication forks created on an intact, double-strand DNA molecule? In this part of the chapter, we discuss how cells initiate DNA replication and how they carefully regulate this process to ensure that it takes place only at the proper time and chromosomal sites. We also discuss special problems that the replication machinery in eukaryotic cells must overcome including the need to replicate the enormously long DNA molecules found in eukaryotic chromosomes, as well as the need to copy DNA molecules that are tightly complexed with nucleosomes.

### DNA Synthesis Begins at Replication Origins

As discussed previously, the DNA double helix is normally very stable: the two DNA strands are locked together firmly by the hydrogen bonds formed between the bases on each strand. To begin DNA replication, the double helix must first be opened up and the two strands separated to expose unpaired bases. As we shall see, the process of DNA replication is begun by special *initiator proteins* that bind to double-stranded DNA and pry the two strands apart, breaking the hydrogen bonds between the bases.

The positions at which the DNA helix is first opened are called **replication origins** (Figure 5-24). In simple cells like those of bacteria or budding yeast, origins are specified by DNA sequences several hundred nucleotide pairs in



**Figure 5-24 A replication bubble formed by replication-fork initiation.** This diagram outlines the major steps in the initiation of replication forks at replication origins. In the last step, two replication forks move away from each other, separated by an expanding replication bubble.

length. This DNA contains both short sequences that attract initiator proteins and stretches of DNA that are especially easy to open. We saw in Figure 4-5A that an A-T base pair is held together by fewer hydrogen bonds than is a G-C base pair. Therefore, DNA rich in A-T base pairs is relatively easy to pull apart, and regions of DNA enriched in A-T base pairs are typically found at replication origins.

Although the basic process of replication-fork initiation depicted in Figure 5-24 is fundamentally the same for bacteria and eukaryotes, the detailed way in which this process is performed and regulated differs considerably between these two groups of organisms. We first consider the case in bacteria and then turn to the more complex situation found in yeasts, mammals, and other eukaryotes.

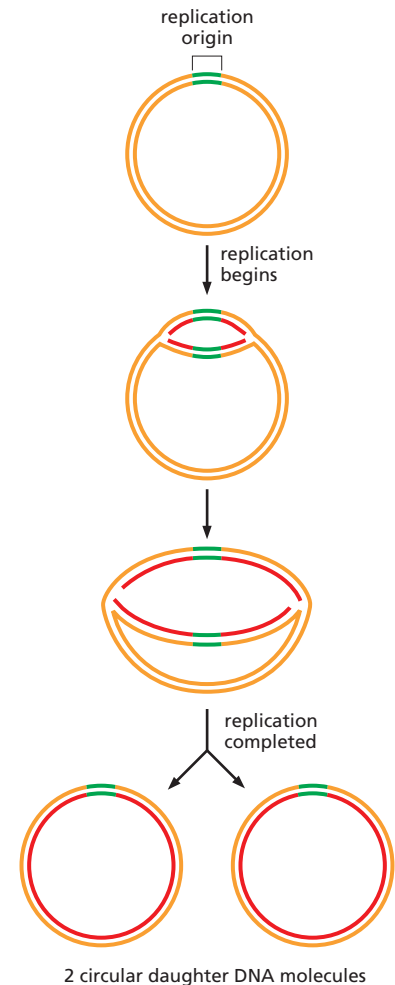
### Bacterial Chromosomes Typically Have a Single Origin of DNA Replication

The genome of *E. coli* is contained in a single circular DNA molecule of  $4.6 \times 10^6$  nucleotide pairs. DNA replication begins at a single origin of replication, and the two replication forks assembled there proceed (at approximately 1000 nucleotides per second) in opposite directions until they meet up roughly half-way around the chromosome (Figure 5-25). The only point at which *E. coli* can control DNA replication is initiation: once the forks have been assembled at the origin, they synthesize DNA at a relatively constant speed until replication is finished. Therefore, it is not surprising that the initiation step of DNA replication is tightly regulated. The process begins when specialized initiator proteins (in their ATP-bound state) bind in multiple copies to specific DNA sites located at the replication origin, wrapping the DNA around the proteins to form a large protein-DNA filament that introduces torsional stress on the DNA double helix (Figure 5-26). This stress is partially relieved by melting of the adjacent AT-rich sequences. The protein-DNA complex then attracts two DNA helicases, each bound to a helicase loader, and these are placed—facing in opposite directions—around adjacent DNA single strands whose bases have been exposed by the assembly of the initiator protein-DNA complex. The helicase loader is analogous to the clamp loader we encountered earlier; it has the additional job of keeping the helicase in an inactive form until it is properly loaded. Once the helicases are properly positioned on DNA, the loaders dissociate and the helicases begin to unwind DNA, exposing enough single-stranded DNA for DNA primases to synthesize the first RNA primers. This quickly leads to the assembly of the remaining replication proteins to create two replication forks that move in opposite directions away from the replication origin, each synthesizing new DNA as they travel.

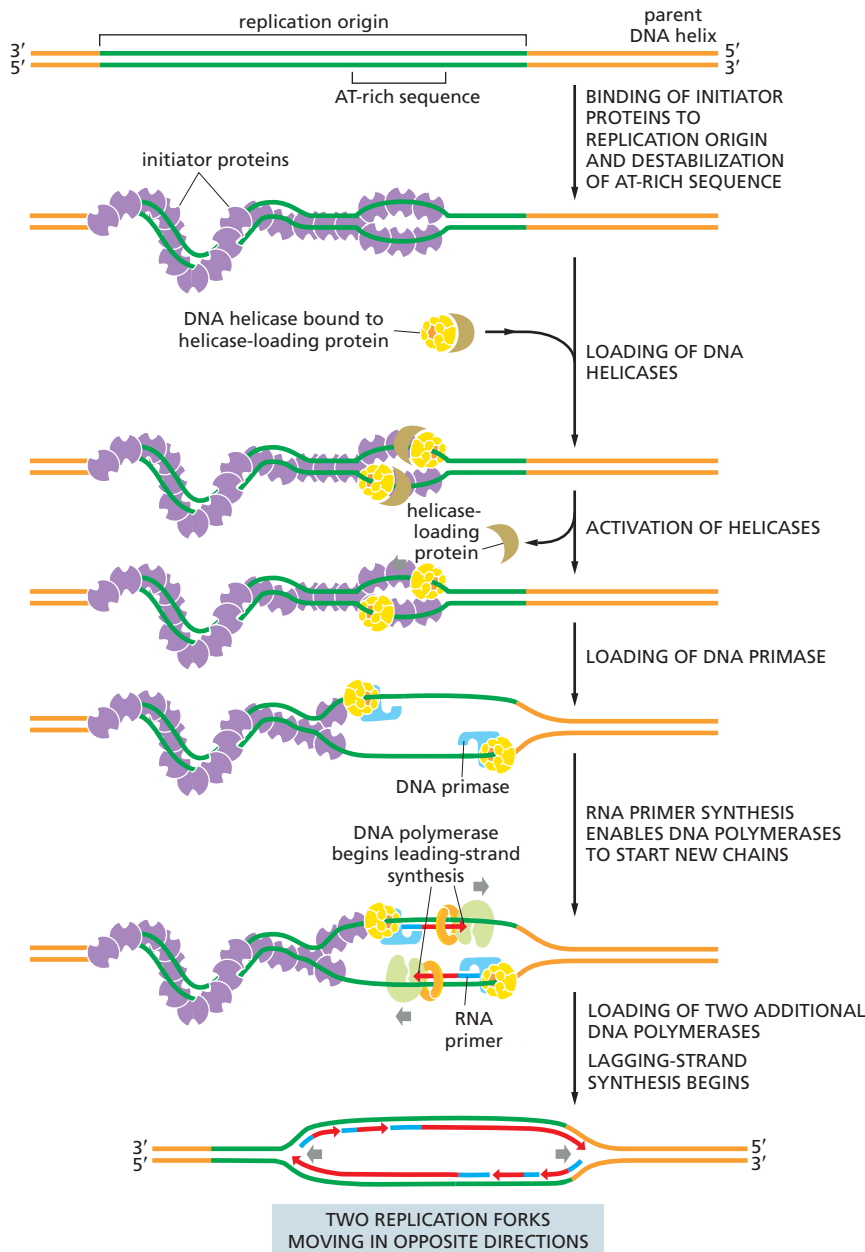
In *E. coli*, the interaction of the initiator proteins with the replication origin is carefully regulated, with initiation occurring only when sufficient nutrients are available for the bacterium to complete an entire round of replication. Initiation is also controlled to ensure that only one round of DNA replication occurs for each cell division. After replication is initiated, the initiator protein is inactivated by hydrolysis of its bound ATP molecule, and the origin of replication experiences a *refractory period*. The refractory period is caused by a delay in the methylation of newly incorporated A nucleotides in the origin (Figure 5-27). Initiation cannot occur again until the A's are methylated and the initiator protein is restored to its ATP-bound state, conditions that are met only when the cell is capable of carrying out a new round of DNA replication.

### Eukaryotic Chromosomes Contain Multiple Origins of Replication

We have seen how two replication forks begin at a single replication origin in bacteria and proceed in opposite directions, moving away from the origin until all of the DNA in the single circular chromosome is replicated. The bacterial genome is sufficiently small for these two replication forks to duplicate the genome in about



**Figure 5-25 DNA replication of a bacterial genome.** It takes *E. coli* about 30 minutes to duplicate its genome of  $4.6 \times 10^6$  nucleotide pairs. For simplicity, Okazaki fragments are not shown on the lagging strand.



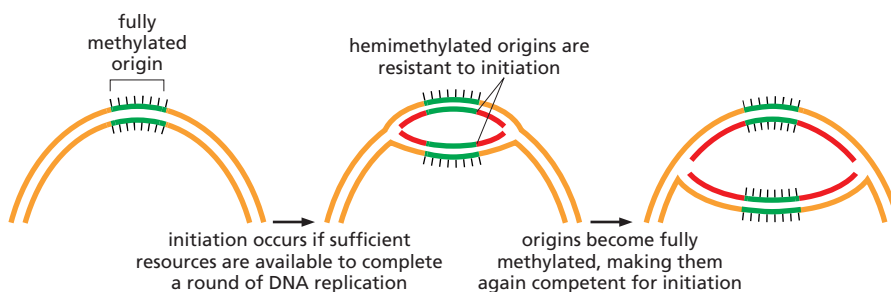
**Figure 5-26 The proteins that initiate DNA replication in bacteria.** The mechanism shown was established by studies *in vitro* with mixtures of highly purified proteins. For *E. coli* DNA replication, the major initiator protein (purple), the helicase (yellow), and the primase (blue) are the *dnaA*, *dnaB*, and *dnaG* proteins, respectively. In the first step, many molecules of the initiator protein bind to specific DNA sequences at the replication origin and destabilize the double helix by forming a filamentous structure in which the DNA is wrapped around the protein. Next, two helicases are brought in by helicase-loading proteins (the *dnaC* proteins; brown), which inhibit the helicases until they are properly loaded at the replication origin. (The helicase-loading proteins prevent the replicative DNA helicases from inappropriately entering other single-strand stretches of DNA in the bacterial genome.) Aided by single-strand binding protein (not shown), the loaded helicases further separate the DNA strands, thereby enabling primases to enter and synthesize initial primers. In subsequent steps, two complete replication forks are assembled at the origin and move in opposite directions away from the replication origin. The initiator proteins are displaced as the left-hand fork moves through them.

**Figure 5-27 Methylation of the *E. coli* replication origin creates a refractory period for DNA initiation.** DNA methylation occurs at GATC sequences, 11 of which are found in the origin of replication (spanning approximately 250 nucleotide pairs). In its hemimethylated state (that is, one strand of the DNA methylated, the other unmethylated), the origin of replication is bound by an inhibitor protein (Seq A, not shown), which blocks the ability of the initiator proteins to unwind the origin DNA. About 15 minutes after replication is initiated, the hemimethylated origins become fully methylated by a DNA methylase enzyme; Seq A then dissociates allowing the origin of replication to become active.

A single enzyme, the *Dam* methylase, is responsible for methylating all *E. coli* GATC sequences. As discussed earlier in the chapter, a lag in methylation after the replication of GATC sequences is also used by the *E. coli* mismatch proofreading system to distinguish the newly synthesized DNA strand from the parent DNA strand; in that case, the relevant GATC sequences are scattered throughout the chromosome, and they are not bound by Seq A.

30 minutes. Because of the much greater size of most eukaryotic chromosomes, a different strategy is required to allow their replication in a timely manner.

A method for determining the general pattern of eukaryotic chromosome replication was developed in the early 1960s that is similar to the strategy we saw earlier for visualizing bacterial replication (see Figure 5-6). Human cells growing

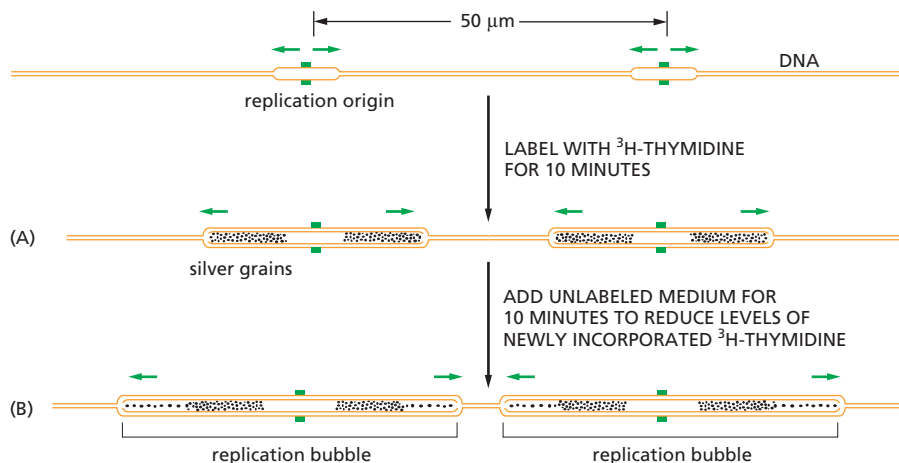


in culture are labeled for a short time with  $^3\text{H}$ -thymidine so that the DNA synthesized during this period becomes highly radioactive. The cells are then gently lysed, and the DNA is stretched on the surface of a glass slide coated with a photographic emulsion. Development of the emulsion in the dark reveals the pattern of labeled DNA through a technique known as *autoradiography*. The time allotted for radioactive labeling is chosen to allow each replication fork to move several micrometers along the DNA, so that the replicated DNA can be detected in the light microscope as lines of silver grains (radioactivity exposes photographic emulsion much as light does), even though the DNA molecule itself is too thin to be visible. In this way, both the rate and the direction of replication-fork movement can be determined (Figure 5–28). From the rate at which tracks of replicated DNA increase in length with increasing labeling time, the eukaryotic replication forks are estimated to travel at about 50 nucleotides per second. This is approximately twentyfold slower than the rate at which bacterial replication forks move, possibly reflecting the increased difficulty of replicating DNA that is packaged in chromatin.

An average-size human chromosome contains a single linear DNA molecule of about 150 million nucleotide pairs. It would take  $0.02 \text{ seconds/nucleotide} \times 150 \times 10^6 \text{ nucleotides} = 3.0 \times 10^6 \text{ seconds}$  (about 35 days) to replicate such a DNA molecule from end to end with a single replication fork moving at a rate of 50 nucleotides per second. As expected, therefore, the autoradiographic experiments just described reveal that many forks, belonging to separate replication bubbles, are moving simultaneously on each eukaryotic chromosome.

Much more sophisticated methods now exist for monitoring DNA replication initiation and tracking the movement of DNA replication forks across whole genomes. If a population of cells can be synchronized so they all begin DNA replication at the same time, the amount of each segment of DNA in the genome can be determined at specific time points using one of the DNA sequencing methods described in Chapter 8. Because a segment of a genome that has been replicated will contain twice as much DNA as an unreplicated segment, replication-fork initiation and fork movement can be accurately monitored across an entire genome.

Experiments of this type have shown the following: (1) Approximately 30,000–50,000 origins of replication are used each time a human cell divides. (2) The human genome has many more (perhaps tenfold more) potential origins than this, and different cell types use different sets of origins. This excess of origins may allow a cell to coordinate its active origins with other features of its chromosomes such as which genes are being expressed. The excess origins also provide “backups” in case a primary origin fails. (3) Origins of replication do not all “fire” simultaneously; rather, they often are activated in a prescribed order in a given cell type. (4) Regardless of when a given origin fires or where on the chromosome it is located, the replication forks all move at approximately the same speed. (5) As in bacteria, replication forks are formed in pairs and create an expanding



**Figure 5–28** The experiments that first demonstrated the pattern in which replication forks are formed and move on eukaryotic chromosomes. The new DNA made in human cells in culture was labeled briefly with a pulse of highly radioactive thymidine ( $^3\text{H}$ -thymidine). (A) In this experiment, the cells were lysed, and the DNA was stretched out on a glass slide that was subsequently covered with a photographic emulsion. After several months, the emulsion was developed, revealing a line of silver grains over the radioactive DNA. The brown DNA in this figure is shown only to help with the interpretation of the autoradiograph; the unlabeled DNA is invisible in such experiments. (B) This experiment was the same except that a further incubation in unlabeled medium allowed additional DNA, with a lower level of radioactivity, to be replicated. The pairs of dark tracks in B were found to have silver grains tapering off in opposite directions, demonstrating bidirectional fork movement from a central replication origin where a replication bubble forms (see Figure 5–24). A replication fork is thought to stop only when it encounters a replication fork moving in the opposite direction or when it reaches the end of the chromosome; in this way, all the DNA is eventually replicated.

replication bubble as they move in opposite directions away from a common point of origin, stopping only when they meet a replication fork moving in the opposite direction or when they reach a chromosome end. In this way, many replication forks operate independently on each chromosome and yet form two complete daughter DNA helices.

### In Eukaryotes, DNA Replication Takes Place During Only One Part of the Cell Cycle

When growing rapidly, bacteria replicate their DNA nearly continually. In contrast, DNA replication in most eukaryotic cells occurs only during a specific part of the cell-division cycle, called the *DNA synthesis phase*, or **S phase** (Figure 5–29). In a mammalian cell, the S phase typically lasts for about 8 hours; in simpler eukaryotic cells such as yeasts, the S phase can be as short as 40 minutes. By its end, each chromosome has been replicated to produce two complete copies, which remain joined together at their centromeres until the *M phase* (M for *mitosis*), which soon follows. Although different origins of replication fire at different times, all DNA replication is begun and completed during S phase. In Chapter 17, we describe the control system that runs the cell cycle, and we explain how entry into each phase of the cycle requires the cell to have successfully completed the previous phase.

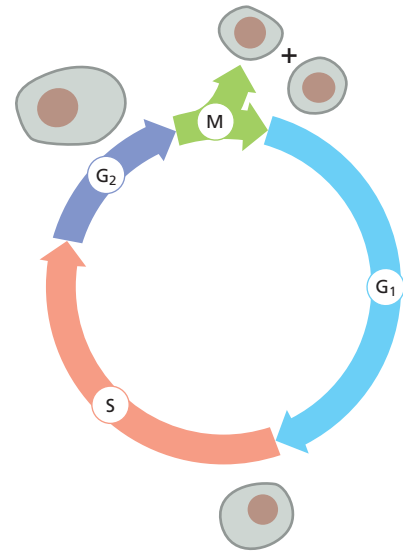
In the following sections, we explore how DNA replication begins on eukaryotic chromosomes and how this event is coordinated with the cell cycle.

### Eukaryotic Origins of Replication Are “Licensed” for Replication by the Assembly of an Origin Recognition Complex

Having seen that a eukaryotic chromosome is replicated using many origins of replication, each of which fires at a characteristic time in S phase of the cell cycle, we turn to the nature of these origins of replication. We saw earlier in this chapter that replication origins have been precisely defined in bacteria as specific DNA sequences that attract initiator proteins, which then assemble the DNA replication machinery. We shall see that this is also the case for the single-cell budding yeast *S. cerevisiae*, but it appears not to be strictly true for many other eukaryotes.

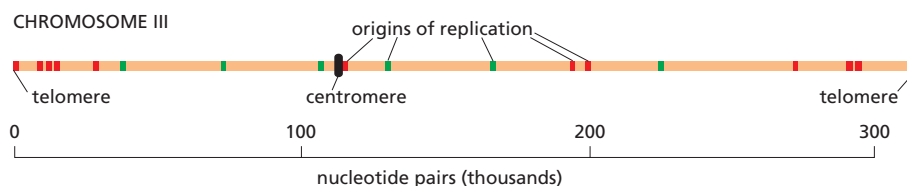
For budding yeast, the location of every origin of replication on each chromosome has been determined. The particular chromosome shown in Figure 5–30—chromosome III from *S. cerevisiae*—is one of the smallest chromosomes known, with a length less than 1/100 that of a typical human chromosome. Its major origins are spaced an average of 30,000 nucleotide pairs apart, but only a subset of these origins is used by a given cell. Nonetheless, this chromosome can be replicated in about 15 minutes.

The minimal DNA sequence required for directing DNA replication initiation in *S. cerevisiae* has been determined by taking a segment of DNA that spans an origin of replication and testing smaller and smaller DNA fragments for their ability to function as origins. These DNA sequences that can serve as an origin of replication are found to contain (1) a binding site for a large, multisubunit initiator protein called ORC, for origin recognition complex; (2) a stretch of DNA that is rich in A's and T's and therefore easy to pull apart; and (3) at least one binding site for proteins that facilitate ORC binding, probably by adjusting the local chromatin structure.



**Figure 5–29** The four successive phases of a standard eukaryotic cell cycle.

During the G<sub>1</sub>, S, and G<sub>2</sub> phases, the cell grows continually. During M phase growth stops, the nucleus divides, and the cell divides in two. DNA replication is confined to the part of the cell cycle known as S phase. G<sub>1</sub> is the gap between M phase and S phase; G<sub>2</sub> is the gap between S phase and M phase. Many eukaryotic cells spend only a small fraction of their time in S phase.



**Figure 5–30** The origins of DNA replication on chromosome III of the yeast *S. cerevisiae*. This chromosome, one of the smallest eukaryotic chromosomes known, carries a total of 180 genes. As indicated, it contains 18 replication origins, although they are used with different frequencies. Those in red are typically used in less than 10% of cell divisions, while those in green are used about 90% of the time.

### Features of the Human Genome That Specify Origins of Replication Remain to Be Fully Understood

Compared with the situation in budding yeast, the determinants of replication origins in humans have been difficult to discover. It has been possible to identify specific human DNA sequences, each several thousand nucleotide pairs in length, that are sufficient to serve as replication origins. These origins continue to function when moved to a different chromosomal region by recombinant DNA methods, as long as they are placed in a region where the chromatin is relatively uncondensed. However, comparisons of such DNA sequences have not revealed DNA sequences in common as in the origins of bacteria and yeasts.

Despite this, a human ORC that is very similar to the yeast ORC binds to origins of replication and initiates DNA replication in humans. Many of the other proteins that function in the initiation process in yeast likewise have central roles in humans. The yeast and human initiation mechanisms are thus similar, although some property of the genome other than a specific DNA sequence has the central role in attracting an ORC to a mammalian origin of replication. Origins of replication are often nucleosome-free, and it has been proposed that DNA that is difficult to fold onto a histone core may help define origins of replication. Nearby transcriptional activity on the genome may also play a role in activating certain origins, by altering the local chromatin structures, as we discuss in Chapter 7. This idea helps to explain why different cell types—which express different sets of genes—often use different origins. Consistent with this idea, origins that fire the earliest in S phase tend to be located near highly transcribed regions of the genome.

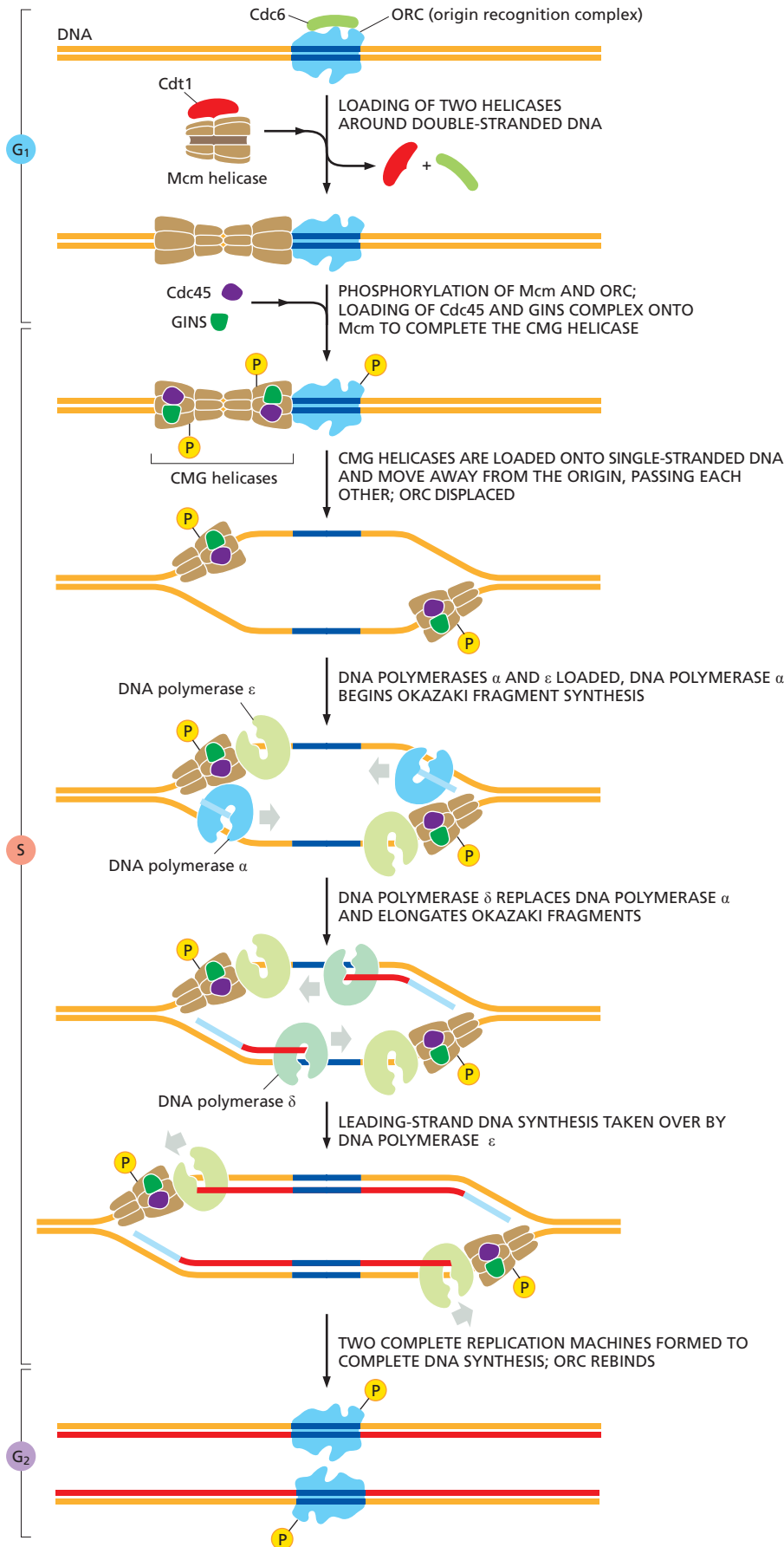
Finally, origins located in proximity to each other tend to fire together, and it seems likely that the three-dimensional structure of chromosomes organizes origins of replication into domains, such that all the origins in a given domain fire simultaneously. All of these influences probably work together to determine how mammalian origins of replication are selected by the cell, thereby explaining the difficulty scientists have had in precisely defining their salient features.

### Properties of the ORC Ensure That Each Region of the DNA Is Replicated Once and Only Once in Each S Phase

In bacteria, once the initiator protein is properly bound to the single origin of replication, the assembly of the replication forks seems to follow more or less automatically. In eukaryotes, the situation is significantly different because of a profound problem eukaryotes have in replicating chromosomes: with so many places to begin replication, how is the process regulated to ensure that all the DNA is copied once and only once?

The answer lies in how the assembly of the replication-fork protein at the origins of replication is regulated. We discuss this process in more detail in Chapter 17, where we consider the machinery that underlies the cell-division cycle. In brief, during G<sub>1</sub> phase, a symmetrical complex of two incomplete helicases is loaded onto DNA by the bound ORC. Then, upon passage from G<sub>1</sub> phase to S phase, specialized protein kinases come into play and direct the final assembly of the two replicative helicases, positioning one on each of the two complementary DNA single strands, where they move in opposite directions to begin opening the DNA double helix. At this point, the additional replication proteins are brought to the DNA, and two complete replication forks move in opposite directions away from the origin of replication (**Figure 5-31**).

The same protein kinases that trigger the final assembly of the helicases prevent the binding of new helicases to that origin until the next M phase resets the entire cycle (for details, see pp. 1043–1045). They do this, in part, by phosphorylating ORC, rendering it unable to accept new helicases. Thus, the kinases specify a single window of opportunity for precursor helicases to be loaded at origins of replication (G<sub>1</sub> phase, when kinase activity is low) and a second window for



**Figure 5–31 DNA replication initiation in eukaryotes.** This mechanism ensures that each origin of replication is activated only once per cell cycle. An origin of replication can be used only if two Mcm helicases (which form the enzymatic cores of the replicative helicases) are loaded in G<sub>1</sub> phase. At the beginning of S phase, specialized kinases phosphorylate both the Mcm helicases and ORC, activating the former and inactivating the latter. These kinases also guide the assembly of additional proteins that complete the helicases to form the fully active replicative helicases, known as the CMG helicases. New Mcm helicases cannot be loaded at the origin until the cell progresses through mitosis to the next G<sub>1</sub> phase, when ORC is dephosphorylated. The name CMG derives from Cdc45, Mcm, and GINS, the components of the active helicase (see Figure 5–19).

them to be assembled into their active form (S phase, when kinase activity is high). Because these two phases of the cell cycle are mutually exclusive and occur in a prescribed order, each origin of replication can fire only once during each cell cycle.

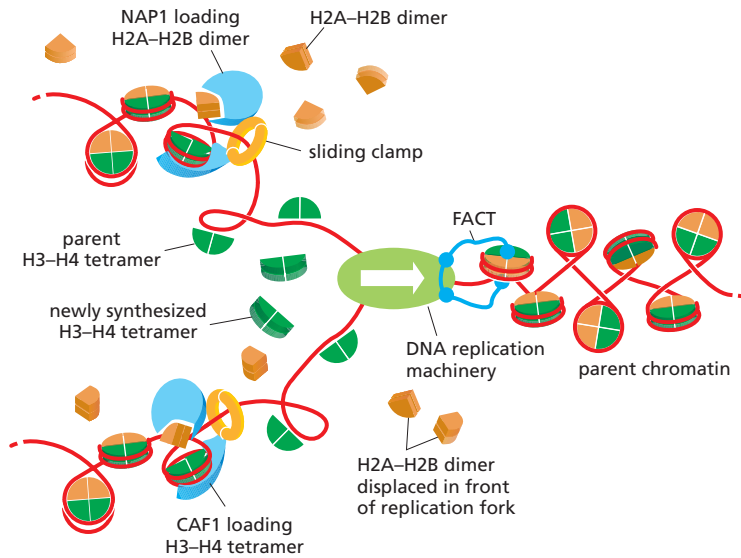
Because there are many more potential replication origins on a eukaryotic chromosome than are actually used in any one cell cycle (see Figure 5-30), the DNA at many ORC-bound replication origins will be replicated by forks formed at a neighboring region of the chromosome. Thus, preventing any single origin from firing more than once during an S phase is not enough to avoid the re-replication of DNA in eukaryotes. In addition, any ORC-DNA complex that is passed by a replication fork must be inactivated, and it is the combination of the two mechanisms that guarantees that each region of the DNA is replicated once and only once in each S phase.

### New Nucleosomes Are Assembled Behind the Replication Fork

Several additional aspects of DNA replication are specific to eukaryotes compared with bacteria. As discussed in Chapter 4, eukaryotic chromosomes are composed of roughly equal mixtures of DNA and protein. Chromosome duplication therefore requires not only the replication of DNA but also the synthesis of new chromosomal proteins and their assembly onto the DNA behind each replication fork. Although we are far from understanding this process in detail, we are beginning to learn how the fundamental unit of chromatin packaging, the nucleosome, is duplicated. The cell requires a large amount of new histone protein, approximately equal in mass to the newly synthesized DNA, each time it divides. For this reason, most eukaryotic organisms possess multiple copies of the gene for each histone. Vertebrate cells, for example, have about 20 repeated gene sets, most containing the genes that encode all five histones (H1, H2A, H2B, H3, and H4).

Unlike most proteins, which are made continually, histones are synthesized mainly in S phase, when the level of histone mRNA increases about fiftyfold as a result of both increased transcription and decreased mRNA degradation. The major histone mRNAs are degraded within minutes when DNA synthesis stops at the end of S phase. The mechanism depends on special properties of the 3' ends of these mRNAs, as discussed in Chapter 7. In contrast to their mRNAs, the histone proteins themselves are remarkably stable and may survive for many generations. The tight linkage between DNA synthesis and histone synthesis appears to reflect a feedback mechanism that monitors the level of free histone to ensure that the amount of histone made exactly matches the amount of new DNA synthesized.

As a replication fork advances it must pass through the parent nucleosomes. In the cell, efficient replication requires *chromatin remodeling complexes* (discussed in Chapter 4) and *histone chaperone proteins* (discussed below) to destabilize the DNA-histone interfaces. Aided by such specialized proteins, replication forks can transit even highly condensed chromatin. As a replication fork passes through chromatin, the histones are transiently displaced leaving about 600 nucleotide pairs of “free” DNA in its wake. The reestablishment of nucleosomes behind a moving fork occurs in an intriguing way. When a nucleosome is traversed by a replication fork, the histone octamer is broken into an H3-H4 tetramer and two H2A-H2B dimers (discussed in Chapter 4), all of which are released from DNA. The H3-H4 tetramers remain in the vicinity of the fork by loosely binding to several of the proteins at the replication fork (primarily the CMG helicase) and are distributed at random to one or the other daughter duplexes as the fork moves forward. In contrast, the H2A-H2B dimers are released completely from the fork and may diffuse to entirely different chromosomes. Freshly made H3-H4 tetramers are added to the newly synthesized DNA to fill in the “spaces,” and H2A-H2B dimers—half of which are old and half new—are then added at random



**Figure 5–32 Formation of nucleosomes behind a replication fork.** Parent H3–H4 tetramers remain associated with the fork and are distributed at random to the daughter DNA molecules, with roughly equal numbers inherited by each daughter. In contrast, H2A–H2B dimers are released completely from the fork as it passes. This release begins just in front of the replication fork and is facilitated by the histone chaperone FACT, which moves with the fork. FACT has several globular protein domains connected by flexible linkers and can make multiple contacts with a nucleosome to aid in its disassembly. Additional histone chaperones (NAP1 and CAF1) restore the full complement of histones to daughter molecules using both parent and newly synthesized histones. Although not shown in the figure, it has been proposed that FACT directly hands off parent H3–H4 tetramers to components of the replication machinery, which in turn hand them off to CAF1 chaperones, which deposit them evenly on the two daughter molecules. The way in which histones are distributed behind a replication fork means that some daughter nucleosomes contain only parent histones or only newly synthesized histones, but most are hybrids of old and new. For simplicity, the DNA double helix is shown as a single red line.

to complete the nucleosomes behind the fork (**Figure 5–32**). The formation of new nucleosomes behind a replication fork has an important consequence for the process of DNA replication itself. As DNA polymerase  $\delta$  discontinuously synthesizes the lagging strand (see **Figure 5–19**), the length of each Okazaki fragment is determined by the point at which DNA polymerase  $\delta$  is blocked by a newly formed nucleosome. This tight coupling between nucleosome duplication and DNA replication probably explains why the length of Okazaki fragments in eukaryotes (~200 nucleotides) is approximately the same as the nucleosome repeat length.

The orderly and rapid addition of new H3–H4 tetramers and H2A–H2B dimers behind a replication fork requires **histone chaperones** (also called *chromatin assembly factors*). These multisubunit complexes bind the highly basic histones and release them on DNA only in the appropriate context. For example, some of the histone chaperones, along with their histone cargoes, are directed to newly replicated DNA through a specific interaction with the sliding clamp (see **Figure 5–32**). As we have seen, these clamps remain on the DNA behind replication forks, and some appear to linger just long enough for the histone chaperones to complete their tasks. Because they bind so well to histones, some histone chaperones also help to disassemble nucleosomes. Of particular importance to DNA replication is the FACT chaperone, which moves at the front of the replication machinery, disassembling nucleosomes as it moves forward (see **Figure 5–32**).

### Termination of DNA Replication Occurs Through the Ordered Disassembly of the Replication Fork

We saw earlier in this chapter that *E. coli* DNA replication begins at a single origin, and two replication forks proceed bidirectionally around the circular genome, meeting at a spot opposite to the origin of replication. Here, the two forks do not simply collide with each other running at full speed; rather, this spot on the *E. coli* genome has a special DNA sequence that slows down and stalls the movement of each fork, causing them to disassemble. The remaining gaps in the daughter DNA molecules are filled in and sealed by repair DNA polymerases and DNA ligase (see **Figures 5–11** and **5–12**), and the two completed bacterial genomes are separated using topoisomerases (see **Figure 5–23**).

As might be expected, the situation in eukaryotes is more complicated. First, each round of replication requires many termination events, roughly as many as there are initiation events at origins of replication. Thus, in mammalian cells, approximately 30,000–50,000 termination events occur in every S phase. Second, the termination of replication forks in eukaryotes is largely independent of any underlying DNA sequence in the genome. Rather, the principal termination

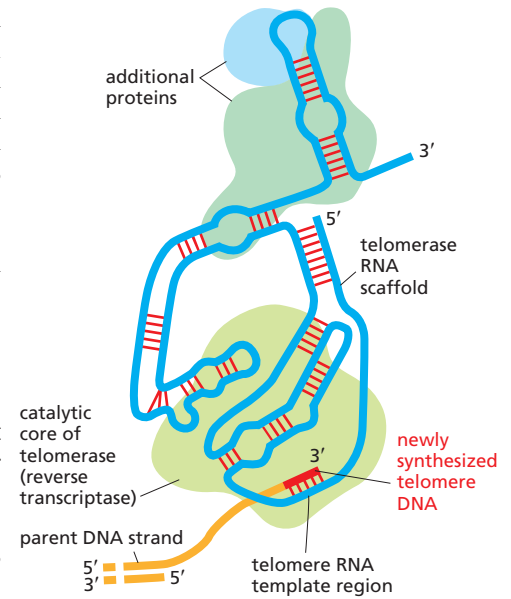
signal is a head-on encounter with a fork moving in the opposite direction. When two forks meet, the CMG helicase at each fork is covalently modified by addition of ubiquitin (see Figure 3-65), which causes its disassembly and removal from DNA. Without the helicase, the other replication proteins rapidly dissociate from the fork. Repair DNA polymerase and DNA ligase subsequently fill in and seal any remaining gaps. Eukaryotic replication forks must also contend with the ends of chromosomes. Here, it is believed that the CMG helicase simply slides off the end of the DNA molecule, leading to the dissociation of the other fork proteins. However, replicating DNA to the very end of a chromosome presents a special challenge to the eukaryotic cell, as we describe next.

### Telomerase Replicates the Ends of Chromosomes

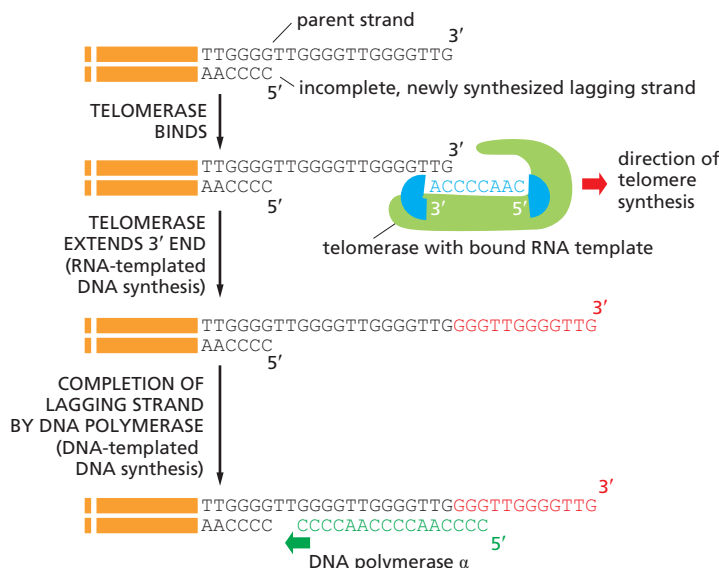
We saw earlier in the chapter that synthesis of the lagging strand at a replication fork must occur discontinuously through a backstitching mechanism that produces short DNA fragments attached to RNA primers. The final RNA primer synthesized on the lagging-strand template cannot be replaced by DNA because there is no primer ahead of it to provide a 3'-OH end for the repair polymerase. Without a mechanism to deal with this problem, DNA would be lost from the ends of all chromosomes each time a cell divides.

Bacteria avoid this “end-replication” problem by having circular DNA molecules as chromosomes, as we have seen. Eukaryotes solve it in a different way: they have specialized nucleotide sequences at the ends of their chromosomes that are incorporated into structures called *telomeres* (discussed in Chapter 4). Telomeres contain many tandem repeats of a short sequence that is similar in organisms as diverse as protozoa, fungi, plants, and mammals. In humans, the sequence of the repeat unit is GGGTTA, and it is repeated roughly a thousand times at each telomere.

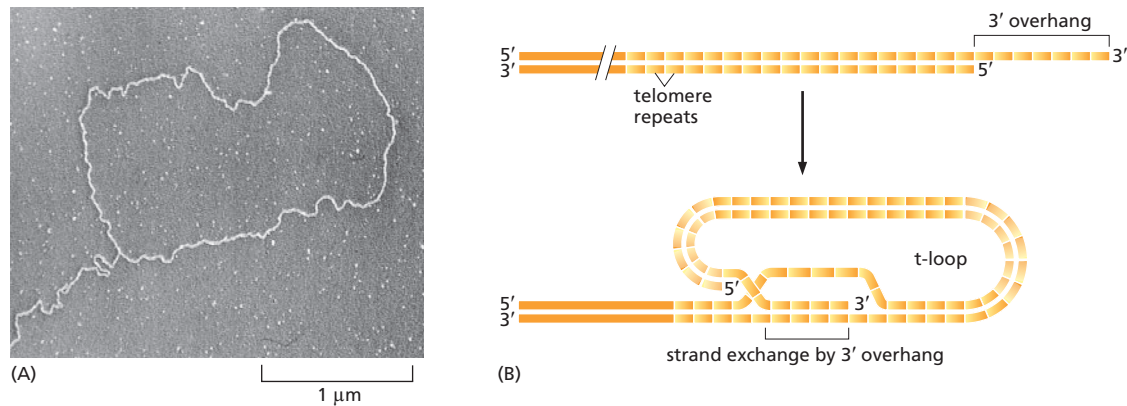
Telomere DNA sequences are recognized by sequence-specific DNA-binding proteins that attract an enzyme, called **telomerase**, that replenishes these sequences each time a cell divides. Telomerase recognizes the tip of an existing telomere DNA repeat sequence and elongates it in the 5'-to-3' direction, using an RNA template that is a component of the enzyme itself to synthesize new DNA copies of the repeat (Figure 5-33). The enzymatic portion of telomerase resembles other *reverse transcriptases*, proteins that synthesize DNA using an RNA template, although, in this case, the telomerase RNA also contributes to the active site and is essential for efficient catalysis. After extension of the parent DNA strand by telomerase, replication of the lagging strand at the chromosome end can be completed by the conventional DNA polymerases, using these extensions as a template to synthesize the complementary strand (Figure 5-34).



**Figure 5-33 Schematic structure of human telomerase.** This large enzyme is composed of 10 protein subunits and an RNA of 451 nucleotides. The RNA forms the scaffold of the complex, provides the template for synthesizing new DNA telomere repeats, and helps form the active site. The synthesis reaction itself is carried out by the reverse transcriptase domain of the protein, shown in *light green*, in conjunction with the RNA. A reverse transcriptase is a special form of polymerase enzyme that uses an RNA template to make a DNA strand; telomerase is unique in carrying its own RNA template with it. Telomerase also contains several additional protein complexes (some of which are shown in *dark green* and *blue*) that are needed to assemble the enzyme and, for many organisms but not humans, to bring it to the ends of chromosomes. (Modified from T.H.D. Nguyen et al., *Nature* 557: 190–195, 2018.)



**Figure 5-34 Telomere replication.** Shown here is the reaction that synthesizes the repeating sequences that form the ends of the chromosomes (telomeres) of eukaryotes. The 3' end of the parent lagging-strand template is extended by RNA-templated DNA synthesis; this allows the incomplete daughter DNA strand that is paired with it to be synthesized to the end of the chromosome. The synthesis of the final bit of lagging strand is carried out by DNA polymerase  $\alpha$ , which carries a DNA primase as one of its subunits (Movie 5.6). DNA polymerase  $\alpha$  is the same enzyme used to begin the synthesis of each Okazaki fragment on the lagging strand; it begins its synthesis with RNA (not shown) and continues with DNA (green). The telomere sequence illustrated is that of the ciliate *Tetrahymena*, in which these reactions were first discovered.



### Telomeres Are Packaged into Specialized Structures That Protect the Ends of Chromosomes

The ends of chromosomes present cells with an additional problem. As we will see in the next part of this chapter, when a chromosome is accidentally broken into two pieces, the break is rapidly repaired. Telomeres must clearly be distinguished from these accidental breaks; otherwise, the cell will attempt to “repair” telomeres, generating chromosome fusions and other genetic abnormalities. Telomeres have several features to prevent this from happening.

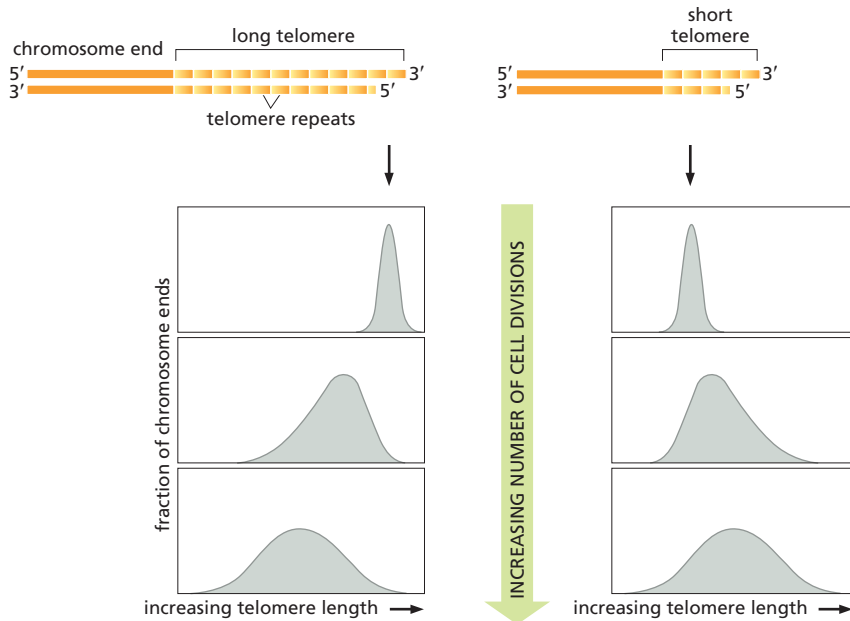
A specialized nuclease chews back the 5′ end of a telomere leaving a protruding, single-strand 3′ end. This protruding end—in combination with the GGGTTA repeats in telomeres—attracts a group of proteins that form a protective chromosome cap known as *shelterin*. In particular, shelterin protects telomeres from being treated as damaged DNA. Another feature of telomeres may offer additional protection. When human telomeres are artificially cross-linked and viewed by electron microscopy, structures known as “t-loops” can be observed in which the protruding single-strand end of the telomere loops back and tucks itself into the duplex DNA of the telomere repeat sequence (Figure 5-35). An attractive idea is that t-loops are orchestrated by shelterin to help “hide” the very ends of chromosomes.

### Telomere Length Is Regulated by Cells and Organisms

Because the processes that grow and shrink each telomere sequence are only approximately balanced, chromosome ends contain variable numbers of telomeric repeats. Not surprisingly, many cells, including stem cells and germ cells, have homeostatic mechanisms that maintain the number of these repeats within a limited range (Figure 5-36).

In most of the dividing somatic cells of humans, however, telomeres gradually shorten, and it has been proposed that this provides a counting mechanism that helps prevent the unlimited proliferation of wayward cells in adult tissues. In its simplest form, this idea holds that our somatic cells start off in the embryo with a full complement of telomeric repeats. These are then eroded to different extents in different cell types. Some stem cells, notably those in tissues that must be replenished at a high rate throughout life—bone marrow or gut lining, for example—retain full telomerase activity. However, in many other types of cells, the level of telomerase is reduced so that the enzyme cannot quite keep up with chromosome duplication. Such cells lose 100–200 nucleotides from each telomere every time they divide. After many cell generations, the descendant cells will inherit chromosomes that lack functioning telomeres, and, as a result of this defect, activate a DNA-damage response causing them to withdraw permanently from the cell cycle and cease dividing—a process called *replicative cell senescence* (discussed in Chapters 17 and 20). In theory, such a mechanism could provide a

**Figure 5-35 A t-loop at the end of a mammalian chromosome.** (A) Electron micrograph of the DNA at the end of an interphase human chromosome. The chromosome was fixed, deproteinated, and artificially thickened before viewing. The loop seen here is approximately 15,000 nucleotide pairs in length. (B) Schematic diagram of t-loop formation. (A, from J.D. Griffith et al., *Cell* 97:503–514, 1999. With permission from Elsevier.)



**Figure 5-36** A demonstration that yeast cells control the length of their telomeres. In this experiment, the telomere at one end of a particular chromosome is artificially made either longer (*left*) or shorter (*right*) than average. After many cell divisions, the chromosome recovers, showing an average telomere length and a length distribution that is typical of the other chromosomes in the yeast cell. A similar feedback mechanism for controlling telomere length has been proposed for the germ-line cells and stem cells of mammals.

safeguard against the uncontrolled cell proliferation of abnormal cells in somatic tissues, thereby helping to protect us from cancer.

The idea that telomere length acts as a “measuring stick” to count cell divisions and thereby regulate the lifetime of the cell lineage has been tested in several ways. For certain types of human cells grown in tissue culture, the experimental results support such a theory. Human fibroblasts normally proliferate for about 60 cell divisions in culture before undergoing replicative cell senescence. Like most other somatic cells in humans, fibroblasts produce only low levels of telomerase, and their telomeres gradually shorten each time they divide. When telomerase is provided to the fibroblasts by inserting a fully active telomerase gene, telomere length is maintained and many of the cells now continue to proliferate indefinitely. Also consistent with these ideas is the observation that, in approximately 90% of cancer cells, the telomerase gene has become reactivated, thereby circumventing the normal safety mechanism (see pp. 1073–1074).

It has been proposed that this type of control on cell proliferation may contribute to the aging of animals like ourselves. These ideas have been tested by producing transgenic mice that lack telomerase entirely. The telomeres in mouse chromosomes are about five times longer than human telomeres, and the mice must therefore be bred through three or more generations before their telomeres have shrunk to the normal human length. It is therefore perhaps not surprising that the first generations of mice develop normally. However, the mice in later generations develop progressively more defects in some of their highly proliferative tissues. In addition, these mice show signs of premature aging and have a pronounced tendency to develop tumors. In these and other respects, these mice resemble humans with the genetic disease *dyskeratosis congenita*. Individuals afflicted with this disease carry one functional and one nonfunctional copy of the telomerase RNA gene; they have prematurely shortened telomeres and typically die of progressive bone marrow failure. These individuals also develop lung scarring and liver cirrhosis and show abnormalities in various epidermal structures including skin, hair follicles, and nails.

The above observations demonstrate that controlling cell proliferation by telomere shortening poses a risk to an organism, because not all of the cells that begin losing the ends of their chromosomes will stop dividing. Some apparently become genetically unstable, but continue to divide, giving rise to variant cells that can lead to cancer. As discussed above, many of these variant cells

ultimately produce high levels of telomerase, thereby ensuring their continued survival. Clearly, the use of telomere shortening as a regulating mechanism is not foolproof and, like many mechanisms in the cell, it must strike a balance between benefit and risk.

## Summary

*The proteins that initiate DNA replication bind to DNA sequences at a replication origin to catalyze the formation of a replication bubble with two outward-moving replication forks. The process begins when an initiator protein–DNA complex is formed that subsequently loads a DNA helicase onto the DNA template. Other proteins are then added to form the multienzyme “replication machine” that catalyzes DNA synthesis at each replication fork.*

*In bacteria and some simple eukaryotes, replication origins are defined by specific DNA sequences that are several hundred nucleotide pairs long. In other eukaryotes, such as humans, features that specify an origin of DNA replication are less well defined, and probably depend more on structural features of chromosomes than on specific DNA sequences.*

*Bacteria typically have a single origin of replication in a circular chromosome. With fork speeds of up to 1000 nucleotides per second, they can replicate their genome in less than an hour. Eukaryotic DNA replication takes place in only one part of the cell cycle, the S phase. The replication fork in eukaryotes moves about 20 times more slowly than the bacterial replication fork, and the much longer eukaryotic chromosomes each require many replication origins to complete their replication in an S phase, which typically lasts for 8 hours in human cells. The different replication origins in these eukaryotic chromosomes are activated in a sequence, determined in part by which genes are currently being transcribed and the structure of chromatin across each chromosome. After the replication fork has passed, chromatin structure is re-formed by the addition of new histones to the old histones that are directly inherited by each daughter DNA molecule.*

*Eukaryotes solve the problem of replicating the ends of their linear chromosomes with a specialized end structure, the telomere, maintained by a special nucleotide-polymerizing enzyme called telomerase. Telomerase extends one of the DNA strands at the end of a chromosome by using an RNA template that is an integral part of the enzyme itself, producing a highly repeated DNA sequence that typically extends for thousands of nucleotide pairs at each chromosome end. Telomeres have specialized structures that distinguish them from broken ends of chromosomes, ensuring that they are not treated as damaged DNA.*

## DNA REPAIR

Maintaining the genetic stability that an organism needs for its survival requires not only an extremely accurate mechanism for replicating DNA but also mechanisms for repairing the many accidental lesions that DNA continually suffers. Most such spontaneous changes in DNA are temporary because they are immediately corrected by a set of processes that are collectively called **DNA repair**. Of the tens of thousands of random changes created every day in the DNA of a human cell by heat, metabolic accidents, radiation of various sorts, and exposure to substances in the environment, only a few (less than 0.02%) accumulate as permanent mutations in the DNA sequence. The rest are eliminated with remarkable efficiency by DNA repair.

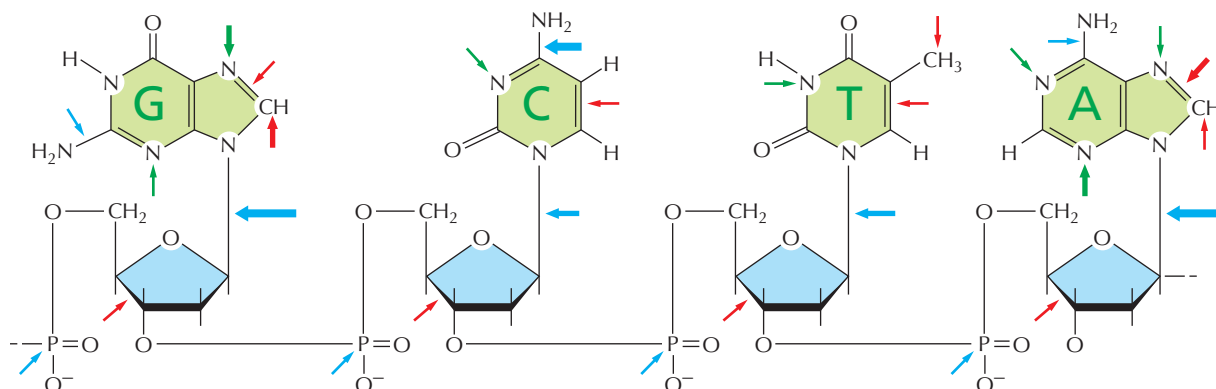
The importance of DNA repair is evident from the large investment that cells make in the enzymes that carry it out: several percent of the coding capacity of most genomes is devoted solely to DNA repair functions. The importance of DNA repair is also demonstrated by the increased rate of mutation that follows the inactivation of a DNA repair gene. Many DNA repair proteins and the genes that encode them—which we now know operate in a wide range of organisms,

TABLE 5–2 Some Inherited Human Syndromes with Defects in DNA Repair

| Name of syndrome or responsible genes                               | Phenotype                                                         | Enzyme or process affected                                          |
|---------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------|
| <i>Msh2</i> , <i>Msh3</i> , <i>Msh6</i> , <i>Mlh1</i> , <i>Pms2</i> | Colon cancer                                                      | Mismatch repair                                                     |
| Polymerase proofreading–associated polyposis                        | Colon cancer                                                      | Proofreading by DNA polymerase $\epsilon$                           |
| Aicardi–Goutières syndrome                                          | Encephalopathy, neurological dysfunction, genome instability      | Removal of misincorporated ribonucleotides in DNA                   |
| Xeroderma pigmentosum (XP) groups A–G                               | Skin cancer, UV sensitivity, neurological abnormalities           | Nucleotide excision repair                                          |
| Cockayne syndrome                                                   | UV sensitivity, developmental abnormalities                       | Coupling of nucleotide excision repair to transcription             |
| XP variant                                                          | UV sensitivity, skin cancer                                       | Translesion synthesis by DNA polymerase $\eta$                      |
| Ataxia telangiectasia (AT)                                          | Leukemia, lymphoma, $\gamma$ -ray sensitivity, genome instability | ATM protein, a protein kinase activated by double-strand DNA breaks |
| Seckel syndrome                                                     | Dwarfism, microcephaly                                            | ATR protein, a protein kinase activated by single-strand DNA breaks |
| <i>Brca1</i>                                                        | Breast and ovarian cancer                                         | Repair by homologous recombination                                  |
| <i>Brca2</i>                                                        | Breast, ovarian, prostate, and pancreatic cancer                  | Repair by homologous recombination                                  |
| Ataxia-telangiectasia-like disorder (ATLD)                          | Leukemia, lymphoma, $\gamma$ -ray sensitivity, genome instability | Mre11 protein, required for processing double-strand DNA breaks     |
| Werner syndrome                                                     | Premature aging, cancer at several sites, genome instability      | Accessory 3'-exonuclease and DNA helicase used in repair            |
| Bloom syndrome                                                      | Cancer at several sites, stunted growth, genome instability       | DNA helicase needed for recombination                               |
| Fanconi anemia groups A–W                                           | Congenital abnormalities, leukemia, genome instability            | DNA interstrand cross-link repair                                   |
| 46BR patient                                                        | Hypersensitivity to DNA-damaging agents, genome instability       | DNA ligase I                                                        |

including humans—were originally identified in bacteria by the isolation and characterization of mutants that displayed an increased mutation rate or an increased sensitivity to DNA-damaging agents.

Studies of the consequences of a diminished capacity for DNA repair in humans have linked many human diseases with decreased repair (Table 5–2). Thus, we saw previously that defects in a human gene whose product normally functions to repair the mismatched base pairs resulting from DNA replication errors can lead to an inherited predisposition to cancers of the colon and some other organs, caused by an increased mutation rate. In another human disease, *xeroderma pigmentosum* (XP), the afflicted individuals have an extreme sensitivity to ultraviolet radiation because they are unable to repair the damage to DNA caused by this component of sunlight. This repair defect results in an increased mutation rate that leads to serious skin lesions and a greatly increased susceptibility to skin cancers. Finally, mutations in the *Brca1* and *Brca2* genes compromise a type of DNA repair known as *homologous recombination* and are a major cause of hereditary breast and ovarian cancers.



### Without DNA Repair, Spontaneous DNA Damage Would Rapidly Change DNA Sequences

Although DNA is a highly stable material—as required for the storage of genetic information—it is a complex organic molecule that is susceptible, even under normal cell conditions, to spontaneous changes that would lead to mutations if left unrepaired (Figure 5–37 and see Table 5–3). For example, the DNA of each human cell loses about 18,000 purine bases (adenine and guanine) every day because their *N*-glycosyl linkages to deoxyribose break, a spontaneous hydrolysis reaction called *depurination*. Similarly, a spontaneous *deamination* of cytosine to uracil in

**Figure 5–37** A summary of spontaneous alterations that require DNA repair.

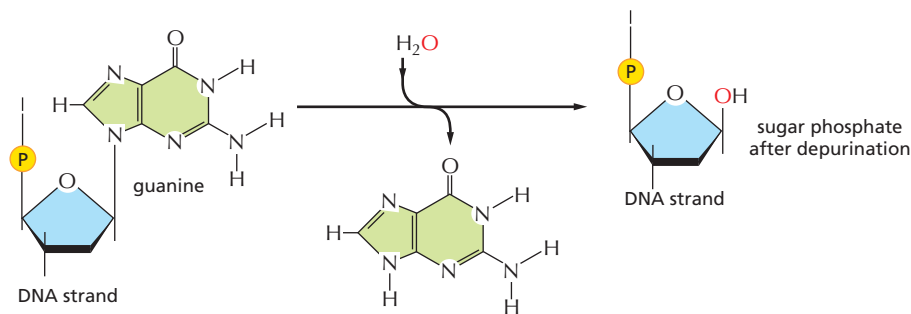
The sites on each nucleotide modified by spontaneous oxidative damage (red arrows), hydrolytic attack (blue arrows), and methylation (green arrows) are shown, with the width of each arrow indicating the relative frequency of each event (see Table 5–3). (After T. Lindahl, *Nature* 362:709–715, 1993.)

**TABLE 5–3** Endogenous DNA Lesions Arising and Repaired in a Diploid Mammalian Cell in 24 Hours

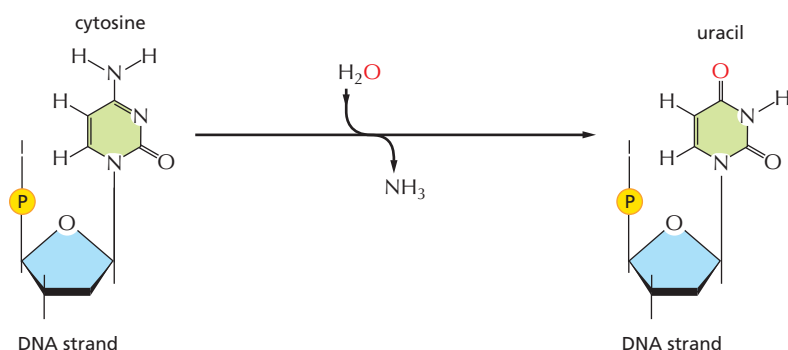
| DNA lesion                                                            | Number repaired in 24 hr |
|-----------------------------------------------------------------------|--------------------------|
| <b>Hydrolysis</b>                                                     |                          |
| Depurination                                                          | 18,000                   |
| Depyrimidination                                                      | 600                      |
| Cytosine deamination                                                  | 100                      |
| 5-Methylcytosine deamination                                          | 10                       |
| <b>Oxidation</b>                                                      |                          |
| 8-oxoguanine                                                          | 1500                     |
| Ring-saturated pyrimidines (thymine glycol, cytosine hydrates)        | 2000                     |
| Lipid peroxidation products (M1G, etheno-A, etheno-C)                 | 1000                     |
| <b>Nonenzymatic methylation by S-adenosylmethionine</b>               |                          |
| 7-Methylguanine                                                       | 6000                     |
| 3-Methyladenine                                                       | 1200                     |
| <b>Nonenzymatic methylation by nitrosated polyamines and peptides</b> |                          |
| O <sup>6</sup> -Methylguanine                                         | 20–100                   |

The DNA lesions listed in the table are the result of the normal chemical reactions that take place in cells. Cells that are exposed to external chemicals and radiation suffer greater and more diverse forms of DNA damage. (From T. Lindahl and D.E. Barnes, *Cold Spring Harb. Symp. Quant. Biol.* 65:127–133, 2000.)

## (A) DEPURINATION



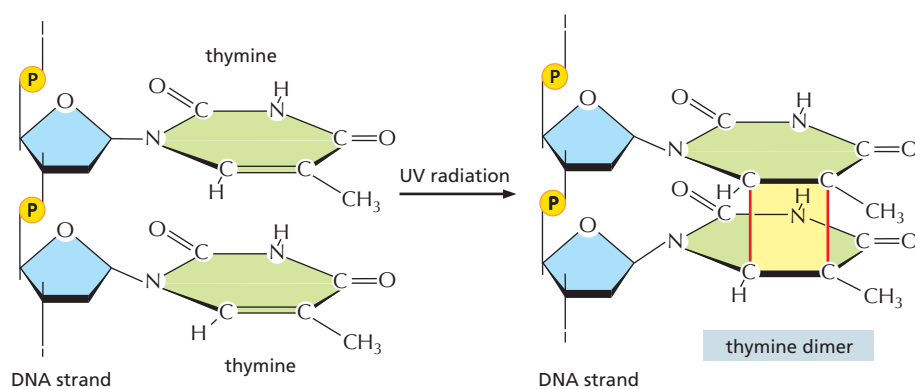
## (B) DEAMINATION



**Figure 5–38 Depurination and deamination are the most frequent spontaneous chemical reactions known to create serious DNA damage in cells.**

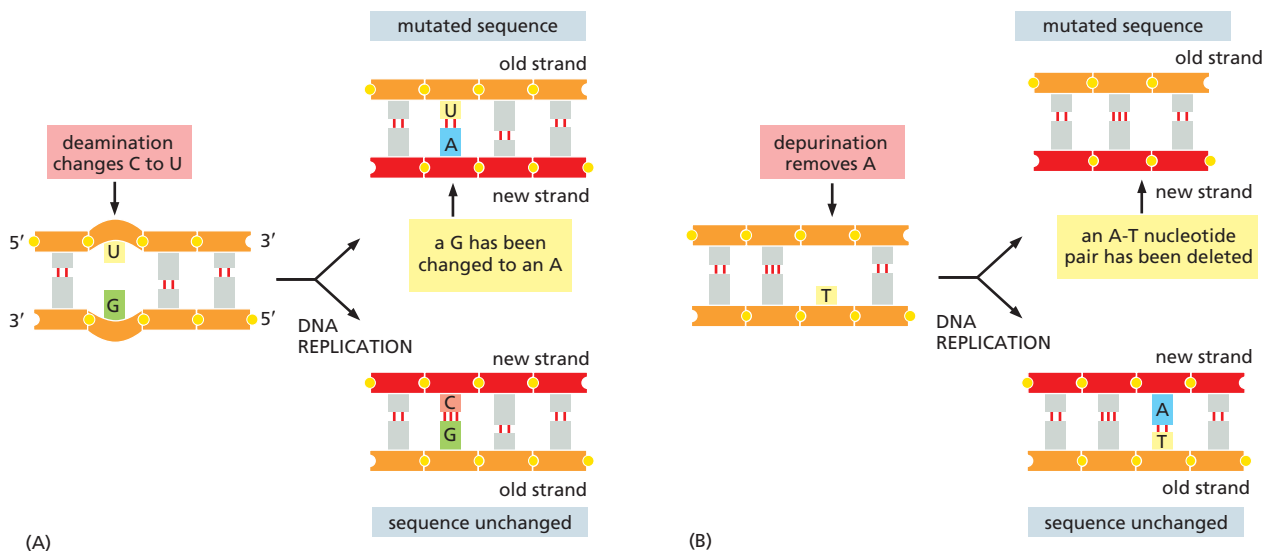
(A) Depurination can remove guanine (or adenine) from DNA. (B) The major type of deamination reaction converts cytosine to uracil, which, as we have seen, is not normally found in DNA. However, deamination can occur on other bases as well. Both depurination and deamination take place on double-helical DNA, and neither reaction breaks the phosphodiester backbone.

DNA occurs at a rate of about 100 bases per cell per day (Figure 5–38). DNA bases are also occasionally damaged by encounters with reactive metabolites produced in the cell (for example, the high-energy methyl donor, *S*-adenosylmethionine) or by exposure to toxic chemicals in the environment. Likewise, ultraviolet radiation from the Sun can produce a covalent linkage between two adjacent pyrimidine bases in DNA to form, for example, thymine dimers (Figure 5–39). If left uncorrected, most of these changes would lead either to the deletion of one or more base pairs or to a base-pair substitution in the daughter DNA chain when the DNA is replicated (Figure 5–40). These mutations would then be propagated throughout all subsequent cell generations. Such a high rate of unrepaired random changes in the DNA sequence would have disastrous consequences, both in the germ line and in somatic tissues.



**Figure 5–39 The ultraviolet radiation in sunlight can cause the formation of thymine dimers.**

Two adjacent thymine bases have become covalently attached to each other to form a thymine dimer. Skin cells that are exposed to sunlight are especially susceptible to this type of DNA damage. Dimers can also form between an adjacent thymine and cytosine.



### The DNA Double Helix Is Readily Repaired

The double-helical structure of DNA is ideally suited for repair because it carries two separate copies of all the genetic information—one in each of its two strands. Thus, when one strand is damaged, the complementary strand retains an intact copy of the same information, and this copy is generally used as the template to restore the correct nucleotide sequences to the damaged strand.

An indication of the importance of a double-strand helix to the safe storage of genetic information is that all cells use it; only a few small viruses use single-stranded DNA or RNA as their genetic material. The types of repair processes described in this part of the chapter cannot operate on such nucleic acids, and once damaged, the chance of a permanent nucleotide change occurring in these single-strand genomes of viruses is thus very high. It seems that only tiny genomes (and therefore tiny targets for DNA damage) can have their genetic information successfully carried in any molecule other than a DNA double helix.

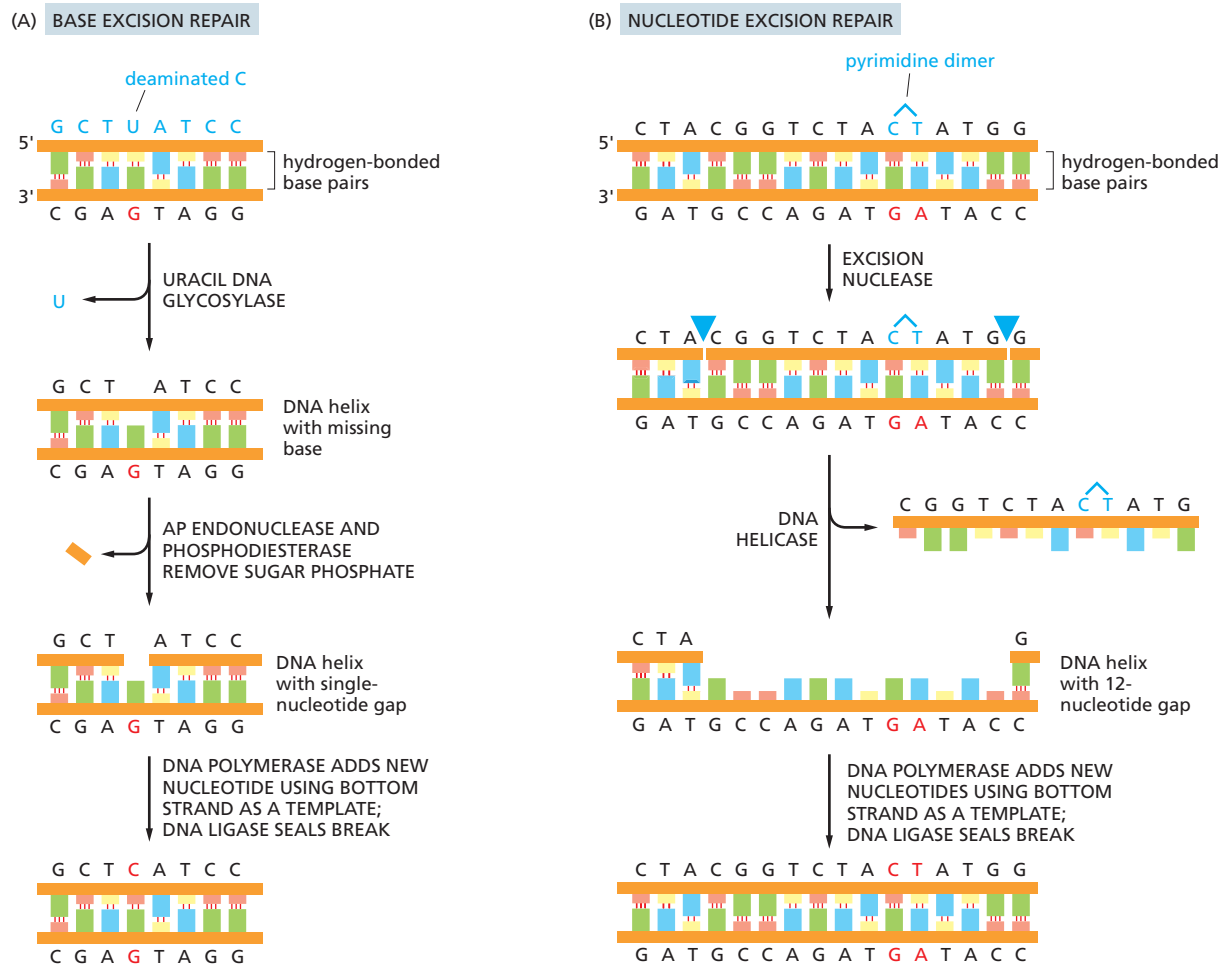
### DNA Damage Can Be Removed by More Than One Pathway

Cells have multiple pathways to repair their DNA using different enzymes that act upon different kinds of lesions. **Figure 5-41** shows two of the most common pathways. In both, the damage is excised, the original DNA sequence is restored by a high-fidelity DNA polymerase using the undamaged strand as its template, and the remaining break in the double helix is sealed by DNA ligase (see **Figure 5-12**).

The two pathways differ in the way in which they remove the damage from DNA. The first pathway, called **base excision repair**, involves a battery of enzymes called *DNA glycosylases*, each of which can recognize a specific type of altered base in DNA and catalyze its hydrolytic removal from the DNA backbone. There are many types of these enzymes, including those that remove deaminated C's, deaminated A's, different types of alkylated or oxidized bases, bases with opened rings, and bases in which a carbon-carbon double bond has been accidentally converted to a carbon-carbon single bond. How are altered bases detected in the double helix? A key step is an enzyme-mediated “flipping-out” of the altered nucleotide from the helix, which allows the DNA glycosylase to probe all faces of the base for damage (**Figure 5-42**). It is thought that these enzymes travel along DNA using base-flipping to evaluate the status of each base. Once an enzyme finds the damaged base that it recognizes, it removes that base from its sugar.

The “missing tooth” created by DNA glycosylase action is recognized by an enzyme called *AP endonuclease* (AP for *apurinic* or *apyrimidinic*, and *endo* to signify that the nuclease cleaves within the polynucleotide chain), which cuts the phosphodiester backbone, after which the resulting gap is repaired (see

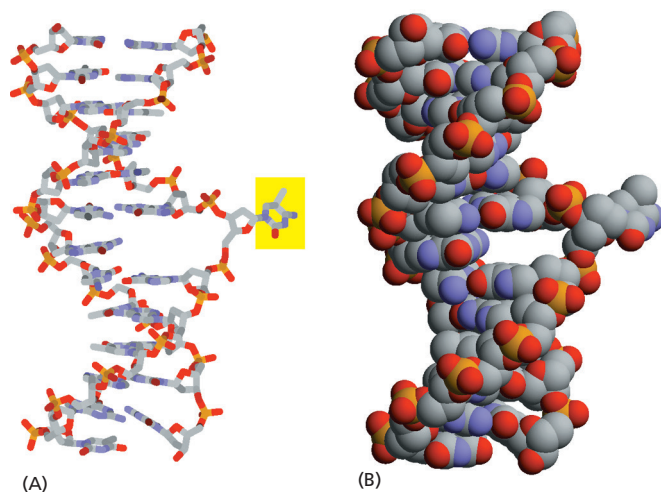
**Figure 5-40** Chemical modifications of nucleotides, if left unrepaired, produce mutations. (A) Deamination of cytosine, if uncorrected, results in the substitution of one base for another when the DNA is replicated. As shown in **Figure 5-43**, deamination of cytosine produces uracil. Uracil differs from cytosine in its base-pairing properties and preferentially base-pairs with adenine. The DNA replication machinery therefore inserts an adenine when it encounters a uracil on the template strand. (B) Depurination, if uncorrected, can lead to the loss of a nucleotide pair. When the replication machinery encounters a missing purine on the template strand, it can skip to the next complete nucleotide, as shown, thus producing a daughter DNA molecule that is missing one nucleotide pair. In other cases, the replication machinery places an incorrect nucleotide across from the missing base, again resulting in a mutation (not shown).



**Figure 5-41 A comparison of two major DNA repair pathways.** (A) *Base excision repair*. This pathway starts with a DNA glycosylase. In the example shown here, the enzyme uracil DNA glycosylase removes an accidentally deaminated cytosine in DNA. After the action of this glycosylase (or another DNA glycosylase that recognizes a different kind of damage), the sugar phosphate with the missing base is cut out by the sequential action of AP endonuclease and a phosphodiesterase. The gap of a single nucleotide is then filled by DNA polymerase and DNA ligase. The net result is that the U that was created by accidental deamination is restored to a C. The loss of a base can occur either from the actions of DNA glycosylases that recognize damaged bases or from spontaneous chemical reactions (see Figure 5-37). AP endonuclease is so named because it recognizes any site in the DNA helix that contains a deoxyribose sugar with a missing base; such sites can arise either by the loss of a purine (apurinic sites) or by the loss of a pyrimidine (apyrimidinic sites). (B) *Nucleotide excision repair*. In bacteria, after a multienzyme complex has recognized a lesion such as a pyrimidine dimer (see Figure 5-39), one cut is made on each side of the lesion, and an associated DNA helicase then removes the entire portion of the damaged strand. The excision repair machinery in bacteria operates as shown. In humans, once the damaged DNA is recognized, a helicase is recruited to locally unwind the DNA duplex. Next, the excision nuclease enters and cleaves on either side of the damage, leaving a gap of about 30 nucleotides that is subsequently filled in. The nucleotide excision repair machinery in both bacteria and humans can recognize and repair many different types of DNA damage.

Figure 5-41A). Depurination, which is by far the most frequent type of damage suffered by DNA, also leaves a deoxyribose sugar with a missing base. Depurinations are directly repaired beginning with AP endonuclease, following the bottom half of the pathway in Figure 5-41A.

The second major repair pathway is called **nucleotide excision repair**. This mechanism can repair the damage caused by almost any large change in the structure of the DNA double helix. Such “bulky lesions” include those created by the covalent reaction of DNA bases with large hydrocarbons (such as the carcinogen benzopyrene, found in tobacco smoke, coal tar, and diesel exhaust), as well as the various pyrimidine dimers (T-T, T-C, and C-C) caused by sunlight. In this pathway, a large multienzyme complex scans the DNA for a distortion in the double helix, rather than for a specific base change. Once it finds a lesion, it cleaves the phosphodiester backbone of the abnormal strand on both sides of



**Figure 5-42 The recognition of an unusual nucleotide in DNA by base-flipping.** The DNA glycosylase family of enzymes recognizes inappropriate bases in DNA in the conformation shown. Each of these enzymes cleaves the glycosyl bond that connects a particular recognized base (yellow) to the backbone sugar, removing it from the DNA. (A) Stick model of the DNA; (B) space-filling model.

the distortion, and a DNA helicase peels away the single-strand oligonucleotide containing the lesion. The large gap produced in the DNA helix is then repaired by DNA polymerase and DNA ligase (see Figure 5-41B).

An alternative to these base and nucleotide excision repair processes is the direct chemical reversal of DNA damage, and this strategy is selectively employed for the rapid removal of certain highly mutagenic or cytotoxic lesions. For example, the lesion  $O^6$ -methylguanine has its methyl group removed by direct transfer to a cysteine residue in the repair protein itself. Because the repair protein is destroyed in the process, each molecule of it can only be used once. In another example, methyl groups in the lesions 1-methyladenine and 3-methylcytosine are “burned off” by an iron-dependent demethylase, with release of formaldehyde from the methylated DNA and regeneration of the native base.

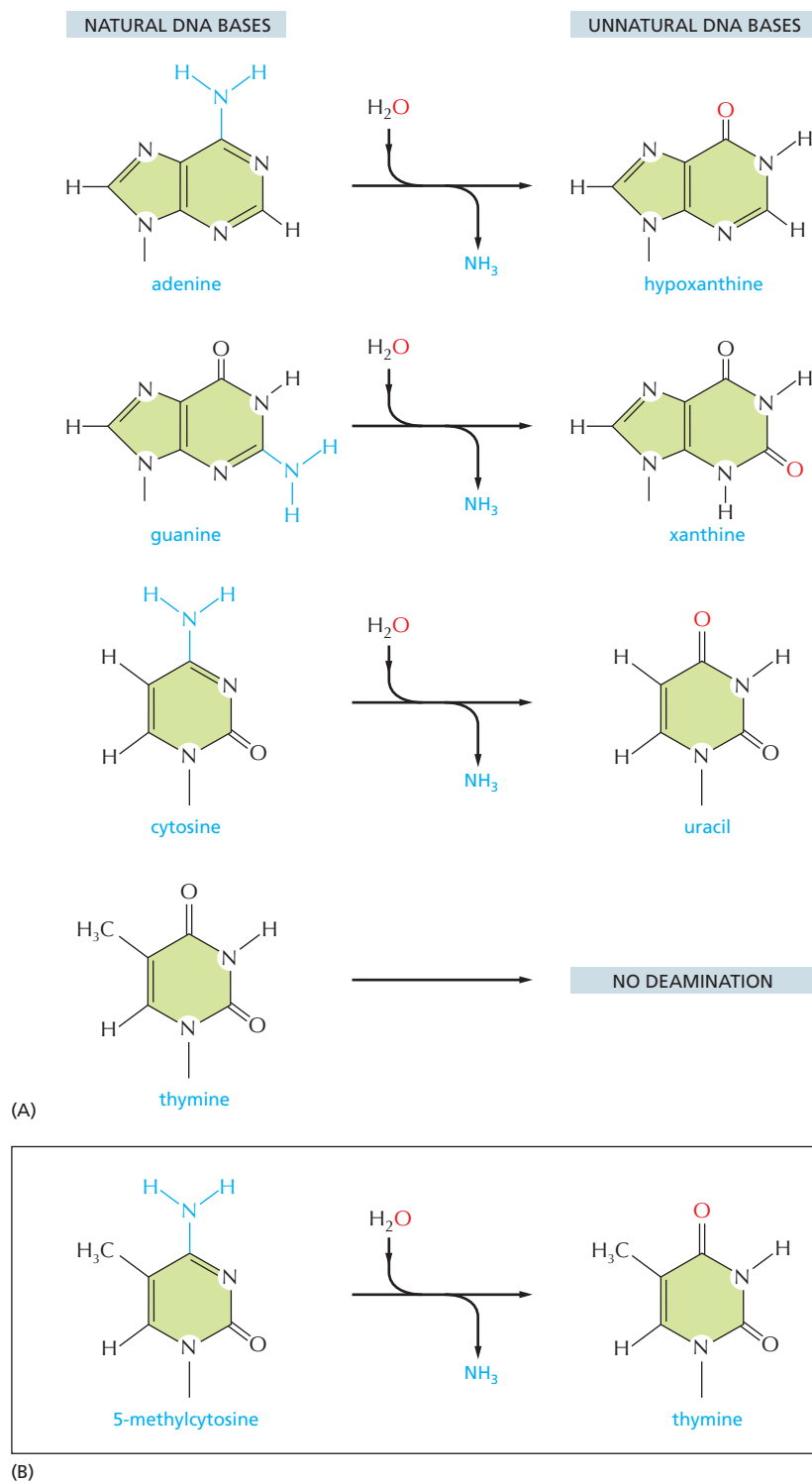
### Coupling Nucleotide Excision Repair to Transcription Ensures That the Cell’s Most Important DNA Is Efficiently Repaired

All of a cell’s DNA is under constant surveillance for damage, and the repair mechanisms we have described act on all parts of the genome. However, cells have a way of directing DNA repair to the DNA sequences that are most needed. They do this by linking RNA polymerase, the enzyme that transcribes DNA into RNA as the first step in gene expression, to the nucleotide excision repair pathway. As discussed above, this repair system can correct many different types of DNA damage. RNA polymerase stalls at DNA lesions and, through the use of coupling proteins, directs the excision repair machinery to those sites, thereby selectively repairing genes that are in current use by the cell. In bacteria, where genes are relatively short, the stalled RNA polymerase can be dissociated from the DNA; the DNA is repaired, and the gene is transcribed again from the beginning. In eukaryotes, where genes can be enormously long, a more complex reaction is used to “back up” the RNA polymerase, repair the damage, and then restart the polymerase.

The importance of transcription-coupled excision repair is seen in people with Cockayne syndrome, which is caused by a defect in this coupling. These individuals suffer from growth retardation, skeletal abnormalities, progressive neural retardation, and severe sensitivity to sunlight. Most of these problems are thought to arise from RNA polymerase molecules that become permanently stalled at sites of DNA damage that lie in important genes.

### The Chemistry of the DNA Bases Facilitates Damage Detection

The DNA double helix is well suited for repair. As noted earlier, it contains a backup copy of all genetic information. Equally importantly, the nature of the four bases in DNA makes the distinction between undamaged and damaged



**Figure 5-43 The deamination of DNA nucleotides.** In each case, the oxygen atom that is added in this reaction with water is colored red. (A) The spontaneous deamination products of A and G are recognizable as unnatural when they occur in DNA and thus are readily found and repaired, as is the deamination of C to U; T has no amino group to remove. (B) About 3% of the C nucleotides in vertebrate DNAs are methylated to help in controlling gene expression (discussed in Chapter 7). When these 5-methyl C nucleotides are accidentally deaminated, they form the natural nucleotide T. This T will be paired with a G on the opposite strand, forming a mismatched base pair.

bases very clear. For example, every possible deamination event in DNA yields an “unnatural” base, which can be directly recognized and removed by a specific DNA glycosylase. Hypoxanthine, for example, is the simplest purine base capable of pairing specifically with C. But hypoxanthine is not used in DNA, presumably because it is the direct deamination product of A. Instead G, with a second amino group, pairs with C: G cannot form from A by spontaneous deamination, and its own deamination product (xanthine) is likewise unique (Figure 5-43).

As discussed in Chapter 6, RNA is thought, on an evolutionary time scale, to have served as the genetic material before DNA, and it seems likely that the genetic code was initially carried in the four nucleotides A, C, G, and U. This raises the question of why the U in RNA was replaced in DNA by T (which is 5-methyl U). We have seen that the spontaneous deamination of C converts it to U, but that this event is rendered relatively harmless by uracil DNA glycosylase. However, if DNA contained U as a natural base, the repair system would not be able to distinguish a deaminated C from a naturally occurring U.

A special situation occurs in vertebrate DNA, in which selected C nucleotides are methylated at specific CG sequences that are associated with inactive genes (discussed in Chapter 7). The accidental deamination of these methylated C nucleotides produces the natural nucleotide T (see Figure 5-43B) in a mismatched base pair with a G on the opposite DNA strand. To help in repairing deaminated methylated C nucleotides, a special DNA glycosylase recognizes a mismatched base pair involving T in the sequence T-G and removes the T. This DNA repair mechanism must be relatively ineffective, however, because methylated C nucleotides are exceptionally common sites for mutations in vertebrate DNA. It is striking that, even though only about 3% of the C nucleotides in human DNA are methylated, mutations in these methylated nucleotides account for about one-third of the single-base mutations that have been observed in inherited human diseases.

### Special Translesion DNA Polymerases Are Used in Emergencies

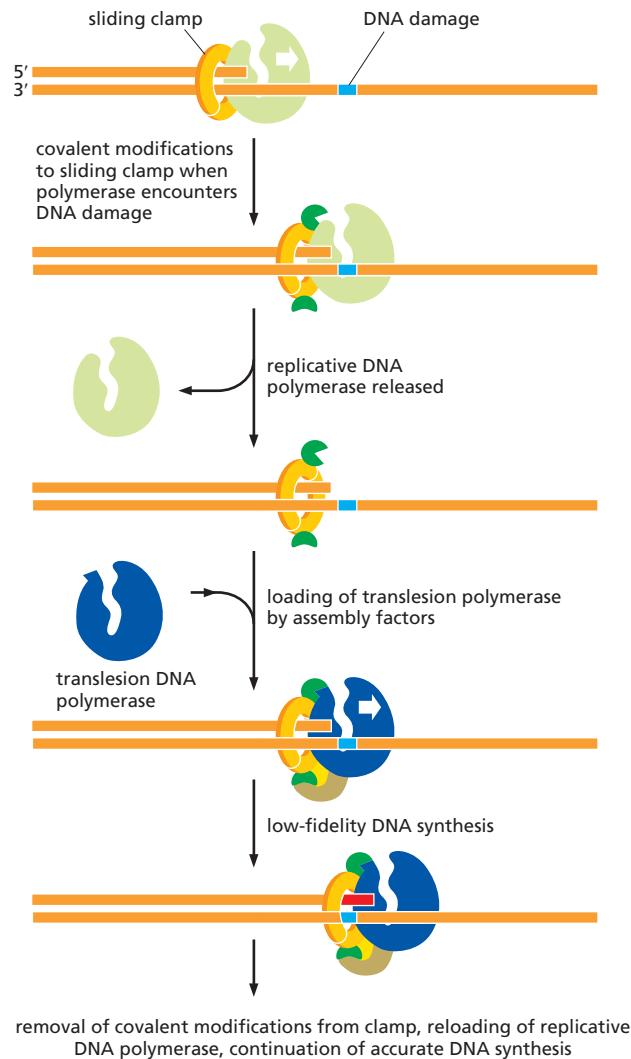
If a cell's DNA suffers heavy damage, the repair mechanisms that we have discussed are often insufficient to cope with it. In these cases, a different strategy is called into play, one that entails some risk to the cell. The highly accurate replicative DNA polymerases stall when they encounter damaged DNA, and in emergencies cells employ versatile, but less accurate, backup polymerases, known as *translesion polymerases*, to replicate through the DNA damage.

Human cells contain seven different translesion polymerases, some of which can recognize a specific type of DNA damage and add the nucleotides required to restore the correct sequence. For example, one such polymerase adds two A's opposite a thymine dimer (see Figure 5-39). Others make only "good guesses," especially when the template base has been extensively damaged. These enzymes are not as accurate as the normal replicative polymerases even when they copy an undamaged DNA sequence. For one thing, they lack exonucleolytic proofreading activity; in addition, many are much less discriminating than the replicative polymerase in choosing which nucleotide to incorporate initially. Each such translesion polymerase is therefore given a chance to add only one or a few nucleotides before a high-fidelity replicative polymerase resumes DNA synthesis.

Despite their usefulness in allowing heavily damaged DNA to be replicated, these translesion polymerases do, as noted above, pose risks to the cell. They are probably responsible for most of the base-substitution and single-nucleotide deletion mutations that accumulate in genomes. Not only do they frequently produce mutations when copying damaged DNA, they probably also generate mutations—at a low level—on undamaged DNA. Clearly, it is important for the cell to tightly regulate these polymerases, activating them only at sites of DNA damage. Exactly how this happens for each translesion polymerase remains to be discovered, but a conceptual model is presented in [Figure 5-44](#). The same principle applies to many of the DNA repair processes discussed in this chapter: because the enzymes that carry out these reactions are potentially dangerous to the genome, they must be brought into play only at the appropriate damaged sites.

### Double-Strand Breaks Are Efficiently Repaired

An especially dangerous type of DNA damage occurs when both strands of the double helix are broken, leaving no intact template strand to enable accurate repair. Ionizing radiation, replication errors, oxidizing agents, and other

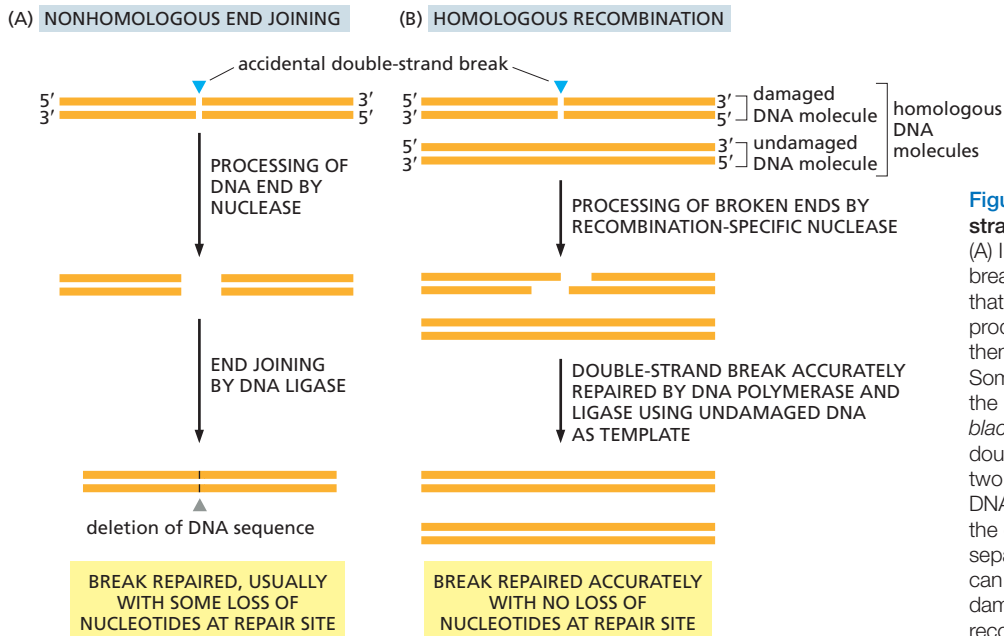


**Figure 5–44 How translesion DNA polymerases are recruited to damaged templates.** According to this model, a replicative polymerase stalled at a site of DNA damage is recognized by the cell as needing rescue. Specialized enzymes covalently modify the sliding clamp (typically, it is ubiquitylated—see Figure 3–65), which releases the replicative DNA polymerase and, together with the damaged DNA, attracts a translesion polymerase specific to that type of damage. Once the damaged DNA is bypassed, the covalent modification of the clamp is removed, the translesion polymerase dissociates, and the high-fidelity replicative polymerase is brought back into play.

metabolites produced in the cell cause breaks of this type. If these lesions were left unrepaired, they would quickly lead to the breakdown of chromosomes into smaller fragments and to loss of genes when the cell divides. However, two distinct mechanisms have evolved to deal with this type of damage by restoring an intact double helix: *nonhomologous end joining* and *homologous recombination* (Figure 5–45).

The simplest to understand is **nonhomologous end joining**, in which the broken ends are processed to remove any damaged nucleotides and simply brought together and rejoined by DNA ligation, generally with the loss of nucleotides at the site of joining (Figure 5–46). This end-joining mechanism, which can be seen as a “quick and dirty” solution to the repair of double-strand breaks, is the predominant way of repairing these lesions in mammalian somatic cells. Although a change in the DNA sequence (a mutation) usually results at the site of breakage, so little of the mammalian genome is essential for life that this mechanism is apparently an acceptable solution to the problem of rejoining broken chromosomes. By the time a human reaches the age of 70, the typical somatic cell contains more than 2000 such “scars,” distributed throughout its genome, representing places where DNA has been inaccurately repaired by nonhomologous end joining.

But nonhomologous end joining presents another danger: nonhomologous end joining can occasionally generate rearrangements in which one broken chromosome becomes covalently attached to another. This can result

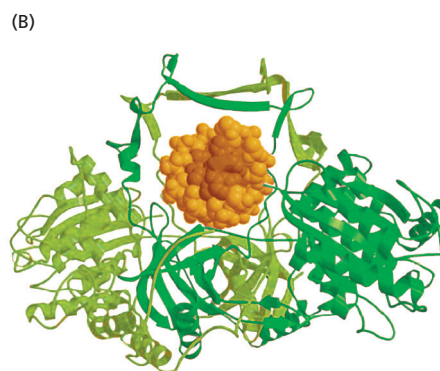
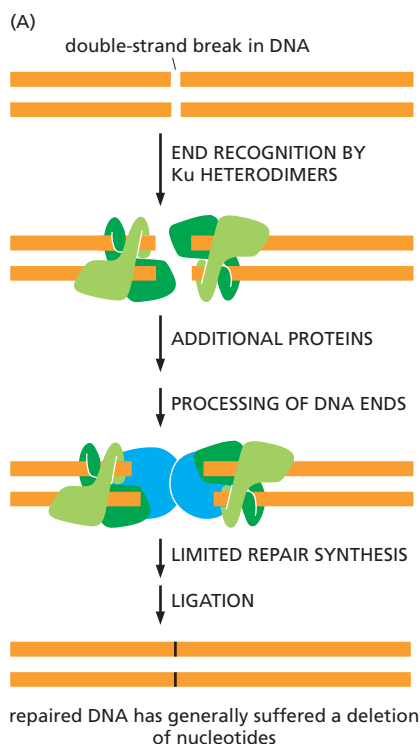


**Figure 5-45 Cells can repair double-strand breaks in one of two ways.**

(A) In nonhomologous end joining, the break is first “cleaned” by a nuclease that chews back the broken ends to produce flush ends. The flush ends are then stitched together by a DNA ligase. Some nucleotides are usually lost in the repair process, as indicated by the *black lines* in the repaired DNA. (B) If a double-strand break occurs in one of two duplicated DNA double helices after DNA replication has occurred, but before the chromosome copies have been separated, the undamaged double helix can be used as a template to repair the damaged double helix through homologous recombination. Although more complicated than nonhomologous end joining, this process accurately restores the original DNA sequence at the site of the break. Homologous recombination is described in detail in the next part of this chapter. Although nonhomologous end joining and homologous recombination are the two principal ways that cells repair double-strand breaks, additional mechanisms exist.

in chromosomes with two centromeres and chromosomes lacking centromeres altogether; both types of aberrant chromosomes are missegregated during cell division. As previously discussed, the specialized structure of telomeres prevents the natural ends of chromosomes from being mistaken for broken DNA and “repaired” in this way.

A much more accurate type of double-strand break repair is also possible (see Figure 5-45B). Here, a damaged DNA molecule is repaired using a second DNA double helix as a template, one with an identical (or nearly identical) DNA sequence. This reaction utilizes *homologous recombination*, a mechanism to be



**Figure 5-46 Nonhomologous end joining.** (A) A central role is played by the Ku protein, a heterodimer that quickly grasps the broken chromosome ends. The additional proteins (shown in blue) are recruited to hold the broken ends together and remove any damaged nucleotides before the two DNA molecules are joined covalently by a specialized ligase that is dedicated to nonhomologous end joining. During this process, any single-strand gaps that arise are “filled in” by specialized repair polymerases. When DNA suffers double-strand breaks through ionizing radiation or chemical attack, the broken ends are often chemically damaged. Nonhomologous end joining is unusually versatile in being able to “clean up” just about any type of damaged end. (B) Three-dimensional structure of a Ku heterodimer bound to the end of a duplex DNA fragment. This Ku protein is also essential for V(D)J joining, a specific process through which antibody and T-cell receptor diversity is generated in developing B and T cells (discussed in Chapter 24). V(D)J joining and nonhomologous end joining share many mechanistic similarities, but the former relies on specific double-strand breaks that are produced deliberately by the cell. (From J. Walker, R. Corpina, and J. Goldberg, *Nature* 412:607–614, 2001. With permission from Springer Nature; PDB codes: 1JEQ, 1JEY.)

described later in this chapter. Most organisms employ both nonhomologous end joining and homologous recombination to repair double-strand breaks in DNA. Nonhomologous end joining predominates in humans; homologous recombination is used only in the S and G<sub>2</sub> cell-cycle phases, when one newly replicated daughter molecule can act as a template to repair damage to the other daughter that remains nearby.

### DNA Damage Delays Progression of the Cell Cycle

We have just seen that cells contain multiple enzyme systems that can recognize and repair many types of DNA damage (**Movie 5.7**). Because of the importance of maintaining intact, undamaged DNA from generation to generation, eukaryotic cells delay the progression of their cell cycle until DNA repair is complete. As discussed in detail in Chapter 17, the orderly progression of the cell cycle is stopped when damaged DNA is detected, and it restarts only when the damage has been repaired. In mammalian cells, the presence of DNA damage can block entry from G<sub>1</sub> phase into S phase, it can slow S phase once it has begun, and it can block the transition from G<sub>2</sub> phase to M phase. These delays facilitate DNA repair by providing the time needed for the repair to reach completion.

DNA damage also results in an increased synthesis of many DNA repair enzymes. This response depends on special signaling proteins that sense DNA damage and synthesize more of the DNA repair enzymes appropriate for the damage. The importance of this mechanism is revealed by the phenotype of humans who are born with defects in the gene that encodes the *ATM* protein. These individuals have the disease *ataxia telangiectasia* (AT), the symptoms of which include neurodegeneration, a predisposition to cancer, and genome instability. The ATM protein is a large protein kinase that generates the intracellular signals needed to halt the cell cycle in response to many types of spontaneous DNA damage (see Figure 17-60), and individuals with defects in this protein suffer from the effects of unrepaired DNA lesions.

### Summary

*Genetic information can be stored stably in DNA sequences only because a large set of DNA repair enzymes continually scans the DNA double helix and replaces any damaged nucleotides. Most types of DNA repair depend on the fact that a DNA molecule carries two copies of its genetic information—one copy on each of its two complementary strands. This allows an accidental lesion on one strand to be removed by a repair enzyme and a corrected strand then resynthesized by reference to the information in the undamaged strand.*

*Most of the damage to DNA bases is excised by one of two major DNA repair pathways. In base excision repair, the altered base is removed by a DNA glycosylase enzyme, followed by excision of the resulting sugar phosphate. In nucleotide excision repair, a small section of the DNA strand surrounding the damage is removed from the DNA double helix. In both cases, the gap left in the DNA helix is filled in by the sequential action of DNA polymerase and DNA ligase, using the undamaged DNA strand as the template. Some types of DNA damage can be repaired by a different strategy—the direct chemical reversal of the damage—which is carried out by specialized repair proteins. Usually, all such corrections are completed prior to DNA replication. But if not, a special class of inaccurate DNA polymerases, called translesion polymerases, is used to bypass the damage, allowing the cell to survive but sometimes creating permanent mutations at the sites of damage.*

*Other critical repair systems—based on either nonhomologous end joining or homologous recombination—are needed to reseal the accidental double-strand breaks that occasionally occur in the DNA helix. In most cells, an elevated level of DNA damage causes a delay in the cell cycle, which helps to ensure that the damage is repaired before the cell divides.*

## HOMOLOGOUS RECOMBINATION

In the preceding parts of this chapter, we discussed the mechanisms that allow the DNA sequences in cells to be maintained from generation to generation with very little change. In this part, we further explore a group of repair mechanisms that depend on a process called *homologous recombination*. The key feature of **homologous recombination** (also known as *general recombination*) is an exchange of DNA strands between a pair of homologous duplex DNA sequences. Such a strand exchange between two regions of double helix that are very similar or identical in nucleotide sequence allows one stretch of duplex DNA to restore lost or damaged information on a second stretch of duplex DNA. Because the DNA sequence information that is used to correct the damage can come from a separate DNA molecule, homologous recombination can repair many types of DNA damage. It makes possible, for example, the accurate repair of double-strand breaks, as mentioned previously (see Figure 5–45B). As pointed out earlier, these double-strand breaks can result from reactive chemicals or radiation (for example, that from radon gas that accumulates in some old basements). But more frequently they arise from DNA replication accidents—when forks become stalled or broken independently of any such external cause. Homologous recombination accurately corrects these accidents, and, because they occur during nearly every round of DNA replication, this repair pathway is essential for every proliferating cell. Homologous recombination can also repair other types of DNA damage (for example, covalent cross-links between the two strands of a DNA double helix), being perhaps the most versatile DNA repair mechanism available to the cell; this probably explains why its mechanism and the proteins that carry it out have been conserved in virtually all cells on Earth.

We shall also see that homologous recombination plays an additional role in sexually reproducing organisms. During meiosis, a key step in gamete (sperm and egg) production, it catalyzes the orderly exchange of blocks of genetic information between corresponding (homologous) maternal and paternal chromosomes. This creates new combinations of DNA sequences in the chromosomes that are passed to offspring, giving the next generation unique characteristics upon which natural selection can act.

### Homologous Recombination Has Common Features in All Cells

The current view of homologous recombination as a critical DNA repair mechanism in all cells developed slowly from its original discovery as a key component in the specialized process of meiosis in plants and animals. The subsequent recognition that homologous recombination also occurs in unicellular organisms made it readily amenable to molecular analyses. Thus, much of what we know about the biochemistry of genetic recombination was derived from studies of bacteria, especially of *E. coli* and its viruses, as well as from experiments with simple eukaryotes such as yeasts. For these organisms with short generation times and relatively small genomes, it was possible to isolate a large set of mutants with defects in their recombination processes. The protein altered in each mutant was then identified and, ultimately, studied biochemically. Very close relatives of these proteins were subsequently found in more complex eukaryotes including flies, mice, and humans, and it is now possible to directly analyze homologous recombination in these species as well. As a result, we now know that the fundamental processes that catalyze homologous recombination are common to all cells.

### DNA Base-pairing Guides Homologous Recombination

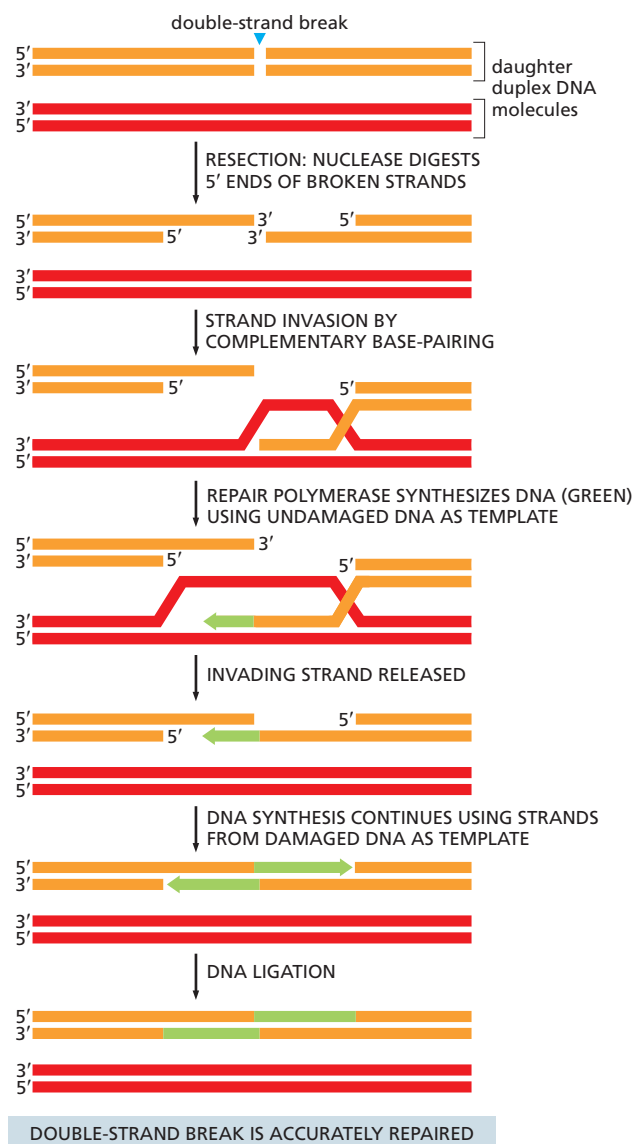
The hallmark of homologous recombination is that it takes place only between DNA duplexes that have extensive regions of sequence similarity (homology). Not surprisingly, base-pairing underlies this requirement: before undergoing homologous recombination, two DNA helices will “sample” each other’s DNA sequence by testing the potential base-pairing between a single strand from one

DNA duplex and a complementary single strand from the other. Recombination is initiated when a match is found; this match need not be perfect, but it must be very close for homologous recombination to succeed. As we shall see, the process is carefully controlled and guided by a group of specialized proteins.

### Homologous Recombination Can Flawlessly Repair Double-Strand Breaks in DNA

Unlike the nonhomologous end joining discussed earlier, homologous recombination repairs double-strand breaks accurately, without any loss or alteration of nucleotides at the site of repair. For homologous recombination to do this repair job, the damaged DNA must first be brought into proximity with a homologous but undamaged DNA double helix, which can then serve as a template for repair. For this reason, homologous recombination often occurs after DNA replication, when the two daughter DNA molecules lie close together and one can serve as a template for repair of the other.

One of the simplest pathways through which homologous recombination can repair double-strand breaks is shown in [Figure 5–47](#). In essence, the broken DNA duplex and the template duplex carry out a “strand dance” so that one of the damaged strands can use the complementary strand of the intact DNA duplex



**Figure 5–47** A mechanism that repairs double-strand breaks by homologous recombination. Homologous recombination can be regarded as a flexible series of reactions, with the exact pathway differing from one case to the next. The pathway shown here represents one of the major forms of recombinational double-strand break repair; however other, closely related pathways also exist. All share the first two steps—resection and strand invasion—but they diverge afterward. For example, recombinational repair of some double-strand breaks proceeds through a *double Holliday junction*, a structure we discuss later in this chapter.

as a template for repair. Once the damaged and template DNA double helices are in proximity (as occurs, for example, after DNA replication), the ends of the broken DNA are chewed back, or “resected,” by specialized nucleases to produce overhanging, single-strand 3′ ends. The next step is **strand exchange** (also called *strand invasion*), during which one of the single-strand 3′ ends from the damaged DNA molecule searches the template duplex for homologous sequences through base-pairing. Once stable base-pairing is established (which completes the *strand-exchange* step), an accurate DNA polymerase extends the invading strand using the information provided by the undamaged template molecule, thus restoring one of the damaged DNA strands. The last steps—strand displacement, further repair synthesis, and ligation—restore the two original DNA double helices and complete the repair process, as illustrated.

Homologous recombination resembles other DNA repair reactions in that a DNA polymerase utilizes a pristine template to restore damaged DNA. However, instead of using the partner strand as a template, as occurs in most DNA repair pathways, homologous recombination makes use of a complementary strand from a separate DNA duplex. In the following sections, we discuss the steps of homologous recombination in more detail with an emphasis on the proteins that guide this remarkable process.

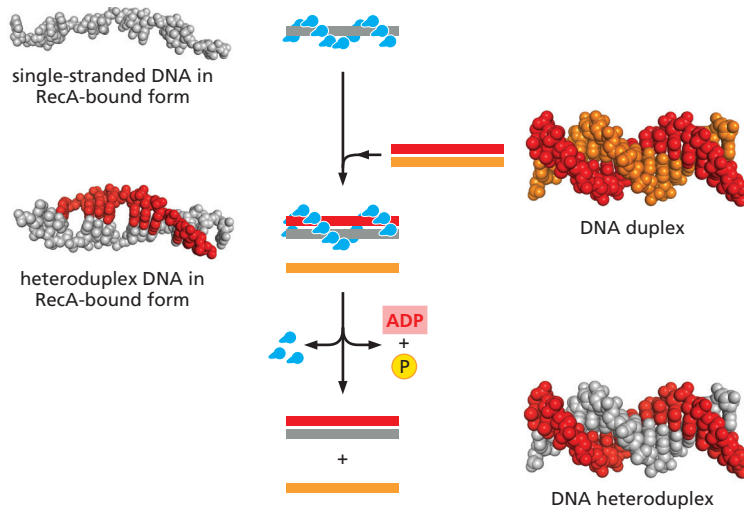
### Specialized Processing of Double-Strand Breaks Commits Repair to Homologous Recombination

Once a double-strand break occurs, nonhomologous end joining and homologous recombination compete to repair the damage. But the specialized nuclease that resects DNA ends to begin homologous recombination becomes highly active during S and G<sub>2</sub> (through its phosphorylation by cell-cycle-controlled kinases), and homologous recombination usually wins out at these times, allowing use of a newly replicated daughter DNA molecule as a template. The initiating nuclease (called the Mre11 complex in eukaryotes) chews back in the 5′→3′ direction leaving protruding 3′ ends on either side of the break that can be as long as several thousand nucleotides. Single-strand binding protein (the same one used at replication forks) then coats the exposed single strands, protecting them from other nucleases in the cell and ensuring that they remain free of intramolecular base-pairing. The formation of these protruding ends prevents nonhomologous end joining from occurring, and it commits the repair pathway to homologous recombination.

### Strand Exchange Is Directed by the RecA/Rad51 Protein

Of all the steps of homologous recombination, strand invasion is the most difficult to imagine. How does the invading single strand rapidly sample a DNA duplex for a complementary sequence? Once the homology is found, how is the structure stabilized? And how is the inherent stability of the template double helix overcome to allow tests for base-pairing during this process?

The answers to these questions come from biochemical and structural studies of the main protein that carries out this feat, called **RecA** in *E. coli* and **Rad51** in virtually all eukaryotic organisms. A special group of accessory proteins loads a set of RecA/Rad51 monomers onto a protruding DNA single strand (such as that in Figure 5-47), forming a cooperatively bound filament that displaces the single-strand binding protein originally present. This orderly loading process produces a protein–DNA filament in which the DNA is held by RecA/Rad51 in an unusual conformation: groups of three consecutive nucleotides are positioned as though they were in a conventional DNA double helix, but, between adjacent triplets, the DNA backbone is untwisted and stretched out (**Figure 5-48**). This unusual protein–DNA structure then grasps a nearby duplex DNA molecule in a way that stretches it, destabilizing it and making it easy to pull the strands apart. The invading single strand then can sample the sequence of the duplex by conventional base-pairing to one of its strands. This sampling occurs in triplet



**Figure 5–48 Strand invasion catalyzed by the RecA protein.** Our understanding of this reaction is based in part on structures determined by x-ray diffraction studies of the bacterial RecA protein bound to single-stranded and double-stranded DNA. These DNA structures (illustrated with the RecA protein removed) are shown on the left side of the diagram. The reaction begins when ATP-bound RecA protein (*blue*) associates with a DNA single strand (typically a protruding 3' end as shown in Figure 5–47), holding it in an elongated form where groups of three bases are separated from each other by a stretched and twisted backbone. The RecA-bound single strand then binds to duplex DNA, destabilizing it to allow the single strand to sample its sequence through base-pairing, three bases at a time. If an extensive match is found, the structure is disassembled through ATP hydrolysis, resulting in protein dissociation and the exchange of one single strand of DNA for another, thereby forming a new *heteroduplex* from the complementary strands of two different DNA molecules. In the vast majority of cases, no match will be found in any one binding event, in which case the RecA-bound DNA single strand rapidly dissociates to begin a new search. (PDB code: 3CMX.)

nucleotide blocks, each of which is already in a “base-pair ready” conformation in the invading strand; when a good triplet match occurs, only then is the adjacent triplet sampled, and so on. In this way, mismatches very quickly cause dissociation, so that millions of possible pairings can be tested. Only an extended stretch of base-pairing (at least 15 nucleotides) can stabilize the invading strand, leading to the next steps in homologous recombination.

RecA/Rad51 is an ATPase, and the steps described above require that each monomer along the filament be in the ATP-bound state. However, the searching itself does not require ATP hydrolysis; instead, the process occurs by simple molecular collisions, allowing an enormous number of potential sequences to be rapidly sampled. Once stable base-pairing occurs and a strand-exchange reaction is completed, ATP hydrolysis is necessary to disassemble RecA from the complex of DNA molecules. At this point, repair DNA polymerases and DNA ligase, which we encountered earlier in this chapter, complete the repair process, as shown previously in Figure 5–47.

### Homologous Recombination Can Rescue Broken and Stalled DNA Replication Forks

Although accurately repairing double-strand breaks is a crucial function of homologous recombination, it can also repair other types of damage. For example, some chemicals cross-link the two strands of DNA together by covalently joining nucleotides on opposite strands. A special set of enzymes unlinks the strands and cuts out the damaged bits on both strands. At this point, the damaged DNA has been converted to a double-strand break, which can be accurately repaired by homologous recombination, as discussed earlier. Similarly, proteins can become accidentally covalently linked to DNA, and these sites can also be converted by nucleases into double-strand breaks, allowing repair by homologous

**Figure 5–49 Repair of a broken replication fork by homologous recombination.** When a moving replication fork encounters a single-strand break, it will collapse but can be repaired by homologous recombination. The process uses many of the same reactions shown in Figure 5–47 and proceeds through the same basic steps. *Green* strands represent the new DNA synthesis that takes place after the replication fork has broken. This pathway allows the fork to move past the break on the damaged template using the undamaged duplex as a template to synthesize DNA. (Adapted from M.M. Cox, *Proc. Natl. Acad. Sci. USA* 98:8173–8180, 2001. Copyright 2001 National Academy of Sciences, USA. With permission from National Academy of Sciences.)

recombination. But perhaps the most important role of homologous recombination is in rescuing broken or stalled DNA replication forks. Many types of events can cause a replication fork to stop, and here we consider two examples. The first arises from an accidental single-strand gap in the parent DNA helix that lies just ahead of a replication fork. When the fork reaches this lesion, it falls apart—resulting in one broken and one intact daughter chromosome. Because this is a “one-sided” double-strand break, it cannot be repaired by nonhomologous end joining, and homologous recombination becomes crucial. The broken fork can be accurately repaired using the same basic reactions we discussed earlier for the repair of double-strand breaks (Figure 5–49). With slight modifications, the set of reactions just depicted can accurately repair many different types of DNA damage, providing that an undamaged duplex DNA template is available.

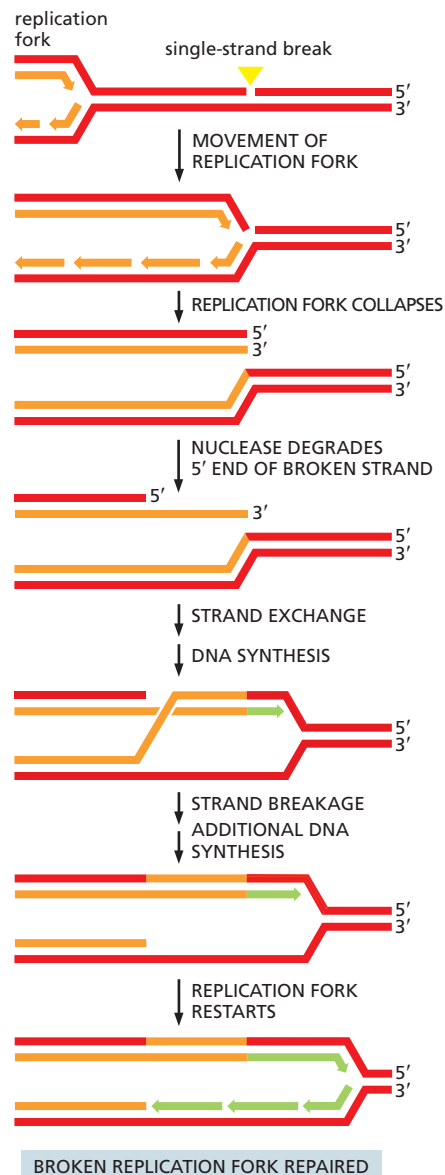
A different type of problem arises when a replication fork attempts to move through certain types of DNA damage that clogs up the replication machinery, stalling the fork. Because such damaged DNA often ends up deeply buried in the core of the replication fork, it cannot be easily repaired. To resolve this problem, the replication machine “backs up” through a series of strand-exchange reactions similar to those we have discussed (Figure 5–50). This maneuver allows one newly synthesized DNA strand to act as a template for synthesis of the other new strand, thereby bypassing the damaged template and allowing replication to proceed.

### DNA Repair by Homologous Recombination Entails Risks to the Cell

Although homologous recombination neatly solves the problem of accurately repairing double-strand breaks and other types of DNA damage, it sometimes “repairs” damage using the wrong bit of the genome as the template. For example, sometimes a broken human chromosome is repaired using the homolog from the other parent instead of the sister chromatid as the template. Because maternal and paternal chromosomes differ in DNA sequence at many positions along their lengths, this type of repair can convert the sequence of the repaired DNA from the maternal to the paternal sequence or vice versa. The result of this type of errant recombination is a **loss of heterozygosity**. It can have severe consequences if the homolog used for repair contains a deleterious mutation, because the recombination event destroys the “good” copy. Loss of heterozygosity, although it happens rarely, is nonetheless a critical step in the formation of many cancers (discussed in Chapter 20).

Cells go to great lengths to minimize the risk of mishaps of these types; indeed, as we have seen, nearly every step of homologous recombination is carefully regulated. Recall that the first step (resection of the broken ends) is coordinated with the cell cycle: it occurs primarily in the S and G<sub>2</sub> phases of the cell cycle, favoring the use of a daughter duplex (either as a partially replicated chromosome or a fully replicated sister chromatid) as a template for repair (see Figure 5–47). The close proximity of the two daughter chromosomes disfavors the use of other genome sequences in the repair process.

The loading of RecA/Rad51 onto the processed DNA ends and the subsequent strand-exchange reaction are also tightly controlled by the cell, and a host of accessory proteins is needed to regulate these steps. There are many such proteins, and exactly how all of them coordinate and control homologous recombination remains a mystery, although we do understand how a few of them work, as described below. We also know that the enzymes that catalyze

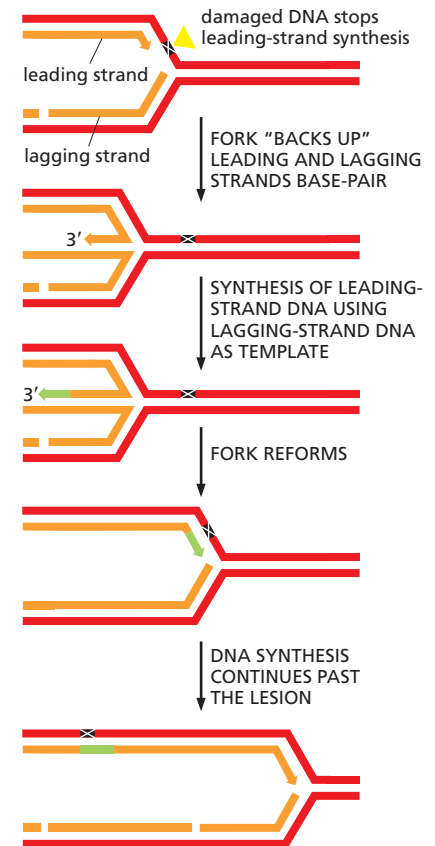


**Figure 5–50 Repair of a stalled replication fork by “fork reversal.”** This mechanism is brought into play when a replication fork stalls when it encounters certain types of damaged nucleotides. A specialized helicase (not shown) peels the newly synthesized DNA strands away from their parent templates, allowing them to form complementary base pairs with each other and backing up the replication fork. At this point two outcomes are possible. In the first, because the damaged DNA has been exposed, it can be repaired by conventional repair mechanisms, and the fork can be restarted. In the second, as shown, DNA synthesis can bypass the damage using newly synthesized daughter DNA (rather than the damaged parent strand) as the template. This scheme allows the replication fork to move through the DNA damage, which can be repaired at a later time. Although the initial steps of replication fork reversal are well understood, exactly how the fork restarts afterward remains a mystery.

recombinational repair are made at relatively high levels in eukaryotes and are dispersed throughout the nucleus in an inactive form. In response to DNA damage, they rapidly converge on the sites of DNA damage, become activated, and form “repair factories” where many lesions are apparently brought together and repaired (**Figure 5–51**). Formation of these factories probably results from many weak interactions between different repair proteins and between repair proteins and damaged DNA, producing the type of biomolecular condensates discussed in Chapter 3 (see Figure 3–77). The high local concentration of the appropriate proteins and their substrates within these condensates is thought to increase the speed and efficiency of the repair process.

In Chapter 20, we shall see that both too much and too little homologous recombination can lead to cancer in humans, the former through repair using the “wrong” template (as described above) and the latter through an increased mutation rate caused by inefficient DNA repair. Clearly, a delicate balance has evolved that keeps this process in check on undamaged DNA, while still allowing it to act efficiently and rapidly on DNA lesions as soon as they arise.

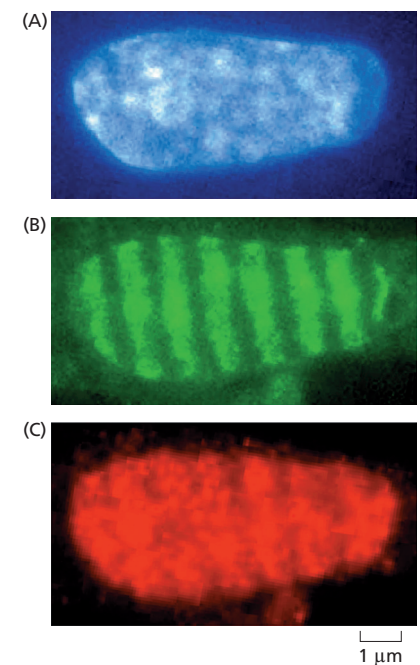
Not surprisingly, mutations in the components that carry out and regulate homologous recombination are responsible for several inherited forms of cancer. Two of these, the *Brca1* and *Brca2* proteins, were first discovered because mutations in their genes lead to a greatly increased frequency of breast cancer. Because these mutations cause inefficient repair by homologous recombination, accumulation of DNA damage can, in a small proportion of cells, give rise to a cancer. *Brca1* regulates an early step in broken-end processing; without it, such ends are not processed correctly for homologous recombination and instead damaged molecules are shunted to the error-prone nonhomologous end-joining pathway (see Figure 5–45). After resection, *Brca2* is needed to correctly load the *Rad51* protein onto the protruding single-strand DNA ends in preparation for strand exchange.

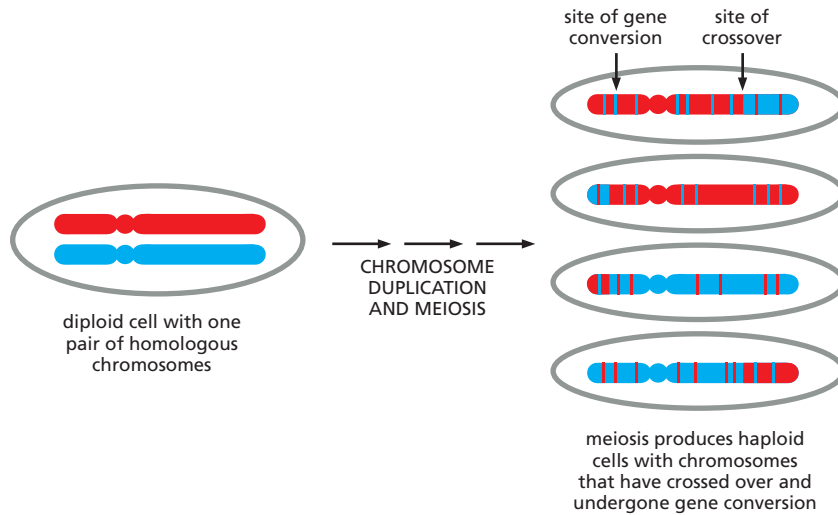


## Homologous Recombination Is Crucial for Meiosis

We have seen that homologous recombination can use a set of reactions—including broken-end resection, strand invasion, limited DNA synthesis, and ligation—to exchange DNA sequences between two double helices with the same nucleotide sequence and thereby repair damaged DNA. We now describe how homologous recombination is used to deliberately exchange material between two different chromosomes in order to generate DNA molecules that carry novel combinations of genes. This is a frequent and necessary part of meiosis, which occurs in sexually reproducing organisms such as fungi, plants, and animals.

**Figure 5–51 Experiment demonstrating the rapid localization of repair proteins to DNA double-strand breaks.** Human fibroblasts were x-irradiated to produce DNA double-strand breaks. Before the x-rays struck the cells, they were passed through a microscopic grid with x-ray-absorbing “bars” spaced 1  $\mu\text{m}$  apart. This produced a striped pattern of DNA damage, allowing a comparison of damaged and undamaged DNA in the same nucleus. (A) Total DNA in a fibroblast nucleus stained with the dye DAPI. (B) Sites of new DNA synthesis due to repair of DNA damage, indicated by incorporation of BrdU (a thymidine analog) and subsequent staining with fluorescently labeled antibodies to BrdU (green). (C) Localization of the Mre11 complex to damaged DNA as visualized by antibodies against the Mre11 subunit (red). Mre11 is the 5'→3' nuclease that produces the protruding single-strand DNA ends needed for strand invasion (see Figure 5–47). A, B, and C were processed 30 minutes after x-irradiation. (From B.E. Nelms et al., *Science* 280:590–592, 1998. With permission from AAAS.)





**Figure 5–52 Chromosome crossing-over occurs in meiosis.** Meiosis is the process by which a diploid cell gives rise to four haploid germ cells, as described in detail in Chapter 17. Meiosis produces germ cells in which the paternal and maternal genetic information (*red* and *blue*) has been reassorted through chromosome crossovers. In addition, many short regions of gene conversion occur, as indicated.

In meiosis, homologous recombination is an integral part of the process that allows chromosomes to be parceled out to germ cells (sperm and eggs in animals). We discuss the process of meiosis in detail in Chapter 17; here we discuss how homologous recombination during meiosis produces chromosome *crossing-over* and *gene conversion*, resulting in hybrid chromosomes that contain genetic information from both the maternal and paternal homologs (Figure 5–52). These mechanisms, at their core, closely resemble those used to repair double-strand breaks.

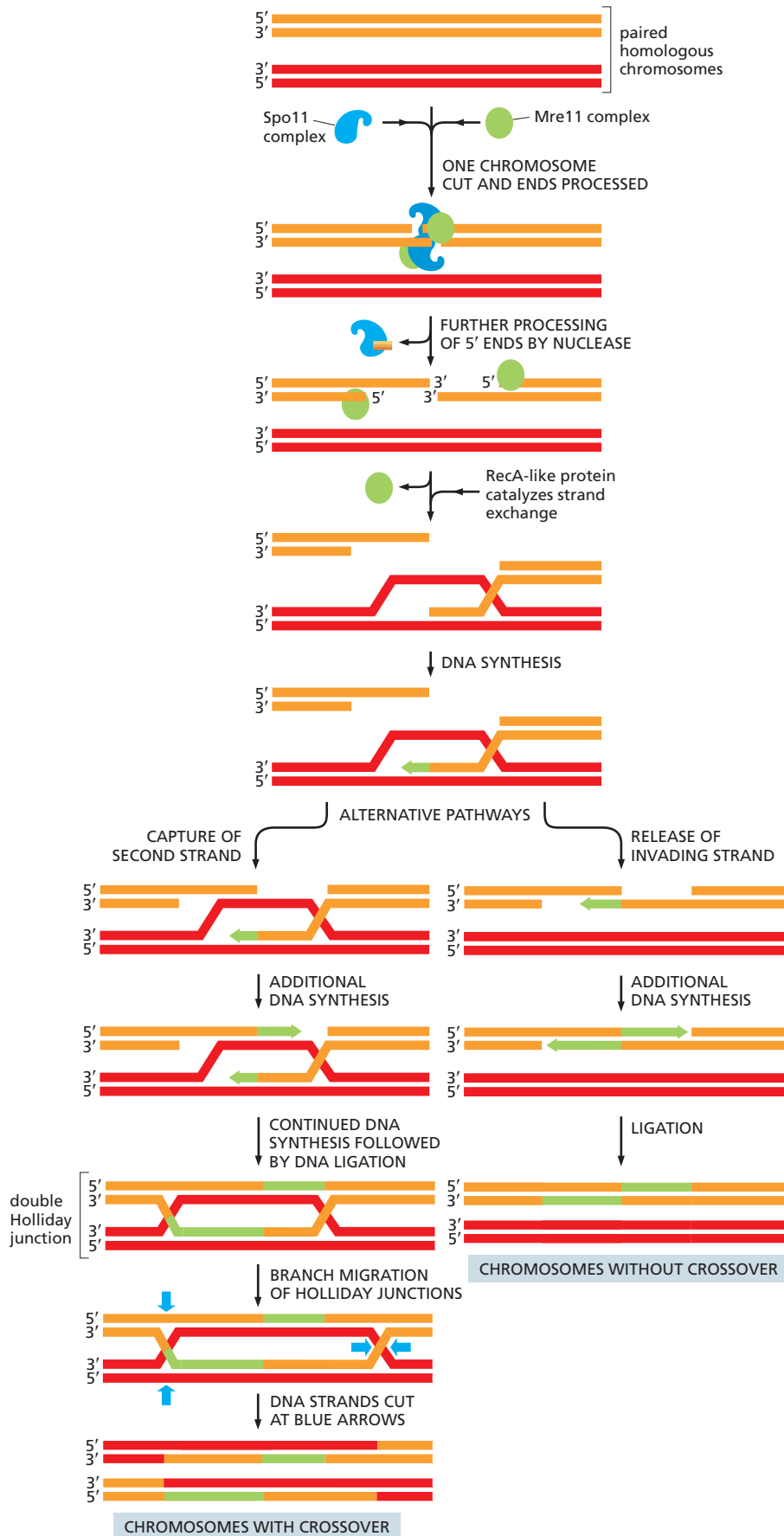
### Meiotic Recombination Begins with a Programmed Double-Strand Break

Homologous recombination in meiosis starts with a bold stroke: a specialized Spo11 protein complex breaks both strands of the DNA double helix in one of the recombining chromosomes (Figure 5–53). After catalyzing this reaction, the protein complex remains covalently bound to the broken DNA, much like the DNA topoisomerase we encountered earlier in this chapter (see Figure 5–22). Many of the subsequent recombination reactions closely resemble those already described for the repair of double-strand breaks; indeed, some of the same proteins are used for both processes. For example, the Mre11 complex, which we encountered earlier, chews back the DNA ends, removing the proteins along with the DNA and leaving the protruding 3' single-strand ends needed for strand invasion.

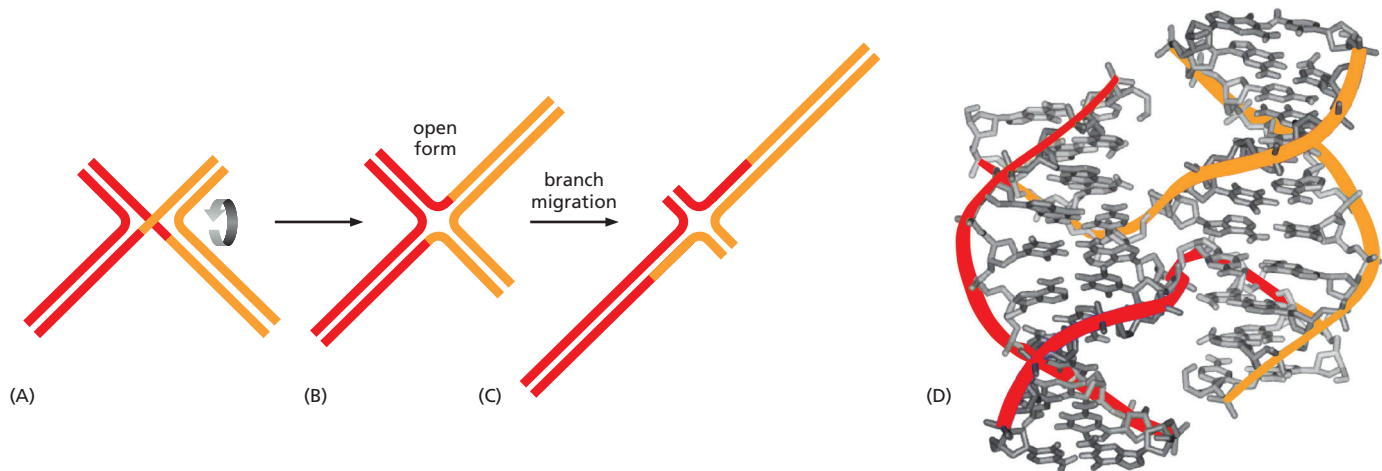
However, several meiosis-specific proteins come into play and guide the reactions somewhat differently, resulting in the distinctive outcomes observed for meiosis. A key difference is that, in meiosis, recombination occurs preferentially between maternal and paternal chromosomal homologs (which are held closely together during meiosis), rather than between newly replicated, identical DNA duplexes as in double-strand break repair. In the sections that follow, we describe in more detail those aspects of homologous recombination that are especially important for meiosis.

### Holliday Junctions Are Recognized by Enzymes That Drive Branch Migration

Of special importance in meiosis is an intermediate structure known as a **Holliday junction**, or *cross-strand exchange*, in which two homologous DNA helices that have paired are held together by the reciprocal exchange of two of the four strands



**Figure 5-53 Homologous recombination during meiosis can generate chromosome crossovers.** Once the Spo11 complex breaks the duplex DNA and the Mre11 complex processes the ends, homologous recombination in meiosis can proceed along alternative pathways (**Movie 5.8**). One (*right side of figure*) closely resembles the double-strand break repair reaction shown in Figure 5-47 and results in chromosomes that have been “repaired” without crossing over. The other (*left side of figure*) proceeds through a double Holliday junction and produces two chromosomes that have crossed over. During meiosis, the maternal and paternal chromosome homologs are held tightly together (see Figure 17-54), and both types of recombination occur between them.

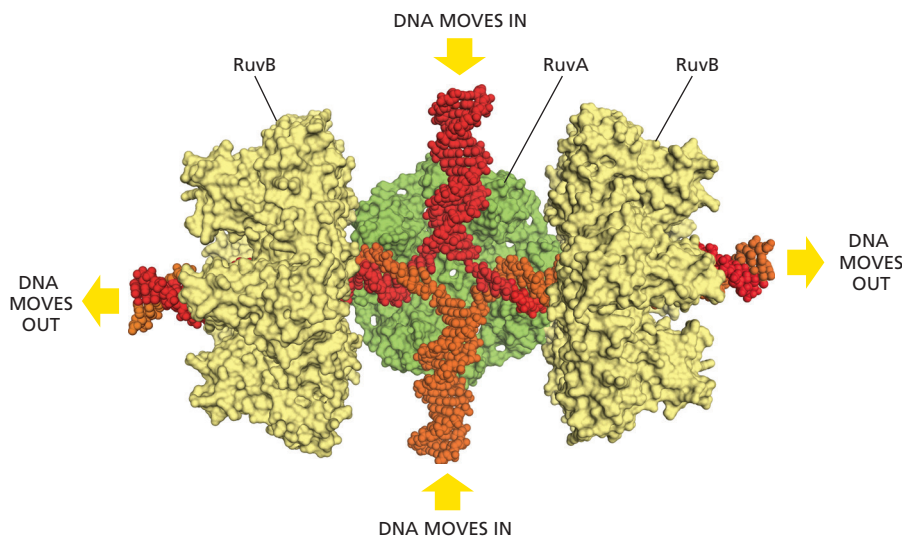


present, one strand originating from each of the helices. This junction can be considered to contain two pairs of strands: one pair of crossing strands and one pair of noncrossing strands (Figure 5-54A). But by undergoing a series of rotational movements, it can isomerize to form an open, symmetrical structure in which both pairs of strands occupy equivalent positions (Figure 5-54B and D). A special set of recombination proteins that binds to this open isomer uses the energy of ATP hydrolysis to catalyze a reaction known as *branch migration* (Figure 5-55), which greatly expands the region of heteroduplex DNA that was initially created by a strand-exchange reaction (Figure 5-54B and C). In meiosis, heteroduplex regions often “migrate” thousands of nucleotides from the original site of the double-strand break. The step where this migration occurs is indicated in Figure 5-53. As shown in the figure, Holliday junctions are often produced in pairs, known as double Holliday junctions.

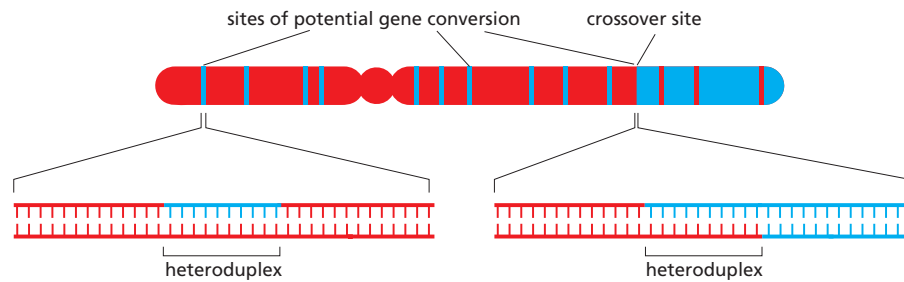
**Figure 5-54 A Holliday junction.** The initially formed structure (A) is usually drawn with two strands crossing, as in Figure 5-53. An isomerization of the Holliday junction (B) produces an open, symmetrical structure that is bound by specialized proteins. (C) These proteins “move” the Holliday junctions by a coordinated set of branch-migration reactions that involve the breaking and formation of base pairs (see Figure 5-55 and Movie 5.8). (D) Three-dimensional structure of the Holliday junction in the open form depicted in B. The Holliday junction is named for the scientist who first proposed its formation. (PDB code: 1DCW.)

### Homologous Recombination Produces Crossovers Between Maternal and Paternal Chromosomes During Meiosis

There are two basic outcomes of homologous recombination during meiosis, as shown previously in Figure 5-53 (Movie 5.8). In humans, approximately 90% of the double-strand breaks produced during meiosis are resolved as non-crossovers (right side of Figure 5-53). Here, the two original DNA duplexes separate from each other in a form unaltered except for a region of heteroduplex that formed near the site of the original double-strand break. As already noted, this set of reactions resembles that described earlier for the repair of double-strand breaks.



**Figure 5-55 Enzyme-catalyzed branch movement at a Holliday junction by branch migration.** A tetramer of the RuvA protein (green) and two hexamers of the RuvB protein (yellow) bind to the open form of the junction. The RuvB protein, which resembles the hexameric helicases used in DNA replication (see Figure 5-14), uses the energy of ATP hydrolysis to spool DNA rapidly through the Holliday junction, extending the heteroduplex region as shown. The RuvA protein coordinates this movement, threading the DNA strands to avoid tangling. This example is from *E. coli*, but similar proteins function in meiosis in sexually reproducing organisms. (PDB codes: 1IXR, 1C7Y.)



**Figure 5-56 Heteroduplexes formed during meiosis.** Heteroduplex DNA is present at sites of recombination that are resolved either as crossovers or non-crossovers. Because the DNA sequences of maternal and paternal chromosomes differ at many positions along their lengths, heteroduplexes often contain a small number of base-pair mismatches, and they are therefore potential sites of gene conversion (see Figure 5-57).

The other outcome is much more profound: a double Holliday junction is formed and is cleaved by specialized enzymes (*blue arrows* on the left side of Figure 5-53) to create a *crossover*. The two original portions of each chromosome upstream and downstream from the two Holliday junctions are thereby swapped, creating two chromosomes that are said to have “crossed over”—each containing a large number of both maternally inherited and paternally inherited genes.

How does the cell decide which double-strand breaks to resolve as crossovers? The answer is not yet known, but we know the decision is not random. The relatively few crossovers that do form are distributed along chromosomes in such a way that a crossover in one position inhibits crossing-over in neighboring regions. Termed *crossover control*, this fascinating but poorly understood regulatory mechanism ensures the roughly even distribution of crossover points along chromosomes. It also ensures that each chromosome—no matter how small—undergoes at least one crossover event every meiosis. For many organisms, roughly two crossovers per chromosome occur during each meiosis, one on each arm. As discussed in detail in Chapter 17, these crossovers, in addition to producing novel DNA molecules, play an important mechanical role in the proper segregation of chromosomes during meiosis.

Whether a meiotic recombination event is resolved as a crossover or a non-crossover, the recombination machinery leaves behind a *heteroduplex region* where a strand with the DNA sequence of the paternal homolog is base-paired with a strand from the maternal homolog (Figure 5-56). These heteroduplex regions can tolerate a small percentage of mismatched base pairs, and because of branch migration, they often extend for thousands of nucleotide pairs. The many non-crossover events that occur in meiosis thereby produce scattered sites in the germ cells where short DNA sequences from one homolog have been pasted into the other homolog. Heteroduplex regions mark sites of potential *gene conversion*—where the four haploid chromosomes produced by meiosis contain three copies of a DNA sequence from one homolog and only one copy of this sequence from the other homolog, as explained next.

### Homologous Recombination Often Results in Gene Conversion

In sexually reproducing organisms, it is a fundamental law of genetics that—aside from mitochondrial DNA, which is inherited only through the mother—each parent makes an equal genetic contribution to an offspring. One complete set of nuclear genes is inherited from the father and one complete set is inherited from the mother. Underlying this law is the accurate parceling out of chromosomes to the germ cells (eggs and sperm) that takes place during meiosis. Thus, when a diploid cell in a parent undergoes meiosis to produce four haploid germ cells, exactly half of the genes distributed among these four cells should be maternal (genes inherited from the mother of this parent) and the other half paternal (genes inherited from the father of this parent). In some organisms (fungi, for example), it is possible to recover and analyze all four of the haploid gametes produced from a single cell by meiosis. Studies in such organisms have revealed rare cases in which the parceling out of genes violates the standard genetic rules. Occasionally, for example, meiosis yields three copies of the maternal version of a gene and only one copy of the paternal version. Alternative versions of the same gene are called **alleles**, and it is the divergence from their expected distribution during

meiosis that is known as **gene conversion** (Movie 5.8). Genetic studies show that only small sections of DNA typically undergo gene conversion, and in many cases only a part of a gene is changed. How is this possible?

We have seen that both crossovers and non-crossovers produce heteroduplex regions of DNA. If the two strands that make up a heteroduplex region do not have identical nucleotide sequences, mismatched base pairs are formed, and these are often repaired by the cell's mismatch repair system (see Figure 5–20). However, unlike what happens after DNA replication, in meiosis the mismatch repair system randomly selects the strand to be used as a template, causing one allele to be lost and the other duplicated (Figure 5–57). Thus, gene conversion (the “conversion” of one allele to the other)—originally regarded as a mysterious deviation from the rules of genetics—can be seen as a straightforward consequence of the mechanisms of homologous recombination during meiosis.

## Summary

*Homologous recombination describes a flexible set of reactions resulting in the exchange of DNA sequences between a pair of identical or nearly identical duplex DNA molecules. Of special importance is a strand-exchange step whereby a single strand from one DNA duplex invades a second duplex and base-pairs with one strand while displacing the other. This reaction, catalyzed by the RecA/Rad51 family of proteins, can only occur if the invading strand can form a short stretch of consecutive nucleotide pairs with one of the strands of the duplex. This requirement ensures that homologous recombination occurs only between identical or very similar DNA sequences.*

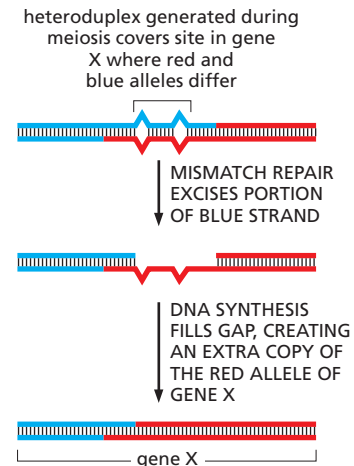
*When used as a DNA repair mechanism, homologous recombination usually occurs between a damaged DNA molecule and its recently duplicated sister molecule, with the undamaged duplex acting as a template to repair the damaged copy flawlessly. In meiosis, homologous recombination is initiated by deliberate, carefully regulated double-strand breaks and occurs preferentially between the homologous chromosomes rather than the newly replicated sister chromatids. The outcome can be either two chromosomes that have crossed over (that is, chromosomes in which the DNA on either side of the site of DNA pairing originates from two different homologs) or two non-crossover chromosomes. In the latter case, the two chromosomes that result are identical to the original two homologs, except for relatively minor DNA sequence changes at the site of recombination.*

## TRANSPOSITION AND CONSERVATIVE SITE-SPECIFIC RECOMBINATION

We have seen that homologous recombination can result in the exchange of DNA sequences between chromosomes. However, the order of genes on the interacting chromosomes typically remains the same after homologous recombination, inasmuch as the recombining sequences must be very similar for the process to occur. In this part of the chapter, we describe two very different types of recombination—**transposition** and **conservative site-specific recombination**—that do not require substantial regions of DNA homology. These two types of recombination reactions can alter the gene order along a chromosome and introduce whole blocks of DNA sequence into the genome.

Transposition and conservative site-specific recombination are largely dedicated to moving a wide variety of specialized segments of DNA—collectively termed *mobile genetic elements*—from one position in a genome to another. We will see that mobile genetic elements can range in size from a few hundred to tens of thousands of nucleotide pairs, and each typically carries a unique set of genes. Often, one of these genes encodes a specialized enzyme that catalyzes the movement of only that element and its close relatives, thereby making this type of recombination possible.

Virtually all cells contain mobile genetic elements, known informally as “jumping genes.” As explained in Chapter 4, over evolutionary time scales, they



**Figure 5–57 Gene conversion caused by mismatch correction.** As shown in the preceding figure, heteroduplex DNA is formed at the sites of homologous recombination between maternal and paternal chromosomes. If the maternal and paternal DNA sequences are slightly different, the heteroduplex region will include some mismatched base pairs, which may then be corrected by the DNA mismatch repair machinery (see Figure 5–20). Because neither strand of DNA is newly synthesized, such repair can “erase” nucleotide sequences on either the paternal or the maternal strand. The consequence of this mismatch repair is gene conversion, detected as a deviation from the segregation of equal copies of maternal and paternal alleles that normally occurs in meiosis.

have had a profound effect on the shaping of modern genomes. For example, nearly half of the DNA in the human genome can be traced to these elements (see Figure 4-63). Over time, random mutation has altered their nucleotide sequences, and, as a result, only a few of the many copies of these elements in our DNA are still active and capable of movement. The remainder are molecular fossils whose existence provides striking clues to our evolutionary history.

Mobile genetic elements are often considered to be molecular parasites (they are also termed “selfish DNA”) that persist because cells cannot get rid of them; they certainly have come close to overrunning our own genome. However, mobile DNA elements can provide benefits to the cell. For example, the genes they carry are sometimes advantageous to the host, as in the case of antibiotic resistance in bacterial cells, discussed later. The movement of mobile genetic elements also produces many of the genetic variants upon which evolution depends, because, in addition to moving themselves, mobile genetic elements occasionally rearrange neighboring sequences of the host genome. Thus, spontaneous mutations observed in bacteria, *Drosophila*, humans, and other organisms are often due to the random movement of mobile genetic elements. While many of these mutations will be deleterious to the organism, some will be advantageous and may spread throughout the population. It is almost certain that much of the variety of life we see around us originally arose from the movement of mobile genetic elements.

In this part of the chapter, we introduce mobile genetic elements and describe the mechanisms that enable them to move from place to place in a genome. As mentioned above, these elements move through a variety of different mechanisms that can be grouped into two broad categories, *transposition* and *conservative site-specific recombination*. We begin with transposition, by far the most predominant of these two processes.

### Through Transposition, Mobile Genetic Elements Can Insert into Any DNA Sequence

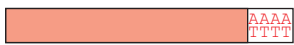
Mobile elements that move by way of transposition are called **transposons**, or **transposable elements**. In transposition, a specific enzyme, usually encoded by the transposon itself and typically called a *transposase*, acts on specific DNA sequences at each end of the transposon, causing it to insert into a new DNA site. Most transposons are only modestly selective in choosing their target site, and they can therefore insert themselves into many different locations in a genome; in particular, there is no general requirement for sequence similarity between the ends of the element and the target sequence. Most transposons move only rarely. In bacteria, the rate is typically one transposition event once every  $10^5$  cell divisions, and significantly more frequent movement would probably destroy the host cell's genome. In plants and animals, the situation is different: it is common for progeny to carry tens to hundreds of new insertions relative to their parents. These high rates are tolerated, in part, because these genomes typically carry vast amounts of nonessential DNA sequences where most of the insertions are likely to occur.

On the basis of their structure and transposition mechanism, transposons can be grouped into three large classes: *DNA-only transposons*, *retroviral-like retrotransposons*, and *nonretroviral retrotransposons*. The differences among them are briefly outlined in Table 5-4, and each class will be discussed in turn.

### DNA-only Transposons Can Move by a Cut-and-Paste Mechanism

**DNA-only transposons**, so named because they exist exclusively as DNA during their movement, predominate in bacteria, and they are largely responsible for the spread of antibiotic resistance in bacterial strains. When antibiotics such as penicillin and streptomycin first became widely available in the 1950s, most bacteria that caused human disease were susceptible to them. Now, the situation

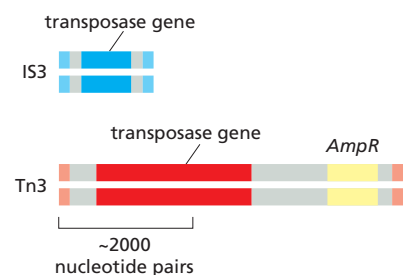
TABLE 5–4 Three Major Classes of Transposable Elements

| Class description and structure                                                                                                                                                                                                                                                                                     | Specialized enzymes required for movement | Mode of movement                                                                    | Examples                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| DNA-only transposons                                                                                                                                                                                                                                                                                                |                                           |                                                                                     |                                                                                                              |
| <br>Short inverted repeats at each end                                                                                                                                                                                             | Transposase                               | Moves as DNA, either by cut-and-paste or replicative pathways                       | P element ( <i>Drosophila</i> ), Ac-Ds (maize), Tn3 ( <i>E. coli</i> ), Tam3 (snapdragon), Helraiser (bat)   |
| Retroviral-like retrotransposons                                                                                                                                                                                                                                                                                    |                                           |                                                                                     |                                                                                                              |
| <br>Directly repeated long terminal repeats (LTRs) at each end                                                                                                                                                                     | Reverse transcriptase and integrase       | Moves via an RNA intermediate whose production is driven by a promoter in the LTR   | Copia and Gypsy ( <i>Drosophila</i> ), Ty1 (yeast), HERVK (human), Bs1 (maize), EVADE ( <i>Arabidopsis</i> ) |
| Nonretroviral retrotransposons                                                                                                                                                                                                                                                                                      |                                           |                                                                                     |                                                                                                              |
| <br>Poly A at 3' end of RNA transcript; 5' end is often truncated                                                                                                                                                                  | Reverse transcriptase and endonuclease    | Moves via an RNA intermediate that is often synthesized from a neighboring promoter | I element ( <i>Drosophila</i> ), L1 (human), Cin4 (maize), Karma (rice)                                      |
| These elements range in length from 1000 to about 12,000 nucleotide pairs. Each family contains many members, only a few of which are listed here. Some viruses can also move in and out of host-cell chromosomes by transpositional mechanisms. These viruses are related to the first two classes of transposons. |                                           |                                                                                     |                                                                                                              |

is different—antibiotics such as penicillin (and its modern derivatives) are no longer effective against many modern bacterial strains, including those causing gonorrhea and bacterial pneumonia. The spread of antibiotic resistance is due largely to genes that encode antibiotic-inactivating enzymes that are carried on transposons (Figure 5–58). Although these mobile elements can transpose only within cells that already carry them, they can be moved from one cell to another through other mechanisms known collectively as horizontal gene transfer (see Figure 1–18). Once introduced into a new cell, a transposon can insert itself into the genome and be faithfully passed on to all progeny cells through the normal processes of DNA replication and cell division.

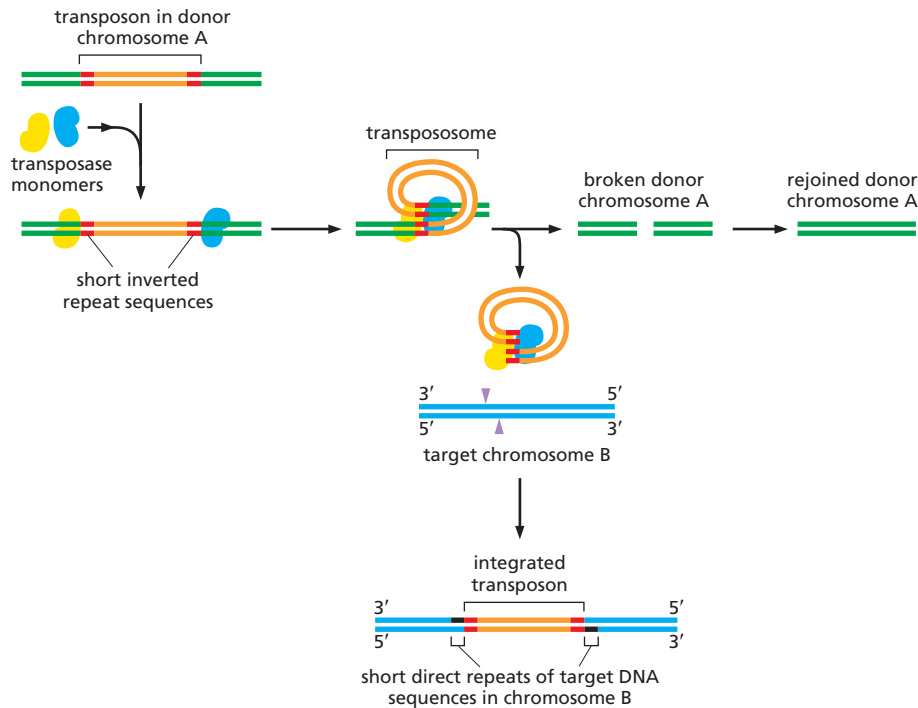
DNA-only transposons can relocate from a donor site to a target site by *cut-and-paste transposition* (Figure 5–59). Here, the transposon is literally excised from one spot on a genome and inserted into another. This reaction produces a short duplication of the target DNA sequence at the insertion site; these direct repeat sequences that flank the transposon serve as convenient records of a prior transposition event. Such “signatures” often provide valuable clues in identifying transposons in genome sequences.

When a cut-and-paste DNA-only transposon is excised from its original location, it leaves behind a “hole” in the chromosome. This lesion can be perfectly healed by recombinational double-strand break repair, provided that the chromosome has recently been replicated so that an identical copy of the damaged host sequence is available. Alternatively, a nonhomologous end-joining reaction can reseal the break; in this case, the DNA sequence that originally flanked the transposon is often altered, producing a mutation at the chromosomal site from which the transposon was excised (see Figure 5–45).



**Figure 5–58 Transposons often code for the components they need for transposition.** Shown here are two types of bacterial DNA-only transposons. Each carries a gene that encodes a transposase (dark blue and red)—the enzyme that catalyzes the element’s movement—as well as short DNA sequences (light blue and pink) that are recognized by the matching transposase. The short sequences (two in each transposon) are usually arranged so that one is an inverted repeat of the other.

Some transposons carry additional genes (yellow) that encode enzymes that inactivate antibiotics such as ampicillin (*AmpR*). The spread of these transposons is a serious problem in medicine, as it has allowed many disease-causing bacteria to become resistant to the antibiotics developed in the twentieth century.



**Figure 5-59 Cut-and-paste transposition.** DNA-only transposons can be recognized in chromosomes by the inverted repeat DNA sequences (red) present at their ends. These sequences, which can be as short as 20 nucleotides, are all that is necessary for the DNA between them to be transposed by the particular transposase enzyme associated with the element. The cut-and-paste movement of a DNA-only transposable element from one chromosomal site to another begins when the transposase enzyme brings the two inverted DNA sequences together, forming a DNA loop. Insertion into the target chromosome, also catalyzed by the transposase, occurs at a random site through the creation of staggered breaks in the target chromosome (purple arrowheads). After the transposition reaction, the single-strand gaps created by the staggered breaks are repaired by DNA polymerase and ligase (black). As a result, the insertion site is marked by a short direct repeat of the target DNA sequence, as shown. Although the break in the donor chromosome (green) is repaired, this process often alters the DNA sequence, causing a mutation at the original site of the excised transposable element (not shown).

Remarkably, the same mechanism used to excise cut-and-paste transposons from DNA has been found to operate in the developing immune system of vertebrates, catalyzing the DNA rearrangements that produce antibody and T-cell receptor diversity. Known as *V(D)J recombination*, this process will be discussed in Chapter 24. Found only in vertebrates, *V(D)J recombination* is a relatively recent evolutionary novelty, but its mechanism was probably derived from the much more ancient cut-and-paste transposons.

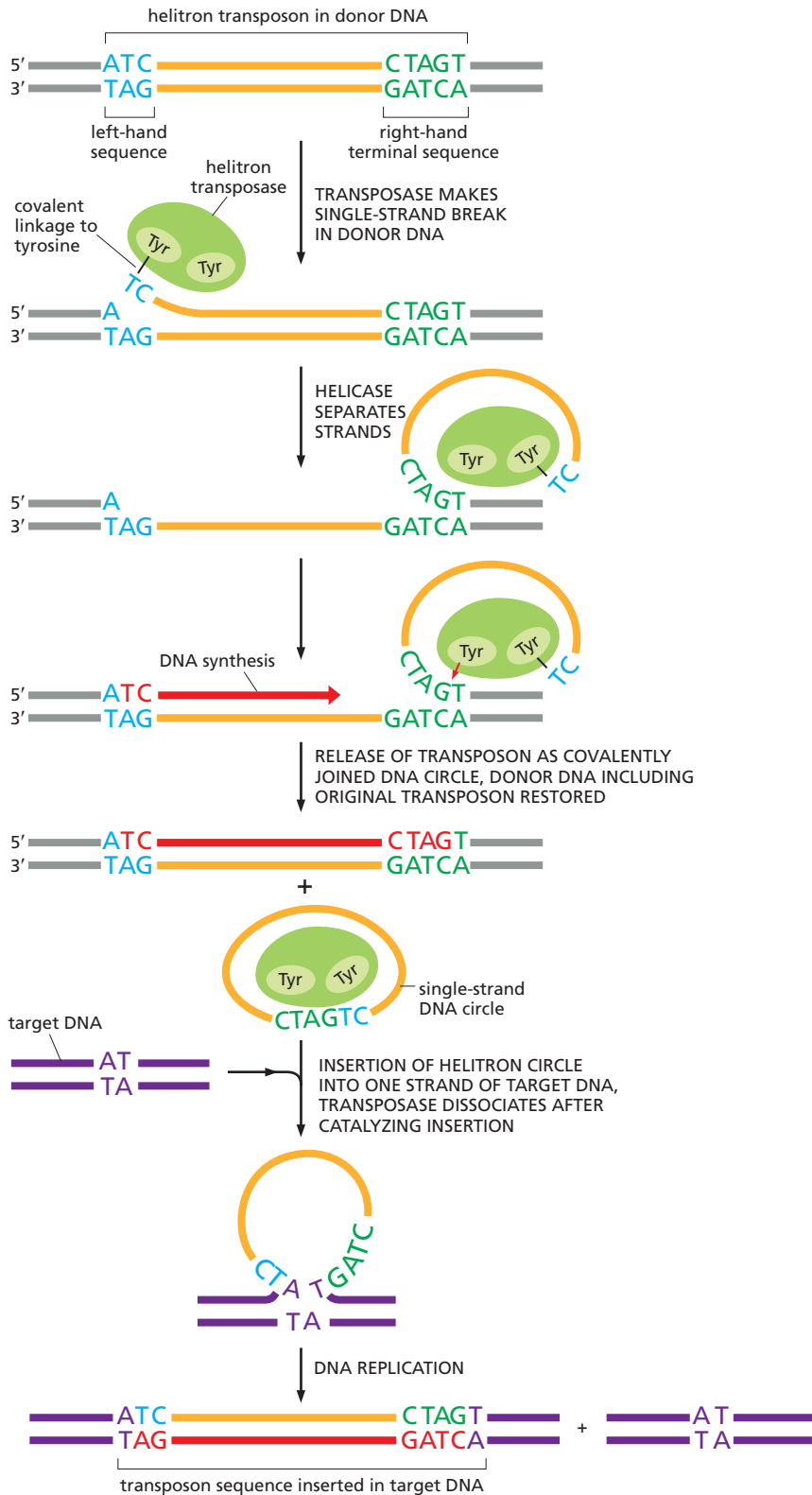
### Some DNA-only Transposons Move by Replicating Themselves

Although cut-and-paste transposition is common, especially in bacteria, there are other ways that DNA-only transposons can move. These involve replicating the transposon and moving the copy to a new position on the genome, leaving the original transposon intact and in its original position. There are several different ways this can occur and we discuss only one here, which is characteristic of a large class of DNA-only transposons known as *helitrons*. Found in all branches of life, these transposons are especially common in plants and animals where they can compose several percent of genomes. They carry a gene for an unusual type of transposase, one that functions as both a sequence-specific nuclease and as a helicase, thereby directing the movement of the transposon (Figure 5-60).

Because of the mechanism behind their movement, helitrons often transfer bits of the genome, along with themselves, to new positions. For this reason, they are thought to be especially important in reshuffling genomic information to produce variant organisms subject to natural selection over evolutionary time scales.

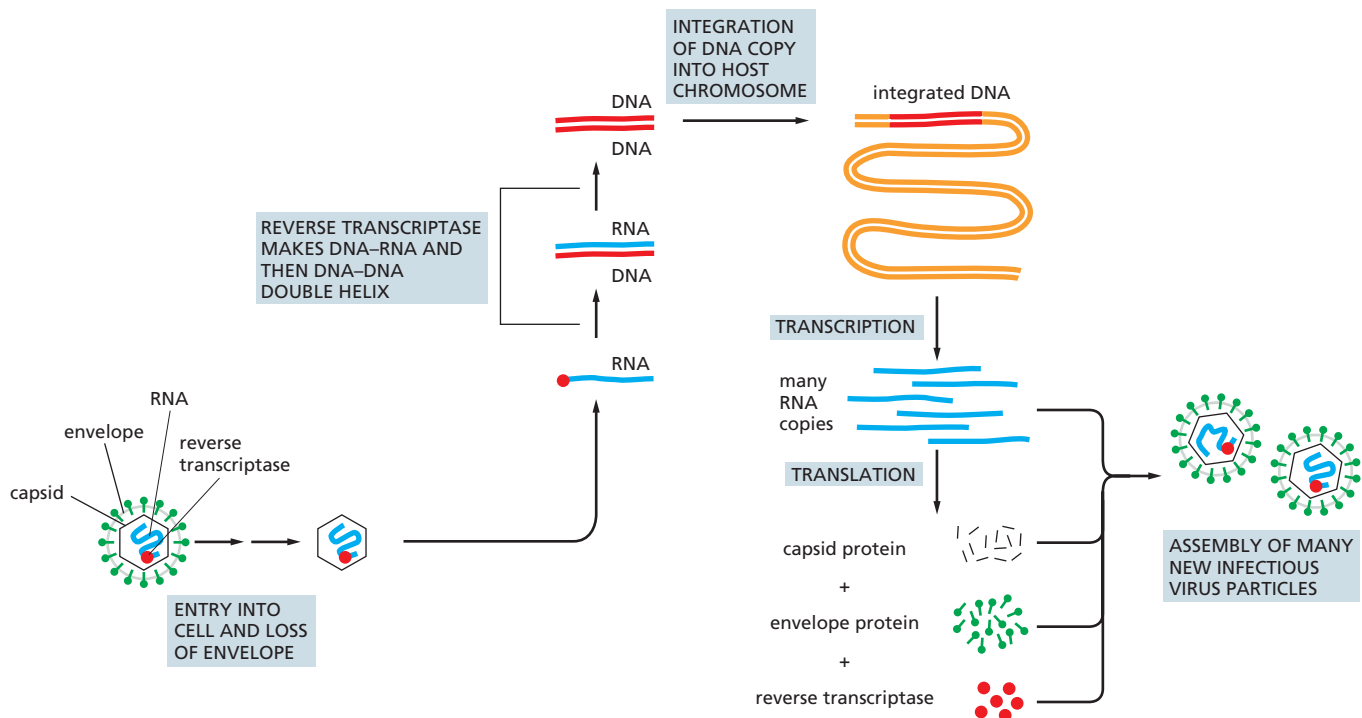
### Some Viruses Use a Transposition Mechanism to Move Themselves into Host-Cell Chromosomes

Certain viruses are considered mobile genetic elements because they use transposition mechanisms to integrate their genomes into that of their host cell. However, unlike transposons, the nucleotide sequences that form these viruses encode proteins that package their genetic information into virus particles that can leave the original host cell to infect other cells. As discussed in Chapter 1, most viruses probably evolved from transposable elements through the capture



**Figure 5–60 Mechanism of transposition by helitrons, a type of DNA-only transposon.** Several models have been proposed for the movement of these recently discovered transposons, and one is shown here. This model is based on studies of a helitron found in bats, called *Helraiser*. The process begins when the transposase (green) makes a single-strand break at one end of the transposon (blue) and, with the aid of a helicase, “peels back” the single strand. A second transposase-mediated reaction releases the transposon in the form of single-stranded DNA, which can move to new positions in the host genome. The transposase (which travels with the single-stranded DNA) can then catalyze the covalent insertion of the transposon into a new location in the host DNA. Transposition by helitrons often moves adjacent host genome sequences along with them. This occurs when, in the third step, the transposase skips over its own CTAGT sequence and cleaves its host DNA downstream at a similar DNA sequence. According to the model, this skipping produces a single-strand DNA circle that includes both helitron and host DNA, and both are inserted into target DNA.

of genes from their host cells. Although originally serving some other purpose in the cell, such captured genes, after a long process of mutation and selection, now code for the structural proteins of viruses, allowing them to escape the cell. Viruses are among the most numerous biological entities on Earth, and we discuss them in more detail in Chapter 23. The viruses that insert themselves into host chromosomes generally do so by employing one of the first two mechanisms



listed in Table 5–4; namely, by behaving like DNA-only transposons or like retroviral-like retrotransposons. Indeed, much of our knowledge of these mechanisms has come from studies of particular viruses that employ them.

Transposition has a key role in the life cycle of many viruses. Most notable are the **retroviruses**, which include the human AIDS virus, HIV. Outside the cell, a retrovirus exists as a single-strand RNA genome packed into a protein shell, or *capsid*, along with a virus-encoded **reverse transcriptase** enzyme. During the infection process, the viral RNA enters a cell and is converted to a double-strand DNA molecule by the action of this crucial enzyme, which is able to polymerize DNA on either an RNA or a DNA template (**Figure 5–61**). The term *retrovirus* refers to the virus's ability to reverse the usual flow of genetic information, which normally is from DNA to RNA (see Figure 1–4).

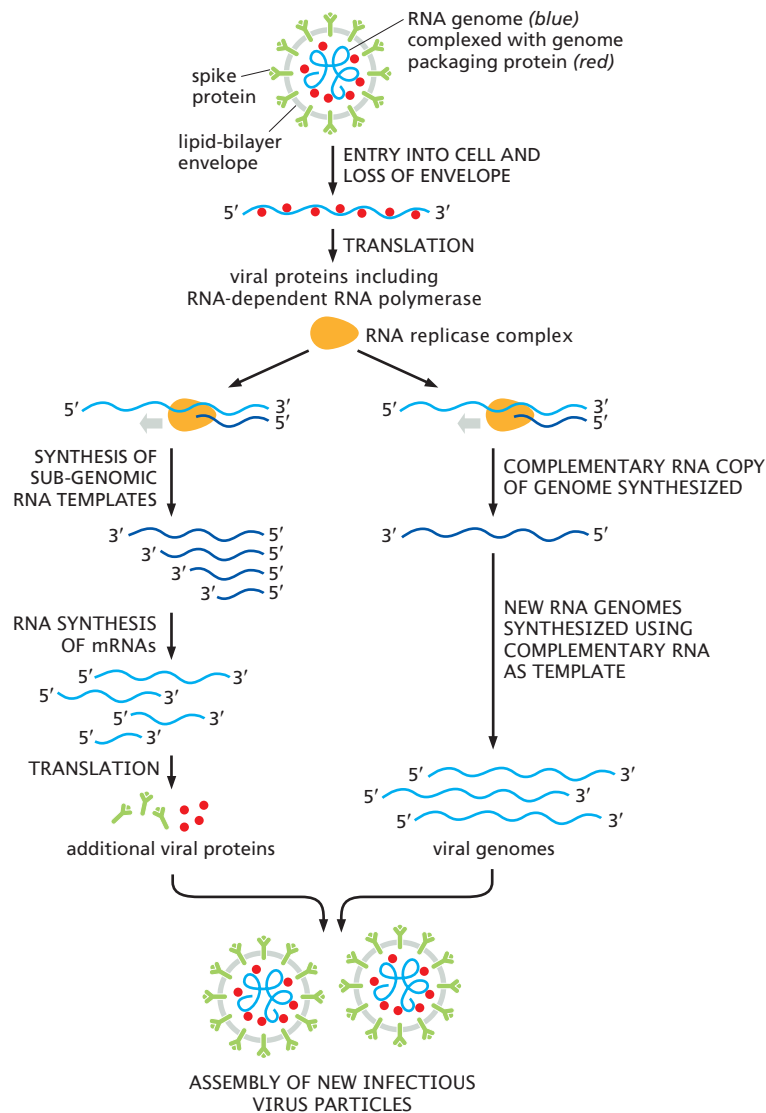
Once the reverse transcriptase has produced a double-strand DNA molecule, specific sequences near its two ends are recognized by a virus-encoded transposase called *integrase*. Integrase then inserts the viral DNA into the chromosome by a mechanism similar to that used by the cut-and-paste DNA-only transposons (see Figure 5–59).

### Some RNA Viruses Replicate and Express Their Genomes Without Using DNA as an Intermediate

Retroviruses are not the only viruses that carry their genomes in the form of RNA. Other viruses also have single-strand RNA genomes, but, unlike retroviruses, many replicate and express their genomes without ever using DNA; that is, they are *RNA-only viruses*. For example, SARS-CoV-2, the coronavirus underlying the COVID-19 pandemic, replicates its single-strand RNA genome using a special, viral-encoded *RNA-dependent RNA polymerase*. Upon entering a cell, the viral genome is directly translated by ribosomes as though it were an mRNA molecule, producing many different viral-encoded proteins, including the RNA-dependent RNA polymerase. (We discuss mRNA and the process of translation in detail in Chapter 6.) The polymerase assembles with several other viral proteins and a few host proteins to form the complete *replicase complex*. This specialized replicase, which does not require a primer to begin synthesis, starts at the 3' end of the viral genome and makes a complementary RNA copy of the entire genome

**Figure 5–61 The life cycle of a retrovirus.**

The retrovirus genome consists of an RNA molecule (*blue*) that is typically between 7000 and 12,000 nucleotides in length. It is packaged inside a virus-encoded protein capsid, which is surrounded by a lipid-based envelope that contains virus-encoded envelope proteins (*green*). Inside an infected cell, the enzyme reverse transcriptase (*red circle*) first makes a DNA copy of the viral RNA molecule and then a second DNA strand, generating a double-strand DNA copy of the RNA genome. The integration of this DNA double helix into the host chromosome is then catalyzed by a virus-encoded integrase enzyme. This integration is required for the synthesis of new viral RNA molecules by the host-cell RNA polymerase, the enzyme that transcribes DNA into RNA (discussed in Chapter 6). As indicated, this viral RNA is then used by host-cell machinery to produce the capsid, envelope, and reverse transcriptase proteins needed to form new virus particles.



**Figure 5-62** Simplified view of the coronavirus life cycle as exemplified by SARS-CoV-2. The viral genome, a single-strand RNA molecule of approximately 30,000 nucleotides, is packaged throughout its length with an RNA-binding protein (red) and enclosed by a lipid bilayer containing the viral spike protein. (The appearance of spike proteins, emanating from the lipid envelope, is responsible for the “corona” moniker.) As described in Chapter 23, the spike proteins bind to a receptor on the surface of susceptible cells and direct the fusion of the viral envelope with the outer cell membrane, releasing the viral genome into the cytoplasm. The genome is then directly translated into protein. Single-strand RNA viruses of this type are called *[+] strand viruses*, denoting the ability of their genomes to be immediately translated by the host-cell machinery. In contrast, the genomes of *[-] strand viruses* must first be used as templates to make complementary RNA strands that are then translated (see Table 23-1).

Among the first proteins made by coronaviruses are those that form the RNA-dependent RNA polymerase, which is responsible for producing new viral genomes by synthesizing RNA using RNA as a template. The replicase complex (which includes the polymerase and several other loosely associated proteins) first synthesizes complete noncoding copies of the viral genome. These complementary copies in turn serve as templates for the replicase complex to synthesize new genomes. The replicase also makes a series of shorter coding RNAs, which are needed to produce additional viral proteins including the spike. Once new viral genomes and proteins have been synthesized, new virus particles are assembled and exit the cell. Although only a few viral proteins are shown in the diagram, the virus codes for at least 27 different proteins; some of these organize the double-membrane structures in which the virus replicates, while others inhibit various immune system responses to the infection.

(Figure 5-62). Using this complementary copy as a template, the replicase then synthesizes new genomes, which are then packaged with newly synthesized viral proteins into complete virus particles. The whole process of viral replication takes about 10 hours, and a single infected cell can produce as many as 1000 new virus particles, which can spread to other cells within the same host or move in aerosols to new hosts. Because coronaviruses do not use DNA, all steps of viral replication can take place outside the nucleus. In the case of SARS-CoV-2, viral replication occurs in the cytoplasm inside double-membrane compartments that are commandeered by the virus from the endoplasmic reticulum, an organelle described in detail in Chapter 12. These virus-induced compartments are believed to protect the virus from the cell’s many antiviral defenses (see pp. 1337–1338) during viral replication and assembly.

Several features of coronaviruses distinguish them from other RNA-only viruses such as those that cause influenza or polio. Perhaps the most unusual is the ability of coronavirus replicase complexes to proofread as they copy their RNA genomes. This proofreading occurs in much the same way that we saw for DNA polymerases earlier in the chapter: An incorrectly added nucleotide is excised by a 3′-to-5′ exonuclease carried in the replicase complex, giving the replicase another chance to add the correct nucleotide. This feature means that coronaviruses do not mutate as rapidly as most other RNA viruses, which lack proofreading ability. As discussed in Chapter 23, the relatively high mutation rate of influenza

virus helps explain why new vaccines are needed every year, and this may not always be the case for coronaviruses.

The proofreading also has other important implications. As discussed earlier in this chapter, the lower the mutation rate, the greater the number of essential proteins that a genome can maintain. Proofreading allows coronaviruses to have a larger genome than is typical for RNA viruses; for example, compare the 30,000 nucleotides of the SARS-CoV-2 genome (coding for at least 27 proteins) to the 13,500 nucleotides of the influenza virus (coding for about 10 proteins). Proofreading also affects the development of antiviral drugs. Viral replicases are attractive targets for such drugs, in part because similar enzymes do not exist in uninfected host cells, reducing the chance of side effects. Drugs of this type (for example, remdesivir) are typically nucleoside triphosphate analogs that “fool” the RNA replicase into adding them to growing RNA chains. Once incorporated, the analogs—which have improper 3′ ends—poison further chain elongation (see Figure 8–42). Coronavirus proofreading can excise many of these analogs and thereby reduce their potency. A related strategy (exemplified by molnupiravir) employs nucleoside triphosphate analogs that are incorporated into RNA by the viral replicase, escape proofreading, but base pair incorrectly in the next round of replication, thereby introducing a lethal number of mutations.

Another striking feature of coronaviruses is the way in which they make many different proteins from a genome carried on a single RNA molecule. As described earlier, the viral genome, once it enters a cell, is treated like an mRNA molecule and translated into protein. We shall see in the next chapter, however, that most eukaryotic mRNAs can code for only a single protein. Coronavirus production of many different proteins from a single RNA genome requires a series of unusual steps, some of which appear unique to coronaviruses. We shall discuss the general topics of mRNA translation and its regulation in Chapters 6 and 7. But first, we return to our discussion of transposons, some of which closely resemble viruses in the way they move from place to place in their host genomes.

### Retroviral-like Retrotransposons Resemble Retroviruses, but Cannot Move from Cell to Cell

A large family of transposons called **retroviral-like retrotransposons** (see Table 5–4) move themselves in and out of chromosomes by a mechanism similar to that used by retroviruses. These elements are present in organisms as diverse as yeasts, flies, and mammals; unlike viruses, they have no intrinsic ability to leave their resident cell but are passed along to all descendants of that cell through the normal processes of DNA replication and cell division. The first step in their transposition is the transcription of the entire transposon, producing an RNA copy of the element that is typically several thousand nucleotides long. This transcript, which is translated as a messenger RNA by the host cell, encodes a reverse transcriptase enzyme. This enzyme makes a double-strand DNA copy of the RNA molecule via an RNA–DNA hybrid intermediate, precisely mirroring the early stages of infection by a retrovirus (see Figure 5–61). Like a retrovirus, the linear, double-strand DNA molecule then integrates into a site on the chromosome using an integrase enzyme that is also encoded by the element. The structure and mechanisms of these integrases closely resemble those of the transposases of DNA-only transposons.

### A Large Fraction of the Human Genome Is Composed of Nonretroviral Retrotransposons

A significant fraction of many vertebrate chromosomes is made up of repeated DNA sequences. In human chromosomes, these repeats are mostly mutated and truncated versions of **nonretroviral retrotransposons**, the third major type of transposon (see Table 5–4). Although most of these transposons in the human genome are immobile, a few retain the ability to move. Movements of the *L1 element* (sometimes referred to as a LINE, or long interspersed nuclear element)

have been identified, some of which result in human disease; for example, a particular type of hemophilia results from an *L1* insertion into the gene encoding the blood-clotting protein Factor VIII (see Figure 6–25).

Nonretroviral retrotransposons are found in many organisms and move via a distinct mechanism that requires a complex of an endonuclease and a reverse transcriptase. As illustrated in Figure 5–63, the RNA and reverse transcriptase have a much more direct role in the recombination event than they do in the retroviral-like retrotransposons described above.

Inspection of the human genome sequence reveals that the bulk of nonretroviral retrotransposons—for example, the many copies of the *Alu* element, a member of the SINE (short interspersed nuclear element) family—do not carry their own endonuclease or reverse transcriptase genes. Nonetheless, they have successfully amplified themselves to become major constituents of our genome, presumably by pirating enzymes encoded by active *L1* elements. Together the LINEs and SINEs make up more than 30% of the human genome (see Figure 4–62); there are 500,000 copies of the former and more than a million of the latter.

### Different Transposable Elements Predominate in Different Organisms

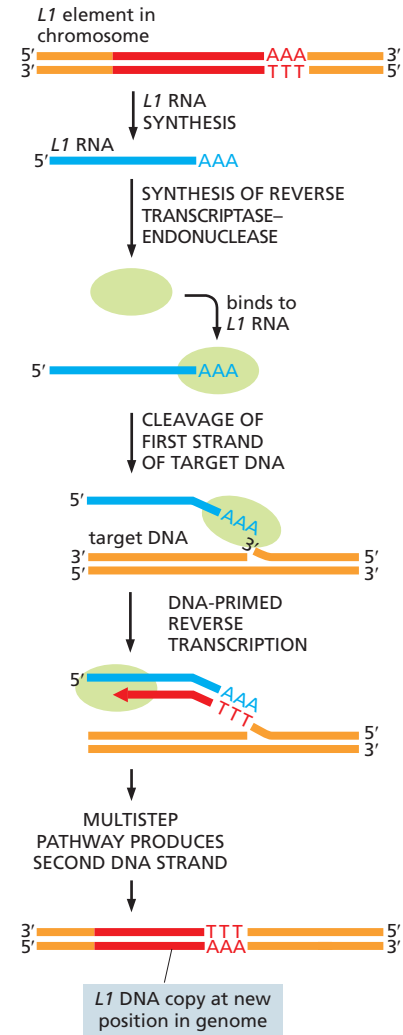
We have described several types of transposable elements: (1) DNA-only transposons, the movement of which is based on DNA breaking and joining reactions; (2) retroviral-like retrotransposons, which also move via DNA breakage and joining, but where RNA has a key role as a template to generate the DNA recombination substrate; and (3) nonretroviral retrotransposons, in which an RNA copy of the element is central to the incorporation of the element into the target DNA, acting as a direct template for a DNA target-primed reverse transcription event.

Intriguingly, different types of transposons predominate in different organisms. For example, the vast majority of bacterial transposons are DNA-only types, with a few related to the nonretroviral retrotransposons also present. In yeasts, the main mobile elements are retroviral-like retrotransposons. In *Drosophila*, DNA-only, retroviral, and nonretroviral transposons are all found. Finally, the human genome contains all three types of transposon, but as discussed below, their evolutionary histories are strikingly different.

### Genome Sequences Reveal the Approximate Times at Which Transposable Elements Have Moved

The nucleotide sequence of the human genome provides a rich fossil record of the activity of transposons over evolutionary time spans. By carefully comparing the nucleotide sequences of the approximately 3 million transposable element remnants in the human genome, it has been possible to broadly reconstruct the movements of transposons in our ancestors' genomes over the past several hundred million years. For example, the cut-and-paste DNA-only transposons appear to have been very active well before the divergence of humans and Old World monkeys (25–35 million years ago), but because they gradually accumulated inactivating mutations, they have been dormant in the human lineage since that time. Likewise, although our genome is littered with relics of retroviral-like retrotransposons, none appear to be active today. Only a single family of retroviral-like retrotransposons is believed to have transposed in the human genome since the divergence of human and chimpanzee approximately 6 million years ago. The nonretroviral retrotransposons are also ancient, but in contrast to other types, some are still moving in our genome, as mentioned previously. For example, it is estimated that *de novo* movement of an *Alu* element occurs once in every 100–200 human births. This movement of nonretroviral retrotransposons is responsible for a small but significant fraction of new human mutations—perhaps two mutations out of every thousand.

The situation in mice is significantly different. Although the mouse and human genomes contain roughly the same density of the three types of transposons, both



**Figure 5–63 Transposition by a nonretroviral retrotransposon.** Transposition of the *L1* element (red) begins when an endonuclease that is part of a complex with the *L1* reverse transcriptase (green) bound to the 3' end of *L1* RNA (blue) nicks the target DNA at the point at which insertion will occur. This cleavage produces a 3'-OH DNA end in the target DNA, which is then used as a primer for the reverse transcription step shown. This generates a single-strand DNA copy of the element that is directly linked to the target DNA. In subsequent reactions, further processing of the single-strand DNA copy results in the generation of a new double-strand DNA copy of the *L1* element that is inserted at the site of the initial nick.

types of retrotransposons are still actively transposing in the mouse genome, being responsible for approximately 10% of new mutations.

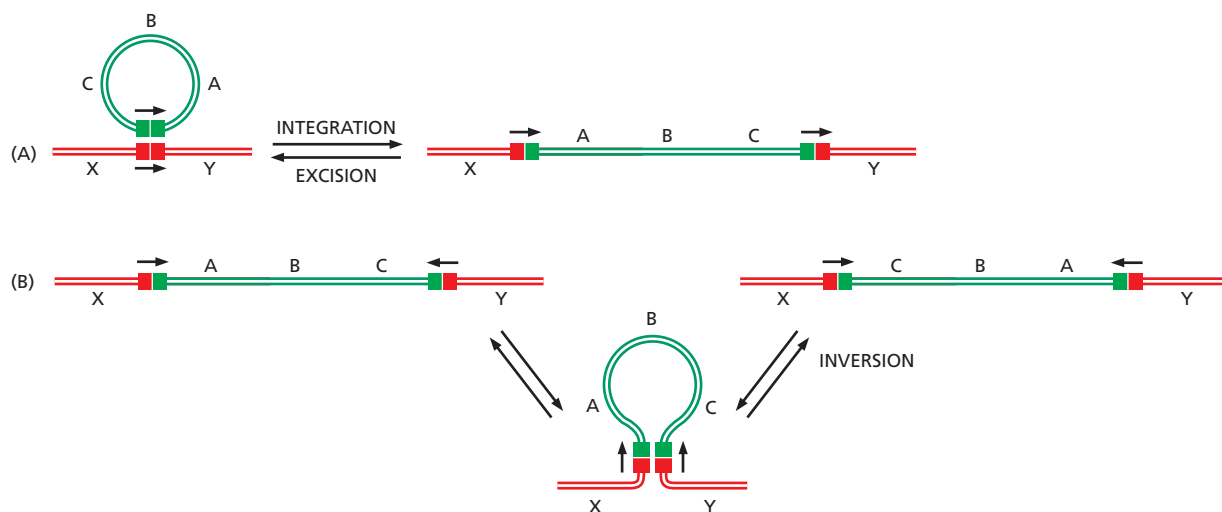
Although we are only beginning to understand how the movements of transposons have shaped the genomes of present-day mammals, it has been proposed that bursts in transposition activity could have been responsible for critical speciation events during the radiation of the mammalian lineages from a common ancestor, a process that began approximately 170 million years ago. At present, we can only wonder how many of our uniquely human qualities have been derived from the past activity of the mobile genetic elements whose remnants are found scattered throughout our chromosomes.

### Conservative Site-specific Recombination Can Reversibly Rearrange DNA

A different kind of recombination mechanism, known as *conservative site-specific recombination*, rearranges other types of mobile DNA elements. In this pathway, breakage and joining occur at two special sites, one on each participating DNA molecule, with the recombination event being carried out by a specialized enzyme that breaks and rejoins the two DNA double helices at these specific sequences. The same enzyme system that joins two DNA molecules can often take them apart again, precisely restoring the sequence of the two original DNA molecules (Figure 5-64A). Alternatively, with a different orientation of these two sequences in a chromosome, conservative site-specific recombination produces a DNA inversion (Figure 5-64B).

The conservative site-specific recombination pathway illustrated in Figure 5-64A is often used by bacterial DNA viruses to move their genomes in and out of the genomes of their host cells. When integrated into its host genome, the viral DNA is replicated along with the host DNA and is faithfully passed on to all descendant cells. If the host cell suffers damage (for example, by UV irradiation), the virus can reverse the site-specific recombination reaction, excise its genome, and package it into a virus particle. In this way, many viruses can replicate themselves passively as a component of the host genome, but can also “leave the sinking ship” by excising their genomes and packaging them in a protective coat until a new, healthy host cell is encountered.

Several features distinguish conservative site-specific recombination from transposition. First, conservative site-specific recombination requires specialized



**Figure 5-64** Two types of DNA rearrangement produced by conservative site-specific recombination. The only difference between the reactions in A and B is the relative orientation of the two short DNA sites (indicated by arrows) at which a site-specific recombination event occurs. (A) Through an integration reaction, a circular DNA molecule can become incorporated into a second DNA molecule; by the reverse reaction (excision), it can exit to re-form the original DNA circle. Many bacterial viruses move in and out of their host chromosomes in this way. (B) Conservative site-specific recombination can also invert a specific segment of DNA in a chromosome. A well-studied example of DNA inversion through site-specific recombination occurs in the bacterium *Salmonella enterica* serovar Typhimurium, as we discuss in the next section.

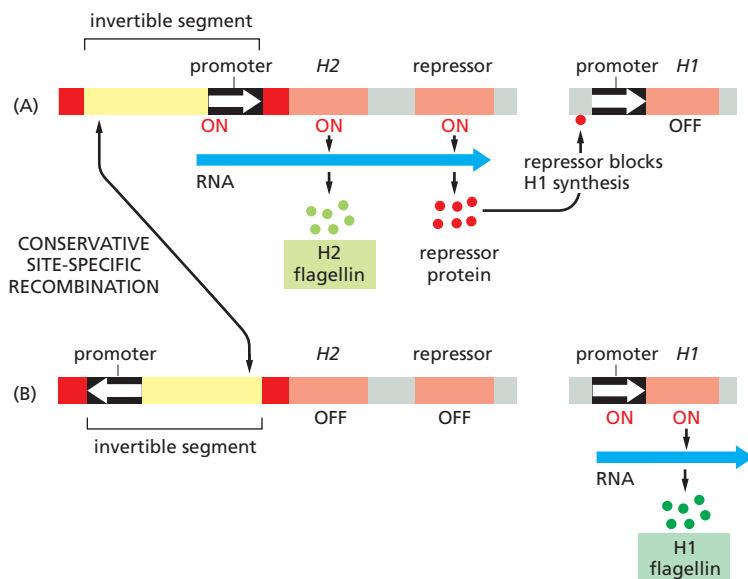
DNA sequences on both the donor and recipient DNA (hence the term “site-specific”). These sequences contain recognition sites for the particular recombinase that will catalyze the rearrangement. In contrast, transposition requires only that the transposon bears a specialized sequence; for most transposons, the recipient DNA can be of nearly any sequence. Second, the reaction mechanisms are fundamentally different. The recombinases that catalyze conservative site-specific recombination resemble topoisomerases in the sense that they form transient high-energy covalent bonds with the DNA and use this energy to complete all the DNA rearrangements without the need for new DNA synthesis (see Figure 5-22). Thus, all the phosphate bonds that are broken during a recombination event are restored upon its completion (hence the term “conservative”). Transposition, in contrast, typically leaves gaps in the DNA that must be repaired by DNA polymerases.

### Conservative Site-specific Recombination Can Be Used to Turn Genes On or Off

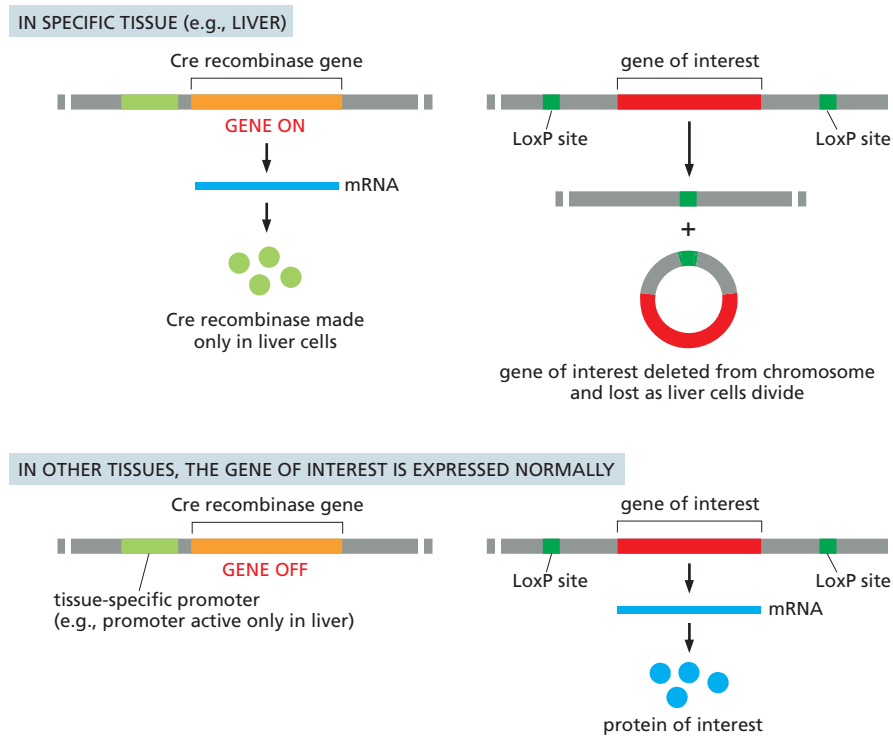
Many bacteria use conservative site-specific recombination to control the expression of particular genes. A well-studied example occurs in *Salmonella* bacteria, an organism that is a major cause of food poisoning in humans. Known as **phase variation**, the switch in gene expression results from the occasional inversion of a specific 1000-nucleotide-pair piece of DNA, brought about by a conservative site-specific recombinase encoded in the *Salmonella* genome. This change alters the expression of the cell-surface protein flagellin, for which the bacterium has two different genes. The DNA inversion changes the orientation of a promoter (a DNA sequence that directs transcription of a gene) that is located within the inverted DNA segment. With the promoter in one orientation, the bacteria synthesize one type of flagellin; with the promoter in the other orientation, they synthesize the other type (Figure 5-65).

The recombination reaction is reversible, allowing bacterial populations to switch back and forth between the two types of flagellin. Inversions occur only rarely, and because such changes in the genome will be copied faithfully during all subsequent replication cycles, entire clones of bacteria will have one type of flagellin or the other.

Phase variation helps protect the bacterial population against the immune response of its vertebrate host. If the host makes antibodies against one type of flagellin, a few bacteria whose flagellin has been altered by gene inversion will still be able to survive and multiply.



**Figure 5-65 Switching gene expression by DNA inversion in bacteria.** Which one of the two flagellin genes in a *Salmonella* bacterium is used to produce its flagellum is controlled by a conservative site-specific recombination event that inverts a small DNA segment containing a promoter. (A) In one orientation, the promoter activates transcription of the *H2* flagellin gene along with the transcription of a repressor protein that blocks the expression of the *H1* flagellin gene. Promoters and repressors are described in detail in Chapter 7; here we note simply that a promoter is needed to express a gene and that a repressor blocks this from happening. (B) When the promoter is inverted, it no longer turns on *H2* or the repressor, and the *H1* gene, which is thereby released from repression, is expressed instead. The inversion reaction requires specific DNA sequences (red) and a recombinase enzyme that is encoded in the invertible DNA segment. Because this conservative site-specific recombination mechanism is activated only rarely (about once in every  $10^5$  cell divisions), the production of one or the other type of flagellin tends to be faithfully inherited in each clone of cells.



**Figure 5–66** How a conservative site-specific recombination enzyme from bacteria is used to delete a specific gene from a particular mouse tissue.

This approach requires the insertion of two specially engineered DNA molecules into the animal's germ line. The first contains the gene for a recombinase (in this case, the Cre recombinase from the bacteriophage P1) under the control of a tissue-specific promoter that ensures the recombinase is expressed only in that tissue. The second DNA molecule contains the gene of interest, flanked by the DNA sequences of the recognition sites for the recombinase (in this case, LoxP sites). The mouse has been engineered to contain only this copy of the gene of interest. Therefore, if the recombinase is expressed only in the liver, the gene of interest will be deleted there, and only there. As described in Chapter 7, many tissue-specific promoters are known; moreover, many of these promoters are active only at specific times in development. Thus, this method makes it possible to study the effects of deleting any gene of interest at specific times during the development of each tissue. For this reason, it is a powerful tool for scientists investigating the role of individual genes in animal and plant development.

### Bacterial Conservative Site-specific Recombinases Have Become Powerful Tools for Cell and Developmental Biologists

Like many of the mechanisms used by cells and viruses, site-specific recombination has been put to work by scientists to aid in the study of a wide variety of problems. To decipher the roles of specific genes in complex multicellular organisms, genetic engineering techniques are used to produce worms, flies, and mice carrying both a gene encoding a site-specific recombination enzyme and a carefully designed target DNA that includes a gene of interest flanked by DNA sites recognized by the recombination enzyme. At an appropriate time, the gene encoding the enzyme can be activated to rearrange the target DNA sequence. Such a rearrangement is widely used to delete a specific gene in a particular tissue of a multicellular organism (Figure 5–66). This strategy is particularly useful when the gene of interest plays a key role in the early development of many tissues, and a complete deletion of the gene from the germ line would cause death very early in development. The same strategy can also be used to artificially express any specific gene in a tissue of interest; here, the triggered deletion joins a strong transcriptional promoter to the gene of interest. With this tool one can in principle determine the influence of any protein in any desired tissue of an intact animal.

### Summary

*The genomes of nearly all organisms contain mobile genetic elements that can move from one position in the genome to another by either transposition or conservative site-specific recombination. In most cases, this movement is random and happens at a very low frequency. There are three classes of transposons: the DNA-only transposons, the retroviral-like retrotransposons, and the nonretroviral retrotransposons. The first two classes have close relatives among the viruses, including the human retrovirus that causes AIDS, HIV. Although mobile genetic elements can be viewed as parasites, many of the new arrangements of DNA sequences that their recombination events produce have been important for creating the genetic variation required for the evolution of cells and organisms.*

## PROBLEMS

Which statements are true? Explain why or why not.

**5-1** The majority of cells in your body have exactly the same nucleotide sequence in their genomes.

**5-2** In a replication bubble, the same parent DNA strand serves as the template strand for leading-strand synthesis at one replication fork and as the template for lagging-strand synthesis at the other fork.

**5-3** In *E. coli*, where the replication fork travels at 500 nucleotide pairs per second, the DNA ahead of the fork—in the absence of topoisomerase—would have to rotate at nearly 3000 revolutions per minute.

**5-4** When bidirectional replication forks from adjacent origins meet, a leading strand always runs into a lagging strand.

**5-5** DNA repair mechanisms all depend on the cell having two homologous chromosomes.

Discuss the following problems.

**5-6** To determine the reproducibility of mutation frequency measurements, you do the following experiment. You inoculate each of 10 cultures with a single *E. coli* bacterium, allow the cultures to grow until each contains  $10^6$  cells, and then measure the number of cells in each culture that carry a mutation in your gene of interest. You were so surprised by the initial results that you repeated the experiment to confirm them. Both sets of results display the same extreme variability, as shown in Table Q5-1. Assuming that the rate of mutation is constant, why do you suppose there is so much variation in the frequencies of mutant cells in different cultures?

TABLE Q5-1 Frequencies of mutant cells in multiple cultures (Problem 5-6)

| Experiment | Culture (mutant cells/ $10^6$ cells) |   |     |   |   |    |    |   |   |    |
|------------|--------------------------------------|---|-----|---|---|----|----|---|---|----|
|            | 1                                    | 2 | 3   | 4 | 5 | 6  | 7  | 8 | 9 | 10 |
| 1          | 4                                    | 0 | 257 | 1 | 2 | 32 | 0  | 0 | 2 | 1  |
| 2          | 128                                  | 0 | 1   | 4 | 0 | 0  | 66 | 5 | 0 | 2  |

**5-7** Discuss the following statement: “Primase is a sloppy enzyme that makes many mistakes. Eventually, the RNA primers it makes are replaced with DNA made by a polymerase with higher fidelity. This is wasteful. It would be more energy-efficient if a DNA polymerase made an accurate copy in the first place.”

**5-8** If DNA polymerase requires a perfectly paired primer in order to add the next nucleotide, how is it that any mismatched nucleotides “escape” this requirement and become substrates for mismatch repair enzymes?

**5-9** DNA repair enzymes preferentially repair mismatched bases on the newly synthesized DNA strand,

using the old DNA strand as a template. If mismatches were instead repaired without regard for which strand served as template, would mismatch repair reduce replication errors? Would such a mismatch repair system result in fewer mutations, more mutations, or the same number of mutations as there would have been without any repair at all? Explain your answers.

**5-10** The laboratory you joined is studying the life cycle of an animal virus that uses circular, double-stranded DNA as its genome. Your project is to define the location of the origin(s) of replication and to determine whether replication proceeds in one or both directions away from an origin (unidirectional or bidirectional replication). To accomplish your goal, you broke open cells infected with the virus, isolated replicating viral genomes, cleaved them with a restriction nuclease that cuts the genome at only one site to produce a linear molecule from the circle, and examined the resulting molecules in the electron microscope. Some of the molecules you observed are illustrated schematically in Figure Q5-1. (Note that it is impossible to distinguish the orientation of one DNA molecule relative to another in the electron microscope because they land on the electron microscope grid in random orientations.)

You must present your conclusions to the rest of the lab tomorrow. How will you answer the two questions your advisor posed for you? Is there a single, unique origin of replication or several origins? Is replication unidirectional or bidirectional?

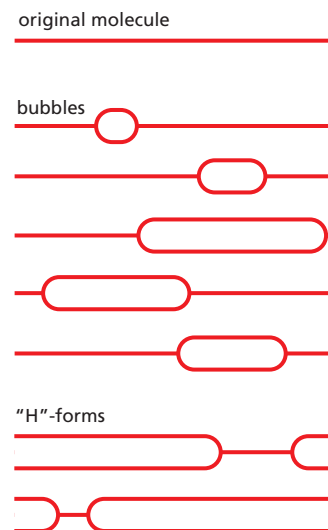
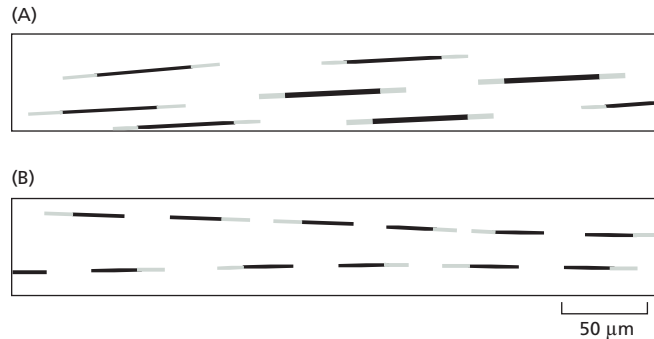


Figure Q5-1 Parent and replicating forms of an animal virus genome (Problem 5-10).

**5-11** You are investigating DNA synthesis in tissue-culture cells, using  $^3\text{H}$ -thymidine to radioactively label the replication forks. By breaking open the cells in a way that allows some of the DNA strands to be stretched out, very long DNA strands can be isolated intact and examined. You overlay the DNA with a photographic emulsion, and expose it for 3–6 months, a procedure known as autoradiography. Because the emulsion is sensitive to radioactive emissions, the  $^3\text{H}$ -labeled DNA shows up as tracks of silver grains. Because the stretching collapses replication bubbles, the daughter duplexes lie side by side and cannot be distinguished from each other.



**Figure Q5-2** Autoradiographic investigation of DNA replication in cultured cells (Problem 5-11). (A) Addition of  $^3\text{H}$ -labeled thymidine immediately after release from the arrest. (B) Addition of  $^3\text{H}$ -labeled thymidine 30 minutes after release from the arrest.

You treat the cells to arrest them at the beginning of S phase. In the first experiment, you release the arrest and add  $^3\text{H}$ -thymidine immediately. After 30 minutes, you wash the cells and change the medium so that the total concentration of thymidine is the same as it was, but only one-third of it is radioactive. After an additional 15 minutes, you prepare DNA for autoradiography. The results of this experiment are shown in **Figure Q5-2A**. In the second experiment, you release the arrest and wait 30 minutes before adding  $^3\text{H}$ -thymidine. After 30 minutes in the presence of  $^3\text{H}$ -thymidine, you once again change the medium to reduce the concentration of radioactive thymidine and incubate the cells for an additional 15 minutes. The results of the second experiment are shown in **Figure Q5-2B**.

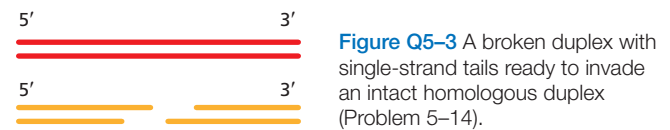
- Explain why, in both experiments, some regions of the tracks are dense with silver grains (dark), whereas others are less dense (light).
- In the first experiment, each track has a central dark section with light sections at each end. In the second experiment, the dark section of each track has a light section at only one end. Explain the reason for this difference.
- Estimate the rate of fork movement ( $\mu\text{m}/\text{min}$ ) in these experiments. Do the estimates from the two experiments agree? Can you use this information to gauge how long it would take to replicate the entire genome?

**5-12** If you compare the frequency of the 16 possible dinucleotide sequences in the *E. coli* and human genomes, there are no striking differences except for one dinucleotide, 5'-CG-3'. The frequency of CG dinucleotides in the human genome is significantly lower than in *E. coli* and significantly lower than expected by chance. Why do you suppose that CG dinucleotides are underrepresented in the human genome?

**5-13** With age, somatic cells are thought to accumulate genomic “scars” as a result of the inaccurate repair of double-strand breaks by nonhomologous end joining (NHEJ). Estimates based on the frequency of breaks in primary human fibroblasts suggest that by age 70, each human somatic cell may carry some 2000 NHEJ-induced mutations due to inaccurate repair. If these mutations were distributed randomly around the genome, how many protein-coding genes would you expect to be affected? Would

you expect cell function to be compromised? Why or why not? (Assume that 2% of the genome—1.5% protein-coding and 0.5% regulatory—is crucial information.)

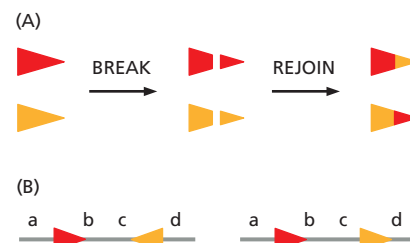
**5-14** Draw a schematic diagram of the double Holliday junction that would result from strand invasion by both ends of the broken duplex into the intact homologous duplex shown in **Figure Q5-3**. Label the left end of each strand in the Holliday junction 5' or 3' so that the relationship to the parent and recombinant duplexes is clear. Indicate how DNA synthesis would be used to fill in any single-strand gaps in your double Holliday junction.



**Figure Q5-3** A broken duplex with single-strand tails ready to invade an intact homologous duplex (Problem 5-14).

**5-15** In addition to correcting DNA mismatches, the mismatch repair system acts to discourage homologous recombination between DNA duplexes that are only moderately similar in sequence. Why would recombination between moderately similar sequences pose a problem for human cells?

**5-16** Cre recombinase is a site-specific enzyme that catalyzes recombination between two LoxP DNA sites. Cre recombinase pairs two LoxP sites in the same orientation, breaks both duplexes at the same point in each LoxP site, and joins the ends with new partners so that each LoxP site is regenerated, as shown schematically in **Figure Q5-4A**. On the basis of this mechanism, predict the arrangement of sequences that will be generated by Cre-mediated site-specific recombination for each of the two DNAs shown in **Figure Q5-4B**.



**Figure Q5-4** Cre recombinase-mediated site-specific recombination (Problem 5-16). (A) Schematic representation of Cre/LoxP site-specific recombination. The LoxP sequences in the DNA are represented by triangles that are colored differently so that the site-specific recombination event can be followed more readily. In reality their DNA sequences are identical. (B) DNA substrates containing two arrangements of LoxP sites.

**5-17** It is thought that a self-correcting polymerase cannot start chains *de novo* because the initial nucleotides will be weakly paired, thus subject to removal by an efficient proofreading exonuclease. This argument implies the converse: an enzyme that starts chains anew cannot be efficient at self-correction. Remarkably, SARS-CoV-2 encodes an RNA replicase complex that is able to do both: using RNA as a template, it starts strands *de novo* and removes errors with an efficient 3'-to-5' proofreading exonuclease. How might this be possible?

## REFERENCES

## General

- Brown TA (2017) *Genomes 4*. New York: Garland Science.
- Friedberg EC, Walker GC, Siede W et al. (2005) *DNA Repair and Mutagenesis*. Washington, DC: ASM Press.
- Haber JE (2013) *Genome Stability: DNA Repair and Recombination*. New York: Garland Science.
- Stent GS (1971) *Molecular Genetics: An Introductory Narrative*. San Francisco: WH Freeman.
- Watson J, Baker T, Bell S et al. (2013) *Molecular Biology of the Gene*, 7th ed. Menlo Park, CA: Benjamin Cummings.

## The Maintenance of DNA Sequences

- Catarina D & Eichler EE (2013) Properties and rates of germline mutations in humans. *Trends Genet.* 29, 575–584.
- Conrad DF, Keebler J, DePristo M . . . 1000 Genomes Project (2011) Variation in genome-wide mutation rates within and between human families. *Nat. Genet.* 43, 712–714.
- Hedges SB (2002) The origin and evolution of model organisms. *Nat. Rev. Genet.* 3, 838–849.
- King MC & Wilson AC (1965) Evolution at two levels in humans and chimpanzees. *Science* 188, 107–116.

## DNA Replication Mechanisms

- Kornberg A (1960) Biological synthesis of DNA. *Science* 131, 1503–1508.
- Li JJ & Kelly TJ (1984) SV40 DNA replication *in vitro*. *Proc. Natl. Acad. Sci. USA* 81, 6973–6977.
- Meselson M & Stahl FW (1958) The replication of DNA in *E. coli*. *Proc. Natl. Acad. Sci. USA* 44, 671–682.
- Modrich P (2016) Mechanisms in *E. coli* and human mismatch repair (Nobel Lecture). *Angew. Chem. Int. Ed. Engl.* 55(30), 8490–8501.
- Okazaki R, Okazaki T, Sakabe K, Sugimoto K & Sugino A (1968) Mechanism of DNA chain growth. I. Possible discontinuity and unusual secondary structure of newly synthesized chains. *Proc. Natl. Acad. Sci. USA* 59, 598–605.
- Yao NY & O'Donnell ME (2021) The DNA replication machine: structure and dynamic function. In *Macromolecular Protein Complexes III: Structure and Function* (Harris JR & Marles-Wright J, eds.), pp. 233–258. *Subcellular Biochemistry*, Vol. 96. Cham, Switzerland: Springer.

## The Initiation and Completion of DNA Replication in Chromosomes

- Bell SP & Labib K (2016) Chromosome duplication in *Saccharomyces cerevisiae*. *Genetics* 203(3), 1027–1067.
- Chan SR & Blackburn EH (2004) Telomeres and telomerase. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 359, 109–121.
- Dewar JM & Walter JC (2017) Mechanisms of DNA replication termination. *Nat. Rev. Mol. Cell Biol.* 18, 507–516.
- On KF, Jaremko M, Stillman B & Leemor JT (2018) A structural view of the initiators for chromosome replication. *Curr. Opin. Struct. Biol.* 53, 131–139.
- Roake CM & Artandi SE (2020) Regulation of human telomerase in homeostasis and disease. *Nat. Rev. Mol. Cell Biol.* 21, 384–397.

- Schvartzman JB, Hernandez P, Krimer DB, Dorier J & Stasiak A (2019) Closing the DNA replication cycle: from simple circular molecules to supercoiled and knotted DNA catenanes. *Nucleic Acids Res.* 47(14), 7182–7198.
- Wang Y, Sušac L & Feigon J (2019) Structural biology of telomerase. *Cold Spring Harb. Perspect. Biol.* 11(12), a032383.

## DNA Repair

- Her J & Bunting SF (2018) How cells ensure correct repair of DNA double-strand breaks. *J. Biol. Chem.* 293(27), 10502–10511.
- Lindahl T (1993) Instability and decay of the primary structure of DNA. *Nature* 362, 709–715.
- Sancar A (2016) Mechanisms of DNA repair by photolyase and excision nuclease (Nobel Lecture). *Angew. Chem. Int. Ed. Engl.* 55(30), 8502–8527.
- Van den Heuvel D, Van der Weegen Y, Boer DEC, Ogi T & Luijsterburg MS (2021) Transcription-coupled DNA repair: from mechanism to human disorder. *Trends Cell Biol.* 31(5), 359–371.
- Yang W & Gao Y (2018) Translesion and repair DNA polymerases: diverse structure and mechanism. *Annu. Rev. Biochem.* 87, 239–261.
- Zhao B, Rothenberg E, Ramsden DA & Lieber MR (2020) The molecular basis and disease relevance of non-homologous DNA end joining. *Nat. Rev. Mol. Cell Biol.* 21, 765–781.

## Homologous Recombination

- Cox MM (2001) Historical overview: searching for replication help in all of the rec places. *Proc. Natl. Acad. Sci. USA* 98, 8173–8180.
- Haber JE (2016) A life investigating pathways that repair broken chromosomes. *Annu. Rev. Genet.* 50, 1–28.
- Holliday R (1990) The history of the DNA heteroduplex. *BioEssays* 12, 133–142.
- Kreuzer KN & Brister JR (2010) Initiation of bacteriophage T4 DNA replication and replication fork dynamics: a review in the Virology Journal series on bacteriophage T4 and its relatives. *Virology* 401, 358.
- Szostak JW, Orr-Weaver TK, Rothstein RJ & Stahl FW (1983) The double-strand break repair model for recombination. *Cell* 33, 25–35.
- Wyatt HDM & West SC (2014) Holliday junction resolvases. *Cold Spring Harb. Perspect. Biol.* 6(9), a023192.
- Zickler D & Kleckner N (2015) Recombination, pairing, and synapsis of homologs during meiosis. *Cold Spring Harb. Perspect. Biol.* 7(6), a016626.

## Transposition and Conservative Site-specific Recombination

- Comfort NC (2001) From controlling elements to transposons: Barbara McClintock and the Nobel Prize. *Trends Biochem. Sci.* 26, 454–457.
- Huang, CR, Burns KH & Boeke JD (2012) Active transposition in genomes. *Annu. Rev. Genet.* 46, 651–675.
- Payer LM & Burns KH (2019) Transposable elements in human genetic disease. *Nat. Rev. Genet.* 20(12), 760–772.
- Varmus H (1988) Retroviruses. *Science* 240, 1427–1435.